

**THE COMPLETE INTRINSIC CONGRUENCE SYSTEM FOR
THE ERDŐS–STRAUS CONJECTURE:
EXACT STRUCTURE AND PROVISIONAL EXCEPTIONAL-SET
BOUNDS FOR GENERAL NUMERATORS**

ANONYMOUS

ABSTRACT. Subject to external verification, we obtain the unconditional bound

$$E_{\text{all}}(N) \ll N \exp\{-c(\log N)^{3/4}\}$$

for Erdős–Straus exceptional denominators. More generally, if $E_m(N)$ counts $n \leq N$ for which m/n is not a sum of three positive unit fractions, then, for every fixed $\varepsilon > 0$, uniformly for $3 \leq m \leq (\log N)^{3-\varepsilon}$, we obtain the second headline

$$E_m(N) \ll_\varepsilon N \exp\left\{-c_\varepsilon \left(\frac{\eta_2(m)(\log N)^3}{\varphi(m)}\right)^{1/4}\right\}, \quad \eta_2(m) = \prod_{p|m} \frac{p^2}{p^2+p-1}.$$

Both 3/4 theorems are explicitly CLAIMED/PROVISIONAL; their load-bearing pruned cubic prime-slice input, Theorem 9.4 (the analogue of Theorem 34.8 in the source notes), is likewise CLAIMED/PROVISIONAL, and the second headline inherits the three qualifications of the source §39.7. No assertion in this draft should be cited as an established improvement before independent expert review. A separate, completely written internal layer gives

$$E_m(N) \ll_m N \exp\left\{-c \left(\frac{\eta_1(m)}{\varphi(m)}\right)^{1/3} (\log N)^{2/3} (\log \log N)^{1/3}\right\};$$

because this too would improve the published general-numerator benchmark, it remains CLAIMED/PROVISIONAL pending external priority and referee checks, although it assumes no unproved hypothesis. We further blind-adjudicate an exact, record-adjacent CLAIMED/PROVISIONAL effectivity perimeter, prove a two-scale tail for the minimal Type-II witness modulus W (cubic variant CLAIMED/PROVISIONAL), and prove fixed-slice and stacked Type-I almost-all theorems. Effectively, (50.7) fails on at most an $\exp\{-c_A(\log \log X)^{3/2}/\sqrt{\log \log \log X}\}$ proportion of hard primes for fixed $0 < A < 1$. In the opposite direction, we prove effectively that $W(p) > c \log p$ and $ck_{\min}(p) > c \log p$ infinitely often; hence both $H_{\text{MOD}}(A)$ and $H_{\text{SPF}}(A)$ are false for every $0 < A < 1$, leaving exactly $A \geq 1$ open in the two pointwise frames. We also prove that a complete congruence-class certificate through depth T has $\log Q \geq (1/2 + o(1))T$ in Type II and $\log Q \geq T + o(T)$ in Type I. These are effective ceilings on the certificate-plus-least-prime architecture, not pointwise upper bounds on either depth statistic. Exact finite censuses through the stated endpoints remain informational only.

For each $M \equiv 3 \pmod{4}$, put $A = (M+1)/4$. The congruence identities studied here give the intrinsic forbidden set

$$\mathcal{R}(M) = \{-4D \pmod{M} : D \mid A^2\}.$$

We prove this description with honest deduplication and show that its total class mass through X is of exact order $(\log X)^3$. The prime-modulus subsystem has mass $\Theta((\log X)^2)$ and avoids density $\exp\{-\Theta((\log X)^2)\}$. Every square avoids the full intrinsic system.

Separately, a weighted rough-modulus estimate and the Lovász local lemma refute the literal all-range partition-free hypothesis H_{PF} by proving

$$\delta_X \geq \exp\left\{-O\left(\frac{(\log X)^3 \log \log \log X}{\log \log X}\right)\right\} = \exp\{-o((\log X)^3)\}.$$

That refutation does not apply in the critical window $\log N \asymp (\log X)^4$. The provisional 3/4 argument instead uses a fixed c -free atom family, residue-resolved factorial moments, a uniform conditioned void supplied by Theorem 9.4, and an even-Bonferroni majorant with the honest coefficient ledger $\exp\{O((\log X)^4)\}$. The literal $\exp\{O(J)\}$ ledger clause of the earlier critical-window hypothesis is not proved and is deliberately bypassed. At the separate residue-dispersion frontier, a nested divisor cube refutes the literal raw factorial-moment and codegree targets, while a complete bipartite grid of maximal $D = 2$ semiprime atoms refutes their implication-antichain replacements. These failures concern the proposed moment formulations, not the critical-window hypothesis, the pair endpoint, or the conjecture; the complete-system moment axis is now dormant. The endpoint has a unique short cofactor, and its closed prefix reaches $4c^2s \ll (\log X)^5 \log \log \log X / (\log \log X)^2$, with an additional large- c wedge. Its exact residual distinct-cell collision sum remains OPEN. On the pointwise side we prove a one-prime-factor slice criterion and isolate a bespoke unproved hypothesis implying Type-I representations for all sufficiently large hard primes. The accompanying GRH, Bateman–Horn, Chebotarev, and Duke audit is a map of the missing input, not an unconditional pointwise theorem.

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1. INTRODUCTION AND STATEMENT MAP

The Erdős–Straus conjecture asks whether, for every integer $n \geq 2$, there are positive integers x, y, z such that

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}. \quad (1)$$

It remains open. The solution parametrizations and average counts of Elsholtz–Tao [7], the divisor criterion presented by Bloom–Elsholtz [2], and the exceptional-set argument reconstructed by Pomerance–Weingartner [13] form the nearest literature context. Yamamoto’s reciprocity analysis [20], Salez’s computation [15], and the later counting work of Elsholtz–Planitzer [6] provide additional context; none is used to promote a conditional or provisional statement below.

The two headline assertions of this draft are

$$\boxed{E_{\text{all}}(N) \ll N \exp\{-c(\log N)^{3/4}\}}, \quad \boxed{E_m(N) \ll_{\varepsilon} N \exp\left\{-c_{\varepsilon} \left(\frac{\eta_2(m)(\log N)^3}{\varphi(m)}\right)^{1/4}\right\}}$$

where the second bound is uniform, for every fixed $\varepsilon > 0$, in $3 \leq m \leq (\log N)^{3-\varepsilon}$; both are CLAIMED/PROVISIONAL. Here $E_m(N)$ has the meaning given in the abstract. The assertions are unconditional in the narrow sense that no unproved hypothesis is assumed, but they are subject to external verification. Their load-bearing prime-slice input, Theorem 9.4, has the same provisional status, and the general form inherits the source §39.7 qualifications. Its exact statement is Theorem 11.8; Theorem 11.7 records the weaker linear-thinning replay. Beyond the records, Theorems 14.2 and 14.6 give the exact W windows and effectivity perimeter, while Theorem 15.1 and Corollary 16.5 give the fixed-slice and stacked almost-all frontiers. These upper tails are not pointwise. Theorems 17.1 and 17.3 supply the opposite, pointwise lower tail effectively. Theorems 18.1 and 18.2 then give the exact proved exponential modulus costs for complete-class certificates, while Corollary 18.3 records the scoped least-prime consequence. Theorem 19.2 records the status-preserving general- m truncation.

The paper has five layers.

- (i) Section 3 derives a complete two-case divisor criterion for prime denominators and the identity families that lead to the intrinsic congruence system.
- (ii) Sections 4–5 determine that system exactly. The distinct-modulus class mass is cubic, while the full prime slice and its avoidance exponent are quadratic. Squares survive every modulus, and a multi-shift certificate gives a subcubic-cost lower bound for a growing part of the composite system.
- (iii) Sections 6–7 state the literal cubic-mass majorant hypothesis H_{PF} , prove its formerly desired conditional consequence, and then refute the hypothesis with all quantifiers visible. The analytic inputs are upper mean-value estimates of Nair–Tenenbaum [12] and Henriot [9]; Shiu’s theorem [17] is used in the prime-harvesting arguments.
- (iv) Section 8 separates that all-range refutation from the critical window and records the cylinder, tensor, finite-transfer, and factorial-moment walls.
- (v) Section 9 states the provisional record inputs. Section 10 then gives the complete fixed- c -free assembly leading, provisionally, to the 3/4 exponent, followed by an explicit internal-verification pedigree. Section 11 proves the m -uniform identity and the two general-numerator layers, including the attested-blind thinned-window strengthening, its fixed-gap range, and the post-adjudication two-regime comparison with Pomerance–Weingartner. Section 12 records the moving genus obstruction, the exact fixed-divisor progression law and its residue-one escape, while keeping pointwise Type-I existence open. Section 13 then proves the one-prime-factor criterion, states the bespoke unproved slice-factor hypothesis, and audits why the named standard hypotheses do not supply its pointwise quantifier. Sections 14–16 add the witness tail, effectivity/priority register, fixed-slice sieve, and effective stacked almost-all theorem. Section 17 then proves the logarithmic lower tail and sharpens both pointwise exponent frontiers. Section 18 proves the complete-class certificate ceilings, states their exact architecture-only perimeter, and extends all three finite depth censuses. Section 19 gives the status-preserving general-numerator tails and their exact effectivity perimeter. Finally, Section 20 gives the endpoint sparsity, Kloosterman-fraction audit, widened prefix and large- c wedge, and exact open collision remainder. The composite frontier records both successive refutations: the divisor cube kills the raw hierarchy, and the

maximal semiprime grid kills its implication-antichain repair. Scott–Sokal’s partition-function correspondence [16] does not itself provide the required pointwise minorant.

Labels are part of the assertions. “Proved” means proved internally in this draft. “Open” means no proof is claimed. The record results in Sections 9–11, the cubic witness tail, Theorem 14.6, and the cubic row of Theorem 19.2 remain CLAIMED/PROVISIONAL as labelled. The logarithmic lower-tail and congruence-certificate theorems are proved and effective. The normalized census panels are informational, the GRH least-prime remark is conditional, and the congruence wall is an architecture assessment rather than a universal pointwise theorem. Writing out internal arguments and reviews for the provisional results does not replace independent external priority and referee checks. In particular, no result of this paper proves the Erdős–Straus conjecture.

Related formalization and frontier. A publicly available Lean development titled “auro-zero” formalizes a sound algebraic identity and finite case-split layer, then assumes one axiom, `good_divisor_exists`, for the remaining hard classes. That axiom is the open divisor-existence core rather than a proof of it; local compilation of the file was not checked. Its cited support is Dyachenko [5], whose own Conclusion says that the infinite general case is not proved constructively and that the relevant finite-covering argument is conditional. The development is nevertheless useful as a formal isolation of the same frontier: Sections 14–18 quantify, almost everywhere, on the effective lower tail, and at the complete-class certificate perimeter, the witness existence that its axiom assumes. No result here depends on that file or on Dyachenko’s conditional existence claim.

The historical comparison is also provisional. Vaughan’s 1970 bound [18] is the record identified by the search. Pomerance–Weingartner’s 2025 preprint [13, §4] cites Vaughan as the state of the art, and a fresh search found no post-1970 improvement of the exceptional-set exponent. Access to Vaughan’s primary paper was blocked, although its DOI was confirmed; claims about its method have therefore been checked only through Pomerance–Weingartner. Thus, if the $m = 4$ theorem survives external review, it would be the first exponent improvement found since 1970. The general- m comparison is made only with the precise 2025 theorem quoted in (43.22). These are search results, not literature guarantees.

2. NOTATION AND ELEMENTARY TOOLS

All logarithms are natural. We write $\tau_k(n)$ for the ordered k -fold divisor function, $\tau = \tau_2$, $\omega(n)$ for the number of distinct prime factors, $P^-(n)$ for the least prime factor, and φ for Euler’s function. Constants implicit in $O, \ll, \asymp$ are absolute unless a subscript says otherwise.

For $X \geq 3$, put $L = \log X$. Once the intrinsic sets have been defined, their full avoider set is

$$\text{Av}_X(N) = \{n \leq N : n \bmod M \notin \mathcal{R}(M) \text{ for every } M \leq X, M \equiv 3 \pmod{4}\}. \quad (2)$$

For fixed X this predicate is periodic, so its natural density δ_X exists. The superscript “prime” restricts the moduli to primes $\ell \equiv 3 \pmod{4}$.

We use three standard inputs: Bombieri–Vinogradov in a fixed power-saving form, Brun–Titchmarsh, and the quantitative Lovász local lemma. The latter says

that events A with a dependency graph have positive joint avoidance probability at least $\prod_A(1 - x_A)$ if

$$\Pr(A) \leq x_A \prod_{B \sim A} (1 - x_B), \quad 0 \leq x_A < 1.$$

The product-space form and its relation to the independent-set polynomial are discussed in [16].

3. CRITERION AND IDENTITY FAMILIES

3.1. A two-term lemma and the complete prime criterion.

Lemma 3.1 (two-term criterion). *Let a, b be coprime positive integers. The equation $a/b = 1/y + 1/z$ has a solution in positive integers if and only if b^2 has a divisor $d \equiv -b \pmod{a}$. Its solutions are*

$$y = \frac{b+d}{a}, \quad z = \frac{b+b^2/d}{a}.$$

Proof. If $d \mid b^2$ and $d \equiv -b \pmod{a}$, then $d' = b^2/d$ also satisfies $d' \equiv -b \pmod{a}$, since b is invertible modulo a . The displayed y, z are integral and

$$(d+b)(d'+b) = b(d+d'+2b),$$

which gives $1/y + 1/z = a/b$. Conversely, from $ayz = b(y+z)$ put $d = ay - b$ and $d' = az - b$. Positivity follows from $1/y < a/b$ and $1/z < a/b$, and $dd' = a^2yz - ab(y+z) + b^2 = b^2$. Thus $d \mid b^2$ and $d \equiv -b \pmod{a}$. $\square$

If p is an odd prime in a solution of (1), at least one denominator is divisible by p , because $4xyz = p(xy + yz + zx)$, and not all three are, since after division their reciprocal sum would be at most 3. Thus exactly one or exactly two denominators are divisible by p .

Theorem 3.2 (complete divisor criterion). *Let p be an odd prime. Equation (1) is soluble if and only if at least one of the following holds.*

(B):

There is $q > 0$ with $q \equiv -p \pmod{4}$ and a divisor $d \mid x_0^2$, where $x_0 = (p+q)/4$, such that $q \mid d + x_0$.

(A):

There is $m > 0$ with $m \equiv 3p \pmod{4}$ and a divisor $d \mid z_0^2$, where $z_0 = (pm+1)/4$, such that $m \mid d + z_0$.

The corresponding solutions are respectively

$$\left(x_0, \frac{p(x_0+d)}{q}, \frac{p(x_0+x_0^2/d)}{q}\right), \quad \left(\frac{z_0+d}{m}, \frac{z_0+z_0^2/d}{m}, pz_0\right).$$

Proof. If two denominators are divisible by p , write them py', pz' . Then

$$\frac{4}{p} - \frac{1}{x_0} = \frac{1}{p} \left(\frac{1}{y'} + \frac{1}{z'} \right), \quad q := 4x_0 - p > 0,$$

so $q/x_0 = 1/y' + 1/z'$. Also $(q, x_0) = (p, x_0) = 1$, since $p \mid x_0$ would put p in all three denominators. Lemma 3.1 gives (B), both formulas, and the converse.

If exactly one denominator is divisible by p , call it pz_0 . The equation becomes

$$xy(4z_0 - 1) = pz_0(x + y).$$

Since $p \nmid xy$, write $4z_0 - 1 = pm$. Then $m/z_0 = 1/x + 1/y$ and $(m, z_0) = 1$. Lemma 3.1 gives (A), while $m \equiv 3p \pmod{4}$ is exactly the integrality condition for z_0 . Reversing the calculation proves the converse. $\square$

This is an independent derivation of the criterion whose shape agrees with Theorem 1 of Bloom–Elsholtz [2].

3.2. Forced identities. For $D = \prod r^{e_r}$ define

$$R_0(D) = \prod_r r^{\lceil e_r/2 \rceil}. \quad (3)$$

Thus $R_0(D)$ is the least positive integer R for which $R \mid x$ forces $D \mid x^2$.

Fix q, d in case (B) and an R divisible by $R_0(d)$. The simultaneous conditions

$$x_0 \equiv 0 \pmod{R}, \quad x_0 \equiv -d \pmod{q}, \quad p = 4x_0 - q$$

form, when consistent, one progression modulo $4\text{lcm}(R, q)$, and every member is represented by Theorem 3.2. The case-(A) construction is identical with m in place of q . Two classical-looking forms are also immediate.

Proposition 3.3 (elementary identity families). *For positive integers g, u, v ,*

$$p = 4guv - u - v \implies \frac{4}{p} = \frac{1}{guv} + \frac{1}{pgu} + \frac{1}{pgv}.$$

For $j \geq 1$, the choice $q = (x_0 + 1)/j$ and $d = 1$ in case (B) gives

$$jp = (4j - 1)x_0 - 1,$$

and hence covers exactly the class $p \equiv -4 \pmod{4j - 1}$.

Proof. The first right side has common denominator $pguv$ and numerator $p + u + v = 4guv$. For the second, substitute $q = (x_0 + 1)/j$ into $p = 4x_0 - q$. Since $4j \equiv 1 \pmod{4j - 1}$, integrality is equivalent to $p \equiv -4 \pmod{4j - 1}$; then $q \mid x_0 + 1 = d + x_0$ and Theorem 3.2 applies. $\square$

Dyachenko’s ED2 identity family [5] is an algebraic cousin of the identities in Proposition 3.3 and of the multiplier supply below. We cite it only at the identity level: the paper’s own Conclusion characterizes its infinite general case as not constructively proved and the relevant finite-covering steps as conditional.

The identity from which the complete intrinsic system arises is the following multiplier form.

Lemma 3.4 (intrinsic multiplier identity). *Let $M \equiv 3 \pmod{4}$, $A = (M + 1)/4$, and $A = uvw$ a factorization into positive integers. If the positive integer n satisfies $nv \equiv -u \pmod{M}$ and $s = (nv + u)/M$, then*

$$\frac{4}{n} = \frac{1}{suw} + \frac{1}{nsuv} + \frac{1}{nuvw}. \quad (4)$$

In particular $n \equiv -uv^{-1} \pmod{M}$ is a forced-solubility class.

Proof. Since $(A, M) = 1$, the factor v is invertible modulo M , and s is a positive integer. On the common denominator $nsuvw$, the numerator on the right is $nv + u + s = sM + s = s(M + 1) = 4suw$. $\square$

4. THE COMPLETE INTRINSIC SYSTEM

For $M \equiv 3 \pmod{4}$, define

$$\mathcal{R}(M) = \{-uv^{-1} \pmod{M} : uvw = (M+1)/4\}, \quad F(M) = |\mathcal{R}(M)|.$$

“Complete” always means complete for the multiplier identity of Lemma 3.4; it does not assert that all possible representations of (1) arise from one fixed M .

Lemma 4.1 (exact structure and deduplication). *Put $A = (M+1)/4$. Then*

$$\mathcal{R}(M) = \{-4D \pmod{M} : D \mid A^2\}, \quad \frac{\tau(A^2) - 1}{2} \leq F(M) \leq \tau(A^2). \quad (5)$$

The lower-bound classes may be taken from the divisors $D < A$, and they are pairwise distinct.

Proof. A factorization $A = uvw$ gives $D = u^2w \mid A^2$, and $4A \equiv 1 \pmod{M}$ gives

$$-uv^{-1} \equiv -u^2wA^{-1} \equiv -4u^2w = -4D \pmod{M}.$$

Conversely, write $A = \prod p^{e_p}$ and $D = \prod p^{d_p}$ with $0 \leq d_p \leq 2e_p$. Set

$$v_p(u) = \lfloor d_p/2 \rfloor, \quad v_p(w) = d_p - 2\lfloor d_p/2 \rfloor, \quad v_p(v) = e_p - v_p(u) - v_p(w).$$

The last number is nonnegative; hence $uvw = A$ and $u^2w = D$. This proves the set identity and upper bound. The involution $D \mapsto A^2/D$ has unique fixed point A , so exactly $(\tau(A^2) - 1)/2$ divisors have $D < A$. For them $0 < 4D < M$, making the residues $-4D$ distinct. $\square$

The set depends on M , not on a factorization $M = k\ell$. Thus a sum over pairs (k, ℓ) must be quotiented by their product before it represents distinct intrinsic moduli.

Theorem 4.2 (exact cubic supply). *As $Q \rightarrow \infty$,*

$$\sum_{\substack{M \leq Q \\ M \equiv 3 \pmod{4}}} \frac{F(M)}{M} \asymp (\log Q)^3. \quad (6)$$

More precisely,

$$\left(\frac{1}{48\zeta(2)} + o(1) \right) (\log Q)^3 \leq \sum_{\substack{M \leq Q \\ M \equiv 3 \pmod{4}}} \frac{F(M)}{M} \leq \left(\frac{1}{24\zeta(2)} + o(1) \right) (\log Q)^3.$$

Proof. At each prime,

$$\sum_{e \geq 0} (2e+1)z^e = \frac{1-z^2}{(1-z)^3},$$

so

$$\sum_{n \geq 1} \frac{\tau(n^2)}{n^s} = \frac{\zeta(s)^3}{\zeta(2s)}, \quad \tau(n^2) = \sum_{d^2 \mid n} \mu(d) \tau_3(n/d^2).$$

The elementary three-dimensional hyperbola argument gives

$$\sum_{m \leq x} \tau_3(m) = \frac{1}{2}x(\log x)^2 + O(x \log x).$$

Substitution in the coefficient identity, using absolute convergence of the d^{-2} sum, yields

$$\sum_{n \leq x} \tau(n^2) = \frac{x(\log x)^2}{2\zeta(2)} + O(x \log x).$$

Partial summation gives

$$\sum_{n \leq x} \frac{\tau(n^2)}{n} = \frac{(\log x)^3}{6\zeta(2)} + O((\log x)^2). \quad (7)$$

Write $M = 4A - 1$. Since $(4A - 1)^{-1} = (4A)^{-1} + O(A^{-2})$ and $\tau(A^2) \ll_\varepsilon A^\varepsilon$, replacing the denominator costs $O(1)$. Apply the two halves of Lemma 4.1 in (7); the subtracted 1 in the lower bound costs only $O(\log Q)$. $\square$

A mass-only Rankin sieve with auxiliary scale $X = e^t$, class mass $O(t^B)$, and product-tail budget $t^{B+1} \ll \log N$ can save at most $O((\log N)^{B/(B+1)})$. Thus the intrinsic supply has the formal ceiling $3/4$. This is only a ceiling inside that mass/product-tail model; it is not a nonexistence theorem for other uses of the class geometry.

4.1. Shifted-divisor form.

Lemma 4.3 (exact reorganization). *For $A = (M + 1)/4$,*

$$D \mid A^2 \iff R_0(D) \mid A \iff M \equiv -1 \pmod{4R_0(D)}.$$

Consequently $n \notin \text{Av}_X$ if and only if there are $D, M \geq 1$ such that

$$M \mid n + 4D, \quad M \leq X, \quad M \equiv -1 \pmod{4R_0(D)}. \quad (8)$$

If $R = \prod p^{a_p}$, then

$$R_0(D) = R \iff D = R^2/s \text{ for some } s \mid \text{rad}(R). \quad (9)$$

Proof. At a prime, $D \mid A^2$ says $e_p(D) \leq 2v_p(A)$, exactly the assertion $\lceil e_p(D)/2 \rceil \leq v_p(A)$. This proves the first line. Lemma 4.1 changes $n \equiv -4D \pmod{M}$ into (8), in both directions. Finally, if $v_p(R) = a$, the condition $\lceil v_p(D)/2 \rceil = a$ allows $v_p(D) = 2a$ or $2a - 1$, independently at each $p \mid R$, which is (9). $\square$

The congruence in (8) implies $R_0(D) \leq (X + 1)/4$ and hence $D \leq (X + 1)^2/16$; no D -tail is omitted.

5. SQUARES AND THE PRIME SLICE

5.1. The square-class phenomenon.

Lemma 5.1 (quadratic sign). *If $M = 4A - 1$ and $D \mid A^2$, then*

$$\left(\frac{D}{M}\right) = 1, \quad \left(\frac{-4D}{M}\right) = -1.$$

Thus every unit class in $\mathcal{R}(M)$ has Jacobi sign -1 . Every n with $(n/M) \in \{0, 1\}$ avoids $\mathcal{R}(M)$; in particular, every perfect square avoids the complete intrinsic system for every X .

Proof. For an odd prime $p \mid A$, quadratic reciprocity and $M \equiv -1 \pmod{p}$, $M \equiv 3 \pmod{4}$ give

$$\left(\frac{p}{M}\right) = \left(\frac{M}{p}\right) (-1)^{(p-1)(M-1)/4} = \left(\frac{-1}{p}\right) (-1)^{(p-1)/2} = 1.$$

If $2 \mid A$, then $M \equiv 7 \pmod{8}$ and $(2/M) = 1$. Multiplicativity gives $(D/M) = 1$. Since $(-1/M) = -1$ and $(4/M) = 1$, the second identity follows. Also $(A, M) = 1$, so all intrinsic classes are units. $\square$

In particular $|\text{Av}_X(N)| \geq \lfloor \sqrt{N} \rfloor$. Unit squares modulo the odd period $P_X = \text{lcm}\{M \leq X : M \equiv 3 \pmod{4}\}$ also give

$$\delta_X \geq \frac{\varphi(P_X)}{2^{\omega(P_X)} P_X} \geq \exp\{-O(X/\log X)\}.$$

These bounds are too small to test a cubic exponent when $\log N \asymp L^4$.

5.2. Harvesting prime classes. We record the fixed-multiplier harvesting input in a self-contained form.

Lemma 5.2 (prime harvest). *For primes $\ell \equiv 3 \pmod{4}$, there are subsets $\mathcal{C}_\ell \subseteq \mathcal{R}(\ell)$ such that*

$$\sum_{X^{1/2} < \ell \leq X} \frac{|\mathcal{C}_\ell|}{\ell} \asymp L^2. \quad (10)$$

Proof. In a dyadic interval $x < \ell \leq 2x$, put $z = x^{1/6}$ and take coprime $u, v \leq z$ with $\omega(uv) \leq D \log \log X$. The progression $\ell \equiv -1 \pmod{4uv}$ makes $uv \mid (\ell + 1)/4$, so Lemma 3.4 supplies the residue $-uv^{-1} \pmod{\ell}$. Distinct reduced pairs give distinct residues: a collision implies $\ell \mid uv' - u'v$, while $|uv' - u'v| < \ell^{2/3} < \ell$, and hence $u/v = u'/v'$.

The elementary harmonic estimates

$$\sum_{\substack{u, v \leq z \\ (u, v) = 1}} \frac{1}{uv} \asymp (\log z)^2, \quad \sum_{u, v \leq z} \frac{1}{\varphi(u)\varphi(v)} \ll (\log z)^2$$

follow respectively by Möbius inversion and from $\sum_{n \leq z} 1/\varphi(n) \ll \log z$. Moreover, for fixed $1 < t < 2$,

$$\sum_{u, v \leq z} \frac{t^{\omega(u) + \omega(v)}}{uv} \ll_t (\log z)^{2t}.$$

Rankin's inequality therefore shows that a fixed D leaves a positive proportion of the first harmonic sum. A modulus $4uv$ is then generated at most $2^{\omega(uv)} \leq (\log X)^{D \log 2}$ times. Bombieri–Vinogradov applies since $4uv \leq 4x^{1/3}$ and absorbs this fixed logarithmic multiplicity. It gives $\gg x \log x$ retained prime incidences in the block; division by $\ell \asymp x$ gives $\gg \log x$. There are $\asymp L$ dyadic blocks between $X^{1/2}$ and X , proving the lower bound in (10). Brun–Titchmarsh and the second harmonic estimate give $O(x \log x)$ incidences per block and hence the matching upper bound. $\square$

Lemma 5.3 (full prime mass). *There are $c, C > 0$ such that*

$$cL^2 \leq \sum_{\substack{\ell \leq X \\ \ell \equiv 3 \pmod{4}}} \frac{F(\ell)}{\ell} \leq CL^2, \quad (11)$$

where ℓ is prime.

Proof. The lower bound follows from Lemma 5.2. For the upper bound put $A = (\ell + 1)/4$. Lemma 4.1 and $2e + 1 \leq (e + 1)(e + 2)/2$ give $F(\ell) \leq \tau(A^2) \leq \tau_3(A)$. On $A \asymp Y$, write $A = abc$ and designate a largest coordinate as c , at a factor-three cost; then $ab \ll Y^{2/3}$. For fixed a, b , Brun–Titchmarsh for the prime $4abc - 1$ modulo $4ab$ gives

$$\#\{c : A \asymp Y, 4abc - 1 \text{ prime}\} \ll \frac{Y}{\varphi(4ab) \log Y}.$$

Since $\varphi(4ab) \geq \varphi(a)\varphi(b)$ and $\sum_{m \leq y} 1/\varphi(m) \ll \log(2y)$,

$$\sum_{\substack{A \asymp Y \\ 4A-1 \text{ prime}}} \tau(A^2) \ll Y \log Y.$$

After division by $\ell \asymp Y$, each dyadic block costs $O(\log Y)$, and summing proves the upper bound. $\square$

Theorem 5.4 (two-sided prime-slice avoidance). *There are $c_1, C_2, C_3 > 0$ such that, for all large X and $\log N \geq C_3 L^3$,*

$$N e^{-C_2 L^2} \leq |\text{Av}_X^{\text{prime}}(N)| \leq N e^{-c_1 L^2}. \quad (12)$$

Its natural density is $e^{-\Theta(L^2)}$.

Proof. Chinese remaindering over distinct prime moduli gives exact density

$$V_X = \prod_{\substack{\ell \leq X \\ \ell \equiv 3 \pmod{4}}} \left(1 - \frac{F(\ell)}{\ell}\right).$$

Lemma 5.1 gives $F(\ell)/\ell < 1/2$. Lemma 5.3 and $-2u \leq \log(1 - u) \leq -u$ imply $V_X = e^{-\Theta(L^2)}$.

For the finite interval put $a_\ell = F(\ell)/\ell$ and $\mu = \sum a_\ell = \Theta(L^2)$. Odd and even Bonferroni truncations at degrees $r = O(\mu)$ bracket the avoider indicator. Their omitted CRT main tails are bounded by $\sum_{j > r} \mu^j/j!$; taking the constant in r large makes each tail at most $V_X/4$. An intersection indexed by a product d of j primes occupies $\prod_{\ell|d} F(\ell)$ classes modulo d , so its finite count is

$$N \prod_{\ell|d} a_\ell + O\left(\prod_{\ell|d} F(\ell)\right).$$

The preceding proof of Lemma 5.3, before division by ℓ , gives $W := \sum_{\ell \leq X} F(\ell) \ll XL$. Hence all rounding errors through degree r total at most $\sum_{j \leq r} W^j/j! = \exp\{O(L^3)\}$. If C_3 is large this is at most $NV_X/4$. The two Bonferroni bounds now give (12). $\square$

Since the full avoiders are contained in the prime-slice avoiders, we obtain the promised unconditional full-system upper bound.

Corollary 5.5. *If $\log N \geq CL^3$, then*

$$|\text{Av}_X(N)| \ll N \exp\{-cL^2\}.$$

The same upper bound holds for δ_X without a condition on N .

5.3. A growing multi-shift certificate. For $Y \geq 1$, let $\text{Av}_X^{(R \leq Y)}$ denote avoidance of every pair in (8) with $R_0(D) \leq Y$.

Theorem 5.6 (multi-shift prime certificate). *Fix $\varepsilon > 0$. For all sufficiently large X , uniformly for $1 \leq Y \leq L^{3-\varepsilon}$ and $\log N \geq L^4$,*

$$|\text{Av}_X^{(R \leq Y)}(N)| \geq N \exp\{-C_\varepsilon Y \log(2Y) \log L\}.$$

The corresponding natural density has the same lower bound. At $Y = L^{3-\varepsilon}$ the exponent is $o(L^3)$.

Proof. Let

$$\mathcal{D}_Y = \{(R, D) : R \leq Y, D = R^2/s, s \mid \text{rad}(R)\}, \quad H = |\mathcal{D}_Y|.$$

The standard Euler-product estimate

$$\sum_{R \leq Y} 2^{\omega(R)} \ll Y \log(2Y)$$

gives $H \ll Y \log(2Y)$. For each prime $p \leq X$, $p \equiv 3 \pmod{4}$, forbid

$$B_p = \{-4D \pmod{p} : (R, D) \in \mathcal{D}_Y, p \nmid R\}.$$

Every B_p is nonzero and has size $g(p) \leq \min(H, p-1)$. Avoiding all B_p certifies the desired avoidance: a hit in (8) has $(M, R) = 1$, and some prime $p \equiv 3 \pmod{4}$ divides M to odd exponent; then $p \leq X$, $p \nmid R$, and $n \equiv -4D \pmod{p}$, a contradiction.

For $p \leq 2H$, choose the allowed residue $n \equiv 0 \pmod{p}$, and let Q_0 be their product. Then $\log Q_0 = O(H)$. In $n = Q_0 m$, every remaining forbidden set has $g(p) \leq H < p/2$, and

$$\mu_Y = \sum_{2H < p \leq X} \frac{g(p)}{p} \ll H \log L.$$

Odd Bonferroni at degree $r \asymp \mu_Y$ retains a fixed fraction of the Euler product, at least $\exp(-2\mu_Y)$. Since $W_Y = \sum g(p) \leq H\pi(X)$, the total rounding error is

$$\exp\{O(r \log W_Y)\} = \exp\{O(HL \log L)\}.$$

For $Y \leq L^{3-\varepsilon}$ this is $\exp\{o(L^4)\}$ and is negligible under $\log N \geq L^4$. Restoring Q_0 proves the claimed bound. On complete periods there is no rounding error, proving the density statement. $\square$

6. THE CUBIC-MASS HYPOTHESIS AND ITS CONDITIONAL CONSEQUENCE

Put

$$\mu_X = \sum_{\substack{M \leq X \\ M \equiv 3 \pmod{4}}} \frac{F(M)}{M} \asymp L^3.$$

The following is the literal hypothesis proposed for partition-free assembly. Its all-range quantifier in N is essential.

Hypothesis 6.1 (the literal cubic-mass majorant H_{PF}). *There are absolute $c, C > 0$ such that, for every sufficiently large X , every $J \geq C\mu_X$, and every N satisfying*

$$J \log X \leq \frac{1}{2} \log N,$$

there is a nonnegative arithmetic function ν on $[1, N]$ which is at least 1 on every member of $\text{Av}_X(N)$ and satisfies

$$\frac{1}{N} \sum_{n \leq N} \nu(n) \ll \exp(-c\mu_X).$$

Its evaluation uses exact congruence counts modulo lcms of at most J participating moduli; every such lcm is at most $X^J \leq N^{1/2}$. The coefficient sum and number of congruence-count terms are at most $\exp(O(J))$, so aggregate rounding errors are required to be negligible under the displayed degree budget.

This hypothesis concerns the complete system, not arbitrary subfamilies. The latter formulation is already false: all $D = 1$ classes with $3 \mid M$ lie in $n \equiv 2 \pmod{3}$, while the complement has density $2/3$ despite growing class mass.

Theorem 6.2 (formal conditional $3/4$ bound). *If H_{PF} holds, then for some $c > 0$ the number $E_{\text{all}}(N)$ of exceptional denominators $n \leq N$ satisfies*

$$E_{\text{all}}(N) \leq N \exp\{-c(\log N)^{3/4}\}.$$

Proof. Put $t = \alpha(\log N)^{1/4}$, $X = e^t$. By Theorem 4.2, $\mu_X \asymp t^3$. Take $J = C\mu_X$. Then $J \log X \ll \alpha^4 \log N$, so small fixed α meets the degree condition. Hypothesis 6.1 bounds the avoiders by $N \exp(-c\mu_X)$. Every exceptional denominator is an avoider, because every intrinsic class is forced soluble by Lemma 3.4. $\square$

The implication is valid, but its antecedent will now be disproved.

7. ROUGH MODULI AND REFUTATION OF H_{PF}

For primes q define

$$b_q = \frac{L^3}{q \log z}, \quad W_{\mathbf{a}}(M) = \mathbf{1}_{P^-(M) > z} \exp\left(2 \sum_{p|M} a_p\right).$$

The sum in the exponential is over distinct prime factors. An *atom* is a pair (M, r) with $M \leq X$, $M \equiv 3 \pmod{4}$, and $r \in \mathcal{R}(M)$; it is the event $n \equiv r \pmod{M}$. Thus a modulus M contributes exactly $F(M)$ atoms.

Theorem 7.1 (weighted rough-modulus mean). *Uniformly for $3 \leq z \leq X^{1/10}$ and charges $0 \leq a_p \leq 1$, put $E = \sum_{z < p \leq X} a_p/p$. Then*

$$\sum_{\substack{M \leq X \\ M \equiv 3 \pmod{4}}} \frac{F(M)}{M} W_{\mathbf{a}}(M) \ll e^{cE} \frac{L^3}{\log z}. \quad (13)$$

In particular, if $a_p = O(L^3/(p \log z))$ and $z = L^3/\sqrt{\log L}$, then $E = o(1)$ and the left side is $o(L^3)$.

Proof. Write $M = 4m - 1$ and define the multiplicative function

$$w(p^\nu) = \begin{cases} 0, & p \leq z, \\ e^{2a_p}, & p > z, \end{cases} \quad w(1) = 1.$$

Lemma 4.1 gives $F(4m - 1)W_{\mathbf{a}}(4m - 1) \leq w(4m - 1)\tau_3(m)$. Apply the Nair–Tenenbaum estimate in Henriot’s discriminant-uniform form [12, 9] to the determinant-one linear forms $Q_1(t) = 4t - 1$, $Q_2(t) = t$ and the multiplicative function

$\mathcal{F}(u, v) = w(u)\tau_3(v)$. The class parameters are uniform in the charges. On a dyadic interval of length V , the local factors give

$$V \prod_{p \ll V} (1 - 2/p)(1 + w(p)/p)(1 + 3/p) \ll V \frac{\log^2(2V)}{\log z} \exp\left(C \sum_{z < p \ll V} \frac{a_p}{p}\right).$$

Indeed the coefficient of $1/p$ in the logarithm is 1 for $p \leq z$ and $2 + O(a_p)$ for $p > z$. The first possible rough value exceeds z , so the lowest nonempty blocks obey the same estimate after changing the constant. Dyadic summation and partial summation yield

$$\sum \frac{F(M)W_{\mathbf{a}}(M)}{M} \ll \frac{e^{CE}}{\log z} \left(L^2 + \int_z^X \frac{(\log t)^2}{t} dt \right),$$

which proves (13). Finally $E \ll L^3/(z \log z) = o(1)$ for the stated charges and cutoff. $\square$

The pointwise neighborhood estimate needs a small maximal-order correction. Fix a sufficiently large absolute C_d and put

$$u_q = \frac{\log(2q)}{q} \exp\left(\frac{C_d \log(2q)}{\log \log(3q)}\right).$$

For an atom modulus divisible by q , set

$$T_q(z; \mathbf{a}) = \sum_{\substack{M \leq X, \ M \equiv 3 \pmod{4} \\ P^-(M) > z, \ q|M}} \frac{F(M)}{M} \exp\left(2 \sum_{p|M} a_p\right).$$

Theorem 7.2 (softened neighborhood bound). *Let*

$$z = \frac{L^3 \log \log L}{\log L}. \quad (14)$$

For all large X there is an absolute $A > 0$ such that $a_q = A(b_q + u_q)$ for primes $z < q \leq X$ satisfies

$$T_q(z; \mathbf{a}) \leq a_q, \quad \max_{q > z} a_q = o(1), \quad \sum_{q > z} \frac{a_q}{q} = o(1),$$

and

$$\sum_{\substack{M \leq X \\ M \equiv 3 \pmod{4}}} \frac{F(M)}{M} W_{\mathbf{a}}(M) \ll \frac{L^3}{\log z} = o(L^3). \quad (15)$$

The same conclusions hold for $z = L^3 g(L)/\log L$ if $g(L) \rightarrow \infty$ and $g(L) = o(\log L)$.

Proof. Use the maximal-order bound

$$\tau_3(n) \leq \exp\left(\frac{C'_d \log(2n)}{\log \log(3n)}\right). \quad (16)$$

Fix $q > z$, write $M = qk$, and choose $r \in \{1, 3\}$ with $qr \equiv 3 \pmod{4}$. With $k = 4t + r$,

$$\frac{qk + 1}{4} = qt + c, \quad c = \frac{qr + 1}{4}, \quad 4c - qr = 1.$$

The cofactor and shifted factor are determinant-one linear forms, and the coefficient norm of their product is at most $13q$. Also $W_{\mathbf{a}}(qk) \leq e^{2a_q w(k)}$ and $F(qk) \leq \tau_3((qk+1)/4)$.

Split at $k = C_0 q^{1/2}$, where C_0 is large enough for Henriot's coefficient-size condition. In the lower range, (16), roughness, and $\sum_{k \leq y} 1/k \ll \log(2y)$ give

$$\frac{1}{q} \sum_{k < C_0 q^{1/2}} \frac{F(qk)W_{\mathbf{a}}(qk)}{k} \ll u_q.$$

Here $a_p = O(1/\log \log L + p^{-1+o(1)})$, and a rough qk has at most $\log(qk)/\log z$ prime factors, so its exponential weight is absorbed by enlarging the constant in (16). This includes $k = 1$.

Decompose the high range dyadically in t . Henriot's theorem applies to the two determinant-one forms and $w(4t+r)\tau_3(qt+c)$, giving

$$\sum_{V < k \leq 2V} w(k)\tau_3((qk+1)/4) \ll V \frac{(\log(2V))^2}{\log z}.$$

After division by qk and summation over blocks, this is $O(L^3/(q \log z)) = O(b_q)$. Thus $T_q \ll b_q + u_q$ uniformly once $\max a_p \leq 1$ and $\sum a_p/p \leq 1$. Choose the fixed A larger than this implicit constant; the estimates below verify those two conditions for all sufficiently large X , closing the argument without circularity.

Indeed

$$\sum_{q > z} \frac{b_q}{q} \ll \frac{L^3}{z \log z} = o(1).$$

For large q , the exponential in u_q is at most $q^{1/4}$, whence

$$\sum_{q > z} \frac{u_q}{q} \ll \sum_{n > z} \frac{\log(2n)}{n^{7/4}} = o(1).$$

The same estimates give $\max a_q = o(1)$. Theorem 7.1 proves (15). Replacing $\log \log L$ by $g(L)$ changes the first displayed bound to $O(1/g(L))$ and nothing else. $\square$

Theorem 7.3 (natural-density lower bound). *With (14),*

$$\delta_X \geq \exp \left\{ -O \left(\frac{L^3 \log \log L}{\log L} \right) \right\} = \exp \{ -o(L^3) \}. \quad (17)$$

More generally, the last cutoff in Theorem 7.2 gives $\delta_X \geq \exp \{ -O(L^3 g(L)/\log L) \}$.

Proof. First impose $n \equiv 0 \pmod{p}$ for every odd prime $p \leq z$. Work on the finite uniform CRT space modulo the lcm of their product and all participating moduli. The conditioning costs $\exp \{ -\vartheta(z) + O(1) \} = \exp \{ -O(z) \}$. If an intrinsic atom has a modulus divisible by such a p , it is now impossible: its residue $-4D$ is a unit modulo M , since $(D, M) = 1$. Every surviving distinct atom has a z -rough modulus M , remains uniform, and has probability exactly $1/M$.

Join two atoms when their moduli share a prime and set

$$x_A = \frac{1}{M} \exp \left(2 \sum_{q|M} a_q \right)$$

for an atom of modulus M . Roughness and $\max a_q = o(1)$ give $x_A < 1/2$. Theorem 7.2 bounds the total activity of the neighbors of A by $\sum_{q|M} a_q$. Therefore

$$\prod_{B \sim A} (1 - x_B) \geq \exp\left(-2 \sum_{B \sim A} x_B\right) \geq \exp\left(-2 \sum_{q|M} a_q\right),$$

so $\Pr(A) \leq x_A \prod_{B \sim A} (1 - x_B)$. The quantitative local lemma gives

$$\begin{aligned} \Pr(\text{no surviving atom} \mid \text{quarantine}) &\geq \prod_A (1 - x_A) \\ &\geq \exp\left(-2 \sum_A x_A\right) \geq \exp\{-O(L^3/\log z)\}. \end{aligned}$$

where the last inequality is (15). Multiplying by the quarantine cost proves (17). $\square$

Corollary 7.4 (refutation of the literal hypothesis). *Hypothesis H_{PF} as stated in Hypothesis 6.1 is false.*

Proof. Theorem 4.2 gives $\mu_X \geq c_0 L^3$. If H_{PF} held with constants c, C , choose $J \geq C\mu_X$. For each fixed sufficiently large X , take arbitrarily large multiples $N = mP_X$ of the full period satisfying $J \log X \leq \frac{1}{2} \log N$. Periodicity gives $|\text{Av}_X(N)|/N = \delta_X$. Since the majorant is at least one on every avoider, H_{PF} would imply

$$\delta_X \ll e^{-c\mu_X} \leq e^{-cc_0 L^3}.$$

This contradicts Theorem 7.3 for all sufficiently large X . $\square$

The scope is exact: this refutes the all-range quantifiers of H_{PF} , not the conclusion of Theorem 6.2 by every possible route and not the critical-window statement below.

8. THE CRITICAL WINDOW AND TRANSFER WALLS

Hypothesis 8.1 (critical-window variant H'_{PF} ; open). *There are $c, C > 0$ and $0 < c_- < c_+$, with c_- large enough for the degree budget, such that the majorant in Hypothesis 6.1 exists whenever*

$$c_- L^4 \leq \log N \leq c_+ L^4, \quad J \geq C\mu_X, \quad J \log X \leq \frac{1}{2} \log N.$$

No assertion is made outside this window.

This remains open and would still imply Theorem 6.2. Corollary 7.4 uses arbitrarily large complete-period multiples at fixed X , which need not lie in this window.

8.1. Cylinder and tensor obstructions.

Theorem 8.2 (all-avoider cylinder wall). *If a nonempty union $\mathcal{U}$ of complete classes modulo Q is contained in the full intrinsic avoider set, then*

$$\prod_{\substack{\ell \leq X, \ell \equiv 3 \pmod{4} \\ \ell \text{ prime}}} \ell \mid Q, \quad \log Q \geq (1/2 + o(1))X.$$

Proof. Choose one class $a \pmod{Q}$ in $\mathcal{U}$. For each prime $\ell \equiv 3 \pmod{4}$, the divisor $D = 1$ gives the forbidden class $-4 \pmod{\ell}$. If $\ell \nmid Q$, the Chinese remainder theorem produces an integer lying both in $a \pmod{Q}$ and in that forbidden class, a contradiction. Thus every such prime divides Q , and the prime number theorem in the progression $3 \pmod{4}$ gives the estimate. $\square$

The CRT space is a product over prime powers, but the avoider indicator is not a tensor product: its atomic factors overlap.

Lemma 8.3 (explicit non-tensor obstruction). *Already at $X = 15$, the complete avoider indicator modulo $1155 = 3 \cdot 5 \cdot 7 \cdot 11$ does not factor into functions of the four prime coordinates.*

Proof. The residues

n	$n \bmod 3$	$n \bmod 5$	$n \bmod 7$	$n \bmod 11$	$\mathbf{1}_{\text{Av}_{15}}(n)$
231	0	1	0	0	1
616	1	1	0	0	1
693	0	3	0	0	1
1078	1	3	0	0	0

form a two-coordinate rectangle with the other coordinates fixed. A tensor product containing the first three corners contains the fourth. To check the final column without an enumeration, Lemma 4.1 gives

$$\mathcal{R}(3) = \{2\}, \quad \mathcal{R}(7) = \{3, 5, 6\}, \quad \mathcal{R}(11) = \{7, 8, 10\}, \quad \mathcal{R}(15) = \{7, 11, 13, 14\}.$$

The first three rows miss these four sets. The last row is $13 \pmod{15}$ and is killed by $D = 8 \mid 4^2$. $\square$

Thus Fourier transformation produces a hypergraph convolution, not a product of local Fourier transforms.

8.2. The corrected finite-transfer criterion. After the odd-prime quarantine, write $n = P_z m$. For $U \subseteq \mathbb{Z}/d\mathbb{Z}$, put $\mathbf{1}_{U,d}(m) = \mathbf{1}_{m \bmod d \in U}$ and $\rho(U) = |U|$.

Lemma 8.4 (residue-set transfer). *Let $\mathcal{F}_X$ be a specified family of surviving atoms and I_X a finite index set. Suppose*

$$B_X(m) = \sum_{\alpha \in I_X} c_\alpha \mathbf{1}_{U_\alpha, d_\alpha}(m) \leq \mathbf{1}_{\{\text{no atom of } \mathcal{F}_X \text{ at } m\}}$$

pointwise, where each d_α divides the surviving CRT period and $(d_\alpha, P_z) = 1$. If

$$\begin{aligned} \mathbb{E}_{\text{CRT}} B_X &\geq e^{-o(L^3)}, & \log \max_{\alpha} d_\alpha &= o(L^4), \\ \log \mathcal{R}_X &= o(L^4), & \mathcal{R}_X &= \sum_{\alpha} |c_\alpha| \rho(U_\alpha). \end{aligned}$$

then, uniformly for fixed $0 < c_- < c_+$ and $c_- L^4 \leq \log N \leq c_+ L^4$,

$$|\text{Av}(\mathcal{F}_X; N)| \geq N e^{-o(L^3)}.$$

Proof. Put $H = \lfloor N/P_z \rfloor$. A set of $\rho(U_\alpha)$ classes modulo d_α has count

$$H \frac{\rho(U_\alpha)}{d_\alpha} + O(\rho(U_\alpha)).$$

Hence $\sum_{m \leq H} B_X(m) = H \mathbb{E}_{\text{CRT}} B_X + O(\mathcal{R}_X)$. The main term has logarithm $\log N - O(z) - o(L^3) = \log N - o(L^3)$, while the error has logarithm $o(L^4)$ and is negligible because $\log N \geq c_- L^4$. The pointwise inequality and $P_z = \exp\{O(z)\} = \exp\{o(L^3)\}$ finish the proof. $\square$

For an atom monomial, compatibility gives one residue class and incompatibility gives none, so $\rho \leq 1$. The following is the isolated low-degree criterion; the factor $\rho(U)$ in the general ledger is essential.

Corollary 8.5 (monomial finite-transfer criterion). *Suppose real coefficients c_S , supported on sets of at most $r = o(L^3)$ surviving atoms, make*

$$B_X(n) = \sum_S c_S \prod_{A \in S} \mathbf{1}_A(n) \leq \mathbf{1}_{\{\text{no surviving atom at } n\}}$$

pointwise. If its exact CRT mean is at least $e^{-o(L^3)}$ and $\log \sum_S |c_S| = o(L^4)$, then $|\text{Av}_X(N)| \geq N e^{-o(L^3)}$ uniformly for $\log N \asymp L^4$.

Proof. Apply Lemma 8.4. Compatible monomials are single classes modulo lcms $d_S \leq X^r = \exp\{o(L^4)\}$, incompatible monomials vanish, and the weighted ledger is at most $\sum_S |c_S|$. $\square$

8.3. The prime side transfers. Let $S_\ell = P_z^{-1} \mathcal{R}(\ell) \pmod{\ell}$. Choose $h(L) \rightarrow \infty$ slowly and put $Y = L^4/h(L)$, so $z < Y < X$. Define

$$d_0 = \prod_{\substack{z < \ell \leq Y \\ \ell \equiv 3 \pmod{4}}} \ell, \quad U_0 = \prod_{\substack{z < \ell \leq Y \\ \ell \equiv 3 \pmod{4}}} ((\mathbb{Z}/\ell\mathbb{Z}) \setminus S_\ell), \quad T_0 = \mathbf{1}_{U_0, d_0}.$$

Here and below ℓ is prime. Then $\log d_0 = O(Y) = o(L^4)$. Let H_P count violated prime atoms with $Y < \ell \leq X$, put $\mu_P = \mathbb{E} H_P = \Theta(L^2)$, let r be the least odd integer at least $8\mu_P$, and define

$$Q_r(H_P) = \sum_{j=0}^r (-1)^j \binom{H_P}{j}, \quad B_P = T_0 Q_r(H_P).$$

Theorem 8.6 (prime-side two-level minorant). *The function B_P is pointwise at most the surviving prime-system avoider indicator. Its tail degree is $O(L^2)$, every term has modulus logarithm $o(L^4)$,*

$$\mathbb{E}_{\text{CRT}} B_P \geq e^{-O(L^2)}, \quad \log \sum_{\alpha} |c_{\alpha}| \rho(U_{\alpha}) = o(L^4).$$

Thus Lemma 8.4 transfers a lower bound $N e^{-o(L^3)}$ in the critical window. The sharper $N e^{-O(L^2)}$ bound is already Theorem 5.4.

Proof. If $T_0 = 0$ the pointwise assertion is immediate; if $T_0 = 1$, it is odd Bonferroni. For $h \geq 1$,

$$\sum_{j=0}^r (-1)^j \binom{h}{j} = -\binom{h-1}{r}.$$

Distinct prime coordinates are independent and atoms at one coordinate are disjoint, so the loss from the exact tail void is at most

$$\mathbb{E} \binom{H_P}{r+1} \leq \frac{\mu_P^{r+1}}{(r+1)!} \leq \left(\frac{e\mu_P}{r+1} \right)^{r+1}.$$

The tail Euler product is at least $e^{-2\mu_P}$; the last display is a small fraction of it. This proves the mean bound, including the independent low tensor.

Expand only the tail factor. If $W_P = \sum_{Y < \ell \leq X} F(\ell)$, the maximal-order divisor bound gives $W_P = \exp\{O(L)\}$. Each term has at most $\rho(U_0)$ residue classes, so

$$\mathcal{R}_X \leq \rho(U_0) \sum_{j \leq r} \frac{W_P^j}{j!}, \quad \log \mathcal{R}_X \leq o(L^4) + O(rL) = o(L^4).$$

Its modulus divides $d_0 X^r$, giving the other budget. Apply Lemma 8.4; the quarantine costs $e^{-O(z)} = e^{-o(L^3)}$. $\square$

8.4. The composite frontier. Delete a composite atom if one of its prime projections is already a prime intrinsic atom; retain the residual family $\mathcal{C}^\circ$. Condition on $T_0 = 1$, and let H count the tail prime atoms together with $\mathcal{C}^\circ$. Put

$$\Lambda = \frac{L^3}{\log L}.$$

For $A \in \mathcal{C}^\circ$, conditioning on $T_0 = 1$ gives

$$\Pr(A \mid T_0 = 1) = \frac{1}{M_A} \prod_{\substack{p \mid M_A \\ z < p \leq Y}} \frac{p}{p - F(p)} = \frac{1 + o(1)}{M_A}$$

uniformly: $F(p) = p^{o(1)}$, $p > z = L^{3+o(1)}$, and $\omega(M_A) \leq L/\log z$. The rough mean therefore gives $\mathbb{E}H = O(\Lambda)$. The conditional full void after the quarantine is at least $e^{-O(\Lambda)}$ by the proof of Theorem 7.3; dividing by $\Pr(T_0 = 1) \leq 1$ gives $\Pr(H = 0 \mid T_0 = 1) \geq e^{-O(\Lambda)}$.

Proposition 8.7 (factorial-moment reduction). *Suppose fixed $C, D > 0$, with D sufficiently large in terms of C and of the constant in the conditioned void lower bound (37.18) of the source notes, and an even $m \in [D\Lambda, D\Lambda + 2]$ satisfy*

$$\mathbb{E}(H)_m \leq (C\Lambda)^m. \tag{18}$$

Then a two-level pointwise minorant satisfies Lemma 8.4 and proves $|\text{Av}_X(N)| \geq Ne^{-o(L^3)}$ uniformly for $\log N \asymp L^4$.

Proof. Use $B = T_0 Q_{m-1}(H)$. Since $m - 1$ is odd, Bonferroni makes it a pointwise minorant. The preceding binomial identity and $\binom{h-1}{m-1} \leq \binom{h}{m}$ give

$$\mathbb{E}(Q_{m-1}(H) \mid T_0 = 1) \geq \Pr(H = 0 \mid T_0 = 1) - \frac{(C\Lambda)^m}{m!}.$$

By Stirling, the second term is at most $(eC/D)^m$, a small fraction of the first for large D . There are $\exp\{O(L)\}$ atoms. Thus every term, after multiplication by T_0 , is a residue set and

$$\begin{aligned} \deg B &= O(\Lambda) = o(L^3), \\ \log(\text{modulus}) &\leq Y + O(\Lambda L) = o(L^4), \\ \log(\text{ledger}) &\leq Y + O(\Lambda L) = o(L^4). \end{aligned}$$

Lemma 8.4 applies. $\square$

Proposition 8.7 is a valid sufficient reduction, but Subsection 8.5 below shows that (18) is false for this unreduced count H . Its pair-dispersion subproblem remains meaningful. For a high prime p and $a \pmod{p}$, set

$$u_{p,a} = \sum_{\substack{A \in \mathcal{C}^\circ \\ p \mid M_A, \, r_A \equiv a \pmod{p}}} \Pr(A \mid T_0 = 1), \quad t_p = \sum_a u_{p,a}.$$

For squarefree pairs whose gcd is p , their pair contribution is bounded by a sub-sum of $p \sum_a u_{p,a}^2$. Even granting the ideal incidence estimate $t_p \ll \Lambda/p$, the available ℓ^1 information yields only

$$\sum_{z < p \leq X} p \sum_a u_{p,a}^2 \leq \sum_{z < p \leq X} p t_p^2 \ll \Lambda^2 \sum_{z < p \leq X} \frac{1}{p} \asymp \Lambda^2 \log L. \quad (19)$$

The live pair-dispersion frontier is

$$\sum_{z < p \leq X} p \sum_a u_{p,a}^2 = O(\Lambda^2). \quad (20)$$

Equation (19) is not a counterexample to (20); it identifies the factor lost by present information. Scott–Sokal’s cluster expansion [16] controls a numerical partition function, not the needed pointwise polynomial. Indeed the literal independent-set polynomial in violated-event indicators fails pointwise: on a five-vertex path its alternating sum is $1 - 5 + 6 - 1 = 1$ although the void indicator is zero.

8.5. Nested implication and the two hierarchy refutations. The more basic failure is logical redundancy among composite atoms. For a compatible nonempty set S , put $Q_S = \text{lcm}_{A \in S} M_A$ and let $\mathcal{L}(S)$ be the sum, over compatible $B \notin S$, of the conditional extension ratios

$$\frac{\Pr(\bigcap_{A \in S \cup \{B\}} A \mid T_0 = 1)}{\Pr(\bigcap_{A \in S} A \mid T_0 = 1)}.$$

The literal source target (40.28) asserted $\mathcal{L}(S) \ll \Lambda$ uniformly for $|S| < m \asymp \Lambda$.

Theorem 8.8 (nested retained-atom obstruction; source Theorem 47.2). *For all sufficiently large X , the family counted by the raw H contains a squarefree atom A such that*

$$\mathcal{L}(\{A\}) \geq \exp\{cL/\log L\}.$$

Moreover, for $m \asymp \Lambda$ and every fixed $C > 0$,

$$\mathbb{E}((H)_m \mid T_0 = 1) > (C\Lambda)^m$$

for all sufficiently large X .

Mechanism sketch. Choose $q \in (Y, 2Y)$ with $q \equiv 3 \pmod{8}$ and distinct $p_1, \dots, p_J \in (Y, 2Y)$ with $p_i \equiv 5 \pmod{8}$, where J is the largest odd integer at most $L/(3 \log(2Y))$; the prime number theorem in fixed progressions supplies them, and $M = q \prod_i p_i < X$. For every odd $I \subseteq \{1, \dots, J\}$, the squarefree $M_I = q \prod_{i \in I} p_i$ is $7 \pmod{8}$, so $D = 2 \mid ((M_I + 1)/4)^2$ supplies the atom $n \equiv -8 \pmod{M_I}$. It survives prime-implied deletion because $(-8/q) = 1$, whereas every intrinsic unit class modulo q has Legendre sign -1 ; the p_i have no prime atom, and $D = 2$ is the least representative. The top event modulo M implies all $2^{J-1} - 1$ proper odd-subset events, which proves the codegree bound. On that event $H \geq 2^{J-1}$ and its conditional probability is $1/M$ (all its primes exceed Y), so $\mathbb{E}(H)_m \geq (2^{J-1})_m/M > (C\Lambda)^m$ for $m \asymp \Lambda$. $\square$

Thus the literal one-step hierarchy (40.28) and the raw- H moment target (37.19), namely (18), are REFUTED. These were internal sufficient mechanisms: Theorem 8.8 does not refute H'_{PF} , (20), or the Erdős–Straus conjecture.

The proposed repair was to pass to the implication antichain $\mathcal{A}^*$: delete an atom A whenever another retained atom B has $M_B \mid M_A$ and $r_A \equiv r_B \pmod{M_B}$. Then $A \subseteq B$, so the union and its void are unchanged. For a candidate B put

$$w_B = \Pr(B \mid T_0 = 1),$$

$$\Gamma_S(B) = \prod_{p \mid (M_B, Q_S)} p^{\min(v_p(M_B), v_p(Q_S))} \prod_{\substack{p \mid (M_B, Q_S) \\ z < p \leq Y, p \equiv 3(4)}} \frac{p - F(p)}{p}.$$

The exact proposed wall (source (47.16)) asked for fixed $C, D > 0$ such that, for every compatible $S \subset \mathcal{A}^*$ with $1 \leq |S| < m$ and even $m \in [D\Lambda, D\Lambda + 2]$,

$$\sum_{\substack{B \in \mathcal{A}^* \setminus S, B \sim S \\ (M_B, Q_S) > 1}} w_B \Gamma_S(B) \leq C\Lambda. \quad (47.16)$$

The coprime part is already $O(\Lambda)$; hence (47.16) would recover the factorial-moment reduction for the reduced count. It was the proposed repair; the following exact structure and grid show that this repair also fails.

8.6. The implication antichain and the semiprime grid. Let $\mathcal{A}_X^\circ$ be the finite retained family counted by H : tail-prime atoms and composite atoms after prime-implied deletion, with classes deduplicated. For an intrinsic class $r \pmod{M}$, put

$$\delta_M(r) = \min\{D : D \mid A_M^2, -4D \equiv r \pmod{M}\}, \quad A_M = \frac{M+1}{4}. \quad (49.1)$$

Theorem 8.9 (exact implication criterion; source Theorem 49.1). *For two distinct retained atoms,*

$$E_{M,r} \subseteq E_{m,s} \iff m \mid M \text{ and } r \equiv s \pmod{m} \iff m \mid M \text{ and } \delta_M(r) \equiv \delta_m(s) \pmod{m}. \quad (49.2)$$

If $M = tm$, the intrinsic conditions unpack exactly as

$$t \equiv 1 \pmod{4}, \quad A_M = tA_m - \frac{t-1}{4}, \quad D = D' + jm \quad (j \in \mathbb{Z}), \quad (49.3)$$

where $D \mid A_M^2$, $D' \mid A_m^2$, and both corresponding classes survive the deletions. There is no further division-by-four condition. At one modulus, distinct retained classes are disjoint and cannot nest.

Proof. A full cylinder $E_{M,r}$ lies in $E_{m,s}$ exactly when reduction modulo m is defined and equals s , giving the first equivalence. Since every participating modulus is odd, multiplication by -4^{-1} gives the second. If $M = tm$, reducing modulo four and comparing $4A_M = M + 1$ with $4A_m = m + 1$ gives (49.3). Equal-modulus nesting is excluded by class deduplication. $\square$

Consequently the antichain, now denoted $\mathcal{A}_X^\dagger$, is exactly

$$\mathcal{A}_X^\dagger = \{(M, r) \in \mathcal{A}_X^\circ : \nexists (m, s) \in \mathcal{A}_X^\circ, m < M, m \mid M, s \equiv r \pmod{m}\}. \quad (49.4)$$

It keeps the inclusion-largest events, equivalently the divisibility-minimal compatible moduli.

Corollary 8.10 (pointwise void identity; source Corollary 49.2). *If $H^\dagger$ counts $\mathcal{A}_X^\dagger$, then, on every integer and also pointwise inside the $T_0 = 1$ fibre,*

$$H = 0 \iff H^\dagger = 0. \quad (49.5)$$

Proof. A deleted cylinder lies in a retained cylinder of smaller modulus; finite descent terminates in $\mathcal{A}_X^\dagger$. Thus the two unions agree. $\square$

Writing $w_A = \Pr(A \mid T_0 = 1)$, $\mu = \sum_{\mathcal{A}_X^\circ} w_A$, and $\mu^\dagger = \sum_{\mathcal{A}_X^\dagger} w_A$, only

$$\Theta(L^2) = \mu_P \leq \mu^\dagger \leq \mu = O(\Lambda) \quad (49.6)$$

is proved. No $\Theta(\Lambda)$ first moment and no constant-factor lower bound $\mu^\dagger \gg \mu$ is available. The void identity and all upper charge, degree, modulus, and ledger estimates survive reduction.

For a divisor g and a reduced class $a \pmod{g}$, define

$$W_{g,a}^\dagger = \sum_{\substack{A \in \mathcal{A}_X^\dagger \\ g \mid M_A, \ r_A \equiv a \pmod{g}}} w_A, \quad W_g^\dagger = \sum_{a \pmod{g}} W_{g,a}^\dagger. \quad (49.8)$$

Theorem 8.11 (uniform residue profile is false; source Theorem 49.4). *There is no absolute C for which*

$$W_{g,a}^\dagger \leq \frac{C}{\varphi(g)} W_g^\dagger \quad (49.9)$$

holds uniformly in X, g, a . In fact, there are $g \asymp X$ and reduced $a \pmod{g}$ for which the ratio between the two sides without C is $g^{1-o(1)}$.

Proof. Choose primes $q \equiv 3 \pmod{8}$ and $p \equiv 5 \pmod{8}$, both in $(3X^{1/2}/4, 4X^{1/2}/5)$, and put $M = qp$. Then $X/2 < M < X$ and, for large X , both primes exceed z and Y . Since $M \equiv 7 \pmod{8}$, the intrinsic $D = 2$ atom is

$$n \equiv -8 \pmod{M}. \quad (49.10)$$

Its projection modulo q has Legendre sign $(-8/q) = 1$, opposite to every intrinsic prime-modulus class; the other prime is $1 \pmod{4}$. Thus the atom survives prime deletion. Its proper modulus divisors $1, q, p$ carry no retained atom in this class, so Theorem 8.9 puts it in $\mathcal{A}_X^\dagger$. Take $g = M$, $a = -8$. Since $2M > X$, every atom in $W_g^\dagger$ has modulus M and weight $1/M$. Their number is at most $F(M) \leq \tau(A_M^2) = M^{o(1)}$, while the displayed bin is nonempty. Hence

$$\frac{\varphi(g)W_{g,a}^\dagger}{W_g^\dagger} \geq \frac{\varphi(M)}{F(M)} = M^{1-o(1)}. \quad (49.11)$$

$\square$

For fixed $q \equiv 3 \pmod{8}$, every surviving semiprime atom with $p \equiv 5 \pmod{8}$ still projects to the one class $-8 \pmod{q}$. Thus the $D = 2$ slice remains residue-concentrated after maximal reduction and cannot support a literal transplant of the $1/\varphi(g)$ cancellation used in the provisional record proof. This does not refute a profile averaged over g or restricted to a sufficiently small range.

Choose distinct primes

$$q_1, \dots, q_t \equiv 3 \pmod{8}, \quad p_1, \dots, p_t \equiv 5 \pmod{8}, \quad (49.12)$$

all greater than z and with every $q_i p_j < X$. By the preceding proof, every edge

$$A_{ij} : \quad n \equiv -8 \pmod{q_i p_j} \quad (49.13)$$

is a retained maximal atom, giving a complete bipartite grid in $\mathcal{A}_X^\dagger$. Let $\mathcal{L}^\dagger(S)$ denote the same conditional extension load as $\mathcal{L}(S)$, but within this antichain.

Theorem 8.12 (collective-closure obstruction; source Theorem 49.5). *For every fixed $C, D > 0$, every sufficiently large X , and every even integer $m \in [D\Lambda, D\Lambda + 2]$, there is a compatible selected set $S \subset \mathcal{A}_X^\dagger$, with pairwise-coprime squarefree moduli and $|S| < m$, such that*

$$\mathcal{L}^\dagger(S) > C\Lambda. \quad (49.14)$$

For the same fixed C, D and every such m , the factorial-moment input also fails for all sufficiently large X :

$$\mathbb{E}((H^\dagger)_m \mid T_0 = 1) > (C\Lambda)^m. \quad (49.15)$$

Proof. For the hierarchy assertion take the diagonal matching $S = \{A_{ii} : 1 \leq i \leq t\}$. Its merged class is -8 modulo

$$Q_S = \prod_{i=1}^t q_i p_i. \quad (49.16)$$

For $i \neq j$, $q_i p_j \mid Q_S$ and the merged class projects to A_{ij} . Thus the intersection of the diagonal events implies every cross edge, and its exact conditional extension ratio is

$$\frac{\Pr(\bigcap_{A \in S} A \cap A_{ij} \mid T_0 = 1)}{\Pr(\bigcap_{A \in S} A \mid T_0 = 1)} = 1. \quad (49.17)$$

Set $t = m - 1$. For either required class $a = 3, 5 \pmod{8}$, and any fixed $B_D > 12D$, the prime number theorem gives

$$\#\{z < r \leq B_D L^3 : r \equiv a \pmod{8}\} = \left(\frac{B_D}{12} + o(1)\right) \Lambda.$$

This supplies the grid for every admissible m ; its products are $O_D(L^6) < X$. The diagonal moduli are pairwise coprime and squarefree, and the $t(t-1)$ forced cross edges prove (49.14).

For the moment assertion instead put

$$t = \left\lfloor \frac{m}{2 \log(2z)} \right\rfloor. \quad (49.18)$$

There are enough primes of each class in $(z, 2z)$, since their number is asymptotic to $z/(4 \log z)$ and exceeds t by a factor tending to infinity. Put $R = \prod_i q_i p_i$ and let C_R be $n \equiv -8 \pmod{R}$. At each conditioned q_i , this class is allowed, so

$$\Pr(C_R \mid T_0 = 1) = \frac{1}{R} \prod_{q_i \leq Y} \frac{q_i}{q_i - F(q_i)} \geq \frac{1}{R}. \quad (49.19)$$

On C_R all t^2 grid atoms occur. Since $R \leq (2z)^{2t}$, $t^2 \geq 2m$ for large X , and $(u)_m \geq (u/2)^m$ for $u \geq 2m$,

$$\mathbb{E}((H^\dagger)_m \mid T_0 = 1) \geq \frac{(t^2)_m}{R} \geq \left(\frac{t^2}{2}\right)^m (2z)^{-2t}. \quad (49.20)$$

After division by $(C\Lambda)^m$, the logarithm is at least

$$m \log \left(\frac{t^2}{2C\Lambda} \right) - 2t \log(2z). \quad (49.21)$$

The second term is at most m , whereas $t^2/\Lambda \gg_D \Lambda/(\log z)^2 \rightarrow \infty$, proving (49.15). $\square$

The point is collective: no diagonal edge contains an off-diagonal edge, but their intersection implies all cross edges. Proposition 47.3's prime-only rung survives. Bounded $\omega(M)$ does not repair these formulations because every grid atom has $\omega(M) = 2$.

The scope is exact. Theorems 8.9 and 8.12 do not rule out a weighted count, a redesigned conditioning selector, a different pointwise minorant, or a further collective void-preserving reduction. Blindly deleting grid atoms can enlarge the void and supplies no pointwise minorant. The endpoint pair estimate (40.19), the corresponding pair target (37.27), the existence of some other minorant proving (33.16), (33.16) itself, and H'_{PF} all remain OPEN. Nothing here proves or refutes the Erdős–Straus conjecture.

Computational 49.7 (exact finite scope). Exact divisor-enumeration diagnostics reduce 1042 retained atoms to 970 at $(X, z, Y) = (1000, 2, 1000)$, retaining conditional first-moment share $0.939936\dots$; the maximum normalized profile there is 220 at $g = 943$. The exact square $(q_1, q_2; p_1, p_2) = (3, 11; 5, 13)$ verifies that selected moduli 15, 143 force the cross atoms at 39, 55 with extension ratio one. These finite fractions are computational diagnostics, not asymptotic evidence.

Assessment 49.8 (wave-17 verdict). The uniform profile, (47.16), and the ordinary factorial-moment target for $H^\dagger$ are REFUTED; the cube killed nesting and the grid kills antichains. The complete-system moment axis is therefore DORMANT: its original stakes have largely collapsed since the CLAIMED/PROVISIONAL unconditional record of Theorem 10.7, and any revival needs a genuinely new pointwise formulation.

9. PROVISIONAL RECORD CONTEXT

This section is deliberately segregated. Both records are CLAIMED/PROVISIONAL; neither is used in the refutation of H_{PF} .

9.1. The provisional exceptional-set record. Let $E(N)$ count exceptional prime denominators $p \leq N$.

Theorem 9.1 (CLAIMED/PROVISIONAL). *There is an absolute $c > 0$ such that*

$$E(N) \ll N \exp\{-c(\log N)^{2/3}(\log \log N)^{1/3}\}.$$

Internal argument; provisional status retained. Put

$$\begin{aligned} \mathcal{K}(K) &= \{k \leq K : k \equiv 1 \pmod{4}\}, & L_K &= \text{lcm}_{k \in \mathcal{K}(K)} k, \\ h(\mathcal{J}) &= \sum_{k \in \mathcal{J}} \frac{\varphi(k)}{k^2}. \end{aligned}$$

The multiplier form of Lemma 3.4, with $M = k\ell$, has the following class-mass consequence. Uniformly for $K \leq (\log X)^A$, every subfamily $\mathcal{J} \subseteq \mathcal{K}(K)$ containing 1, and every $c \pmod{24L_{\mathcal{J}}}$ with $(c, 24L_{\mathcal{J}}) = 1$, there are distinct forced classes modulo primes $X^{1/2} < \ell \leq X$ whose counts $f_c(\ell)$ satisfy

$$\sum_{X^{1/2} < \ell \leq X} \frac{f_c(\ell)}{\ell} \asymp (\log X)^2 h(\mathcal{J}). \quad (21)$$

Here the classes arise from

$$H < u, v \leq \ell^{1/3}, \quad (u, v) = (uv, k) = 1, \quad uv \mid (k\ell + 1)/4, \quad k \mid u + cv,$$

with $H = K^{10}$ and $\omega(uv) \leq D \log \log X$.

For completeness, the harmonic lattice input behind (21) is obtained by partitioning $(H, z]$ into fixed-ratio boxes. Möbius inversion of $(u, v) = 1$ gives, in $I \times J$ with lengths $\eta U, \eta V$,

$$\begin{aligned} \#\{(u, v) \in I \times J : (u, v) = (uv, k) = 1, \quad k \mid u + cv\} &= \eta^2 UV \frac{\varphi(k)}{k^2} \prod_{p \nmid k} (1 - p^{-2}) \\ &\quad + O((U + V)\tau(k) \log(2UV)). \end{aligned}$$

After division by UV and summation over boxes, the boundary/main ratio is $O(K^2/H)$, uniformly in k, c . Shiu's theorem [17], applied successively in the two variables to $t^{\omega(n)}$ and $n/\varphi(n)$, gives

$$\sum_{uv}^* \frac{t^{\omega(u)+\omega(v)}}{uv} \ll_t \frac{\varphi(k)}{k^2} (\log z)^2 (\log z)^{2(t-1)}, \quad \sum_{\varphi(u)\varphi(v)}^* \frac{1}{\varphi(u)\varphi(v)} \ll \frac{\varphi(k)}{k^2} (\log z)^2.$$

Rankin's inequality removes the high- ω tail after fixed large D . Summing over k gives the factor $h(\mathcal{J})$.

In a dyadic prime interval, the divisor condition is the progression $\ell \equiv -k^{-1} \pmod{4uv}$. Colliding classes imply $\ell \mid uv' - u'v$ and hence equality of reduced pairs; different k, k' would force the common uv to divide $(k - k')/4$, impossible because $uv > H^2 > K$. A fixed modulus has multiplicity at most $K 2^{\omega(uv)} = (\log X)^{O(1)}$. Bombieri–Vinogradov absorbs it and gives the lower half of (21); Brun–Titchmarsh and the second weighted lattice estimate give the upper half. This proves (21).

Now choose $K = \lfloor \delta \log N \rfloor$ and partition primes by reduced classes modulo $M_0 = 24L_K$. Since $\log M_0 \leq (1 + o(1))K$, each subsequence has length $Y \geq N^{1-2\delta}$. Put

$$X = \exp\{\alpha(\log N / \log \log N)^{1/3}\}.$$

For a fixed subsequence, the larger-sieve denominator is

$$S_c = \sum_{\substack{s \leq Y^{1/2} \\ s \mid P}} \mu^2(s) \prod_{\ell \mid s} \frac{f_c(\ell)}{\ell - f_c(\ell)}, \quad P = \prod_{X^{1/2} < \ell \leq X} \ell.$$

Here $f_c(\ell) \leq K\ell^{2/3} < \ell/2$ for large N . Writing G_c for the untruncated Euler product, Rankin's trick with $v = 1/\log X$ gives

$$\frac{G_c - S_c}{G_c} \leq \exp\{-\log Y / (2 \log X) + C(\log X)^2 \log K\}.$$

Small α makes this at most $1/2$. Since $h(\mathcal{K}(K)) \asymp \log K$, (21) gives $S_c \geq \frac{1}{2} \exp\{a(\log X)^2 \log K\}$. Summing the uniform bound $O(Y/S_c)$ over the reduced subsequences proves the theorem. $\square$

Remark 9.2 (priority and verification status). *Theorem 9.1 remains CLAIMED/PROVISIONAL pending external checking and primary-literature verification. Vaughan's primary paper [18] was located by its confirmed DOI but its PDF was access-blocked. Pomerance–Weingartner's 2025 reconstruction [13, §4] cites Vaughan as the state of the art, and the fresh search found no post-1970 exponent improvement. The draft*

therefore does not claim a direct verification of Vaughan's argument or a settled record; the search result is not a literature guarantee.

9.2. The pruned cubic prime slice. The current harvesting frontier uses the incidence

$$r_{\mathcal{J}}(u, v; c) = \#\{k \in \mathcal{J} : k \mid u + cv\}.$$

Lemma 9.3 (incidence second moment; provisional context). *For $K \geq K_0$, $H = K^{10}$, and $z > H^2$,*

$$\sum_{\substack{H < u, v \leq z \\ (u, v) = 1}} \frac{r_{\mathcal{J}}(u, v; c)^2}{uv} \ll (\log z)^2 (1 + \log K)^3,$$

uniformly for $\mathcal{J} \subseteq \mathcal{K}(K)$ and $c \pmod{24L_{\mathcal{J}}}$ with $(c, 24L_{\mathcal{J}}) = 1$.

Proof. Expand the square. For $k, k' \in \mathcal{J}$, both divisibilities are the congruence $[k, k'] \mid u + cv$. The preceding box calculation, now with $d = [k, k'] \leq K^2$, gives

$$\sum_{uv}^* \frac{1}{uv} \ll \frac{\varphi(d)}{d^2} (\log z)^2 + \frac{\tau(d)(\log z)^2}{H}.$$

Furthermore

$$\sum_{k, k' \leq K} \frac{\varphi([k, k'])}{[k, k']^2} \leq \sum_{k, k' \leq K} \frac{(k, k')}{kk'} \ll (1 + \log K)^3,$$

by writing $k = da, k' = db$. Since $\tau([k, k']) \leq 2K$, all boundary terms total $O(K^3(\log z)^2/H)$, proving the result. $\square$

Theorem 9.4 (unconditional pruned cubic prime slice; CLAIMED/PROVISIONAL). *Fix $0 < \kappa < 1/240$ and let $K = X^{\kappa}$. For every $\mathcal{J} \subseteq \mathcal{K}(K)$ containing 1 and every $c \pmod{24L_{\mathcal{J}}}$ with $(c, 24L_{\mathcal{J}}) = 1$, retain only triples in (21) satisfying $r_{\mathcal{J}}(u, v; c) \leq (\log X)^4$. Then (i) the restriction of the $H_{k\text{BV}}$ error sum to these retained triples is $o(x \log x h(\mathcal{J}))$ unconditionally, uniformly in all the quantified data; and (ii) their distinct class counts $f_c^{\text{good}}(\ell)$ satisfy, uniformly in these data,*

$$\sum_{X^{1/2} < \ell \leq X} \frac{f_c^{\text{good}}(\ell)}{\ell} \asymp (\log X)^2 h(\mathcal{J}).$$

For the full $\mathcal{J} = \mathcal{K}(X^{\kappa})$, this is $\asymp (\log X)^3$.

Internal argument; provisional status retained. The harmonic lattice proof above remains uniform whenever $z > K^{20}$: its box error is $O(K^2/H)$, and Shiu applies because every variable exceeds $H = K^{10}$. At the lowest dyadic interval, $z \geq X^{1/12} > K^{20}$ follows from $\kappa < 1/240$. Lemma 9.3 and Markov's inequality bound the harmonic mass of discarded triples by

$$\frac{(\log z)^2 (1 + \log K)^3}{(\log X)^4} = o((\log z)^2 h(\mathcal{J})).$$

Thus the retained first moment has full order. For fixed $4uv$, coprimality permits at most $2^{\omega(uv)}$ allocations to (u, v) , and the pruning permits at most $(\log X)^4$ choices of k . The retained progression multiplicity is therefore $(\log X)^{D \log 2 + 4}$. Ordinary Bombieri–Vinogradov absorbs it. Distinctness is unchanged, and Brun–Titchmarsh gives the matching upper bound. Finally $h(\mathcal{K}(X^{\kappa})) \asymp \log X$. $\square$

Theorem 9.4 does not prove the literal weighted error-sum hypothesis from which it originated, does not assemble composite moduli, and does not improve Theorem 9.1 unconditionally. Its soft spots are the power-range extension of the Shiu/box estimates and the uniform second-moment boundary sum; these require external review.

10. THE FIXED- c -FREE CRITICAL-WINDOW ASSEMBLY

CLAIMED/PROVISIONAL. Every result in this section which feeds Theorem 10.7 is subject to hostile external checking. In particular, the load-bearing prime-slice input Theorem 9.4 is likewise provisional. The argument is not presented as an established record.

This section attacks the restricted critical-window assembly. It does *not* prove the literal coefficient-sum clause of the source-notes hypothesis $H'_{\text{PF}}(k\ell; \text{good})$. Instead it removes the moving c -family, uses one fixed unpruned atom family, and proves the weaker ledger $\exp\{O(J \log X)\}$. That ledger is of the same order as the lcm budget and is still absorbed when $\log N \geq C(\log X)^4$. Subject to the caveats in Section 10.7, this gives the CLAIMED/PROVISIONAL unconditional $3/4$ exponent.

Put

$$t = \log X, \quad K = \lfloor X^\kappa \rfloor, \quad H = K^{10}, \quad 0 < \kappa < 1/240, \quad (39.1)$$

and partition $(X^{1/2}, X]$ into disjoint dyadic blocks $(x, 2x]$, truncating the two endpoint blocks. In a block put $z = x^{1/6}$. Let $\mathcal{A}_X$ be the following fixed c -free family. An atom is

$$A = (k, \ell, u, v), \quad k \leq K, \quad k \equiv 1 \pmod{4}, \quad \ell \in (x, 2x] \text{ prime}, \quad H < u, v \leq z, \quad (39.2)$$

with

$$(u, v) = (uv, k) = 1, \quad \omega(uv) \leq D \log \log X, \quad 4uv \mid k\ell + 1. \quad (39.3)$$

Here D is the fixed low- ω cutoff constant used in the source Lemma 16.2 and in the proof of Theorem 9.4. The atom denotes the cylinder

$$E_A = \{n : n \equiv -uv^{-1} \pmod{k\ell}\}. \quad (39.4)$$

The dyadic cutoff is only a canonical subfamily of the multiplier-prime family. The lower proof of Theorem 9.4, specifically the source-notes estimates (34.20)–(34.21), furnishes its retained main mass inside these boxes.

10.1. Removing the moving fibre and deduplicating the atoms.

Lemma 10.1 (source Lemma 39.1: c -free reduction and distinct large coordinates; proved, provisionally checked). *The following hold.*

- (1) *If n is exceptional, then it avoids every atom in $\mathcal{A}_X$. If additionally $c \equiv n \pmod{24L_K}$, then every atom hit by n automatically obeys $k \mid u + cv$. Thus the coupling condition used to define the source family $\mathcal{G}_{X,c}$ is implied by the atom itself; it need not be built into the polynomial.*
- (2) *Distinct atoms which have a compatible intersection have distinct primes ℓ . At a fixed ℓ , all atom classes modulo ℓ are distinct. At a fixed modulus $k\ell$, no two ordered pairs (u, v) produce the same class.*

(3) Consequently, for a compatible ordered set $A_1, \dots, A_j$,

$$\Pr\left(\bigcap_{i \leq j} E_{A_i}\right) = \frac{\mathbf{1}_{\text{compatible}}}{\text{lcm}(k_1, \dots, k_j) \prod_{i \leq j} \ell_i}. \quad (39.5)$$

Proof. Lemma 3.4 proves the first assertion. Modulo k , (39.4) says $u + nv \equiv 0$; since $c \equiv n \pmod{k}$, it also says $u + cv \equiv 0$.

Suppose two atoms at the same ℓ are compatible. Their projections modulo ℓ agree, so

$$\ell \mid uv' - u'v.$$

In the same block the absolute value is at most $z^2 = x^{1/3} < \ell$. The same conclusion holds if two admissible cutoffs are compared across blocks, since every $u, v \leq X^{1/6}$, while $\ell > X^{1/2} > X^{1/3}$. Hence $uv' = u'v$, and reducedness gives $(u, v) = (u', v')$. The apparent swap can agree only when $u = v$, which is impossible here because $(u, v) = 1$ and $u, v > H$. Finally $uv \mid (k\ell + 1)/4$ is exactly

$$k\ell \equiv -1 \pmod{4uv}.$$

Since $(\ell, 4uv) = 1$, this determines $k \pmod{4uv}$, and $4uv > 4H^2 > K$ makes the admissible k unique. This proves both forms of deduplication and the distinct- ℓ assertion. The Chinese remainder theorem now gives (39.5). $\square$

The logical point is as important as the geometry. The majorant below covers the fixed predicate $\mathbf{1}_{(n, P_y)=1} \mathbf{1}_{H_X(n)=0}$, where $H_X = \sum_{A \in \mathcal{A}_X} \mathbf{1}_{E_A}$. It does not majorize every avoider of the smaller, c -dependently pruned source family. It does majorize every exceptional prime, which is all the exceptional-set implication needs.

10.2. Exact mass profiles. For $g \mid k$ and $a \in (\mathbb{Z}/g\mathbb{Z})^\times$, write

$$W_{k,a}(g) = \sum_{\substack{A \in \mathcal{A}_X \text{ with multiplier } k \\ -uv^{-1} \equiv a \pmod{g}}} \frac{1}{k\ell}, \quad W_k = \sum_{A: k_A = k} \frac{1}{k\ell}. \quad (39.6)$$

Lemma 10.2 (source Lemma 39.2: multiplier and residue profiles; proved, provisional). *Uniformly for $k \leq K$, $k \equiv 1 \pmod{4}$, $g \mid k$, and reduced $a \pmod{g}$,*

$$W_{k,a}(g) \ll \frac{t^2}{\varphi(g)} \frac{\varphi(k)^2}{k^3}, \quad W_k \asymp t^2 \frac{\varphi(k)^2}{k^3}. \quad (39.7)$$

Consequently

$$\mu_X := \sum_{A \in \mathcal{A}_X} \frac{1}{k_A \ell_A} \asymp t^2 \sum_{\substack{k \leq K \\ k \equiv 1 \pmod{4}}} \frac{\varphi(k)^2}{k^3} \asymp t^2 \log K \asymp t^3. \quad (39.8)$$

For each fixed odd prime p , the share of the proxy Euler weight $b(k)/k = \varphi(k)^2/k^3$ carried by multipliers divisible by p is exactly

$$\alpha_p = \frac{p-1}{p^2+p-1} \sim \frac{1}{p}. \quad (39.9)$$

Because (39.7) determines W_k only up to constants, (39.9) is a statement about the proxy weight, not an exact share of the atom mass; the moment proof below never uses α_p .

Proof. We give the review-inserted derivation rather than appeal to an average over c . Fix the block and k , and put $F(n) = n/\varphi(n)$ and

$$q_0 = \text{lcm}(g, \text{rad } k) \leq k.$$

Work in multiplicative boxes $u \asymp U$, $v \asymp V$. The conditions $(uv, k) = 1$ and $-uv^{-1} \equiv a \pmod{g}$ restrict $(u \bmod q_0, v \bmod q_0)$ to reduced pairs with a fixed ratio modulo g : for each of the $\varphi(q_0)$ choices of v -class, the u -class is determined modulo g and free at the remaining part, so exactly $\varphi(q_0)^2/\varphi(g)$ ordered class pairs are compatible. Shiu's theorem for the multiplicative function F , on the fixed-relative-length interval $u \asymp U$ in one reduced class modulo q_0 , is applicable since $q_0 \leq k \leq K$ and $U > H = K^{10}$, and gives

$$\sum_{\substack{u \asymp U \\ u \equiv r \pmod{q_0}}} F(u) \ll \frac{U}{\varphi(q_0)} \exp \left\{ - \sum_{p|k} \frac{1}{p-1} \right\} \asymp \frac{U}{\varphi(q_0)} \frac{\varphi(k)}{k}.$$

The last comparison holds because the logarithm of the quotient is $O(\sum_p p^{-2})$. Multiplying the u - and v -class sums by the number of compatible class pairs and dividing by UV yields, per box pair,

$$\sum_{\text{box}} \frac{1}{\varphi(u)\varphi(v)} \ll \frac{1}{\varphi(g)} \left(\frac{\varphi(k)}{k} \right)^2.$$

Summing the $O((\log(z/H))^2) = O(t^2)$ box pairs gives

$$\sum \frac{1}{\varphi(u)\varphi(v)} \ll \frac{(\log z)^2}{\varphi(g)} \frac{\varphi(k)^2}{k^2}. \quad (39.11)$$

For fixed (u, v) , Brun–Titchmarsh in the progression $\ell \equiv -k^{-1} \pmod{4uv}$, whose modulus is $4uv \leq 4x^{1/3}$, gives

$$\sum_{\substack{x < \ell \leq 2x \\ \ell \equiv -k^{-1} \pmod{4uv}}} \frac{1}{\ell} \ll \frac{1}{\varphi(4uv) \log x}, \quad \frac{1}{\varphi(4uv)} \ll \frac{1}{\varphi(u)\varphi(v)},$$

where the last inequality uses $(u, v) = 1$. Combining this with (39.11) over the boxes proves

$$\sum_{\substack{A \text{ in the block} \\ k_A = k, -uv^{-1} \equiv a \pmod{g}}} \frac{1}{\ell} \ll \frac{\log x}{\varphi(g)} \frac{\varphi(k)^2}{k^2}, \quad (39.10)$$

with an absolute constant, uniformly in $k \leq K$, $g \mid k$, and reduced a : no k^ε , divisor, or logarithmic loss enters. After the $O(t)$ blocks and division by k , this is the upper half of (39.7).

For the lower half drop the ratio condition. For one fixed k , a modulus $4uv$ has at most $2^{\omega(uv)} = (\log X)^{O(1)}$ ordered allocations. Thus ordinary Bombieri–Vinogradov, with a fixed sufficiently large logarithmic saving, evaluates the prime progressions without the factor K : k is fixed on this line. The low- ω Rankin argument from the source proof retains a fixed proportion uniformly. At a prime $p \mid k$ the exact coprime-pair local factor differs from $(1 - 1/p)^2$ by $1 + O(p^{-2})$, so the unrestricted harmonic pair mass is $\asymp (\log(z/H))^2 (\varphi(k)/k)^2$ uniformly. This gives the matching lower bound block by block and proves (39.7).

Finally $b(k) = (\varphi(k)/k)^2$ is multiplicative and $\sum b(k)/k \asymp \log K$, also after selecting $k \equiv 1 \pmod{4}$, by the principal/ χ_4 decomposition. At an odd prime its local factor is

$$1 + \sum_{e \geq 1} \frac{(1 - 1/p)^2}{p^e} = 1 + \frac{p-1}{p^2}, \quad (39.12)$$

so the part with $p \mid k$, divided by the full factor, is exactly (39.9). $\square$

Given a reduced fibre c , an atom with multiplier k is active for one of the $\varphi(k)$ unit residues and then has prime-coordinate probability $1/\ell$. Averaging those conditional masses produces the factor $1/\varphi(k)$, whereas the unconditioned cylinder has probability $1/(k\ell)$. Formula (39.7), not the conditional mass profile used in the earlier moving- c family, is the correct profile for factorial moments in the c -free CRT space.

10.3. Collapse versus consistency. The rough collapse bound which discards residue consistency would charge a shared prime p by $p\alpha_p^2 \asymp 1/p$, producing a false $\log \log K$ loss. The required cancellation is exact at the level of a new compatible atom.

For a cutoff y , put

$$q_y(k) = \prod_{\substack{p \mid k \\ p \leq y}} \frac{p}{p-1}, \quad b_y(g) = \prod_{\substack{p^e \parallel g \\ p \leq y}} \varphi(p^e) \prod_{\substack{p^e \parallel g \\ p > y}} p^e. \quad (39.13)$$

Condition the CRT space by $(n, P_y) = 1$, where $P_y = \prod_{p \leq y} p$. An atom then has probability $q_y(k)/(k\ell)$. If a compatible previously selected set has multiplier lcm R , adding an atom of multiplier k , with $g = (k, R)$, multiplies its intersection probability by

$$\frac{q_y(k)}{k\ell} b_y(g). \quad (39.14)$$

Compatibility fixes one unit residue modulo g . Lemma 10.2 says that the atom mass in that residue is at most $1/\varphi(g)$ of its multiplier mass. Thus the collapse left after summing the compatible new atoms is

$$\frac{b_y(g)}{\varphi(g)} = \prod_{\substack{p \mid g \\ p > y}} \frac{p}{p-1}; \quad (39.15)$$

small conditioned primes cancel exactly, and an unconditioned shared prime costs only $p/(p-1)$, not p .

Lemma 10.3 (source Lemma 39.3: uniform multiplier Euler bound; proved). *Uniformly in $y \geq 2$, every integer R , and $K \geq 3$,*

$$\sum_{\substack{k \leq K \\ k \equiv 1(4)}} \frac{\varphi(k)^2}{k^3} q_y(k) \prod_{\substack{p \mid (k, R) \\ p > y}} \frac{p}{p-1} \ll \log K. \quad (39.16)$$

Proof. It is enough to sum over all odd k . The unmodified local nonconstant mass is $(p-1)/p^2$. Multiplication either by q_y for $p \leq y$, or by $p/(p-1)$ when $p > y$ and $p \mid R$, changes this to $1/p$. The quotient

$$\frac{1 + 1/p}{1 + (p-1)/p^2} = 1 + O(p^{-2}) \quad (39.17)$$

has a convergent product, uniformly in the chosen set of modified primes. The remaining harmonic Euler product is $O(\log K)$. $\square$

Theorem 10.4 (source Theorem 39.4: all required upper factorial moments; proved, provisional). *In the CRT space conditioned by $(n, P_y) = 1$, for every integer $m \geq 1$,*

$$\mathbb{E}(H_X)_m \leq (Ct^3)^m, \quad (39.18)$$

with C depending at most on κ , uniformly in $2 \leq y < X^{1/2}$. The upper restriction is necessary: for $y \geq \ell$ the coprimality conditioning would renormalize the ℓ -coordinate itself by $\ell/(\ell-1)$, which (39.14) does not include. The application below has $y = Bt^3 < X^{1/2}$ for all sufficiently large X . In particular the bound holds through every order $m = D_B t^3$.

Proof. Expand the ordered factorial moment into distinct compatible atoms. By Lemma 10.1, every admissible next atom has a new ℓ . For a fixed preceding compatible set, (39.14), the residue-profile upper bound (39.7), and (39.15) bound the sum over all possible next atoms by

$$Ct^2 \sum_{\substack{k \leq K \\ k \equiv 1(4)}} \frac{\varphi(k)^2}{k^3} q_y(k) \prod_{\substack{p|(k,R) \\ p > y}} \frac{p}{p-1} \leq Ct^2 \log K \leq Ct^3. \quad (39.19)$$

Removing the forbidden old ℓ 's only decreases this sum. Iteration proves (39.18). $\square$

At pair level, for fixed k, k' and $g = (k, k')$,

$$\sum_{A:k_A=k} \sum_{\substack{B:k_B=k' \\ A \sim B}} \Pr(E_A \cap E_B) \ll \frac{g}{\varphi(g)} W_k W_{k'}. \quad (39.20)$$

Averaging (39.20) uses $\alpha_p^2(p/(p-1)-1) = O(p^{-3})$, not $p\alpha_p^2 \asymp 1/p$. Hence the pair excess is $O(\mu_X^2)$ with an absolute constant. It is *not* $o(\mu_X)$: a fixed small prime already contributes a constant multiple of μ_X^2 to the usual Janson dependency sum. Therefore the proposed direct condition $\Delta = o(\mu)$ and the naive Janson proof fail quantitatively. The factorial-moment theorem survives because an absolute base C^m , rather than Poisson relative error, is enough after the separate void estimate below.

10.4. The CRT void without partitioning by all of M_0 . A completely unconditioned c -free void still has a quarantine floor: taking $n \equiv 0 \pmod{p}$ for many small p disables every atom whose multiplier contains one of them. Thus $Q_r(H_X)$ alone cannot have cubic mean. Take

$$y = Bt^3 \quad (39.21)$$

with a sufficiently large fixed B , and multiply by the exact small-prime coprimality selector

$$S_y(n) = \mathbf{1}_{(n, P_y)=1} = \sum_{d|P_y} \mu(d) \mathbf{1}_{d|n}. \quad (39.22)$$

Every exceptional prime greater than y has $S_y = 1$. Notice that (39.22) has degree $\pi(y) = O(t^3/\log t)$, not the impossible degree $\pi(K)$.

Lemma 10.5 (source Lemma 39.5: conditioned c -free void; proved, provisional). *In the exact CRT space,*

$$\Pr(H_X = 0 \mid S_y = 1) \leq e^{-ct^3}. \quad (39.23)$$

Proof. Reveal all multiplier coordinates. Let c denote the resulting class modulo $24L_K$, where $L_K = \text{lcm}\{k \leq K : k \equiv 1 \pmod{4}\}$, and put

$$\mathcal{J}_c = \{k \leq K : k \equiv 1 \pmod{4}, (k, c) = 1\}, \quad Z(c) = \sum_{\substack{y < p \leq K, \\ p|L_K \\ p|c}} \frac{1}{p}. \quad (39.24)$$

The restriction $p | L_K$ makes Z well defined: a prime $p \equiv 3 \pmod{4}$ with $3p > K$ divides no admissible multiplier, so it has no CRT coordinate here and can be omitted; it removes nothing from $\mathcal{J}_c$. The coordinates for $p > y$, $p | L_K$, remain independent and $\Pr(p | c) = 1/p$. With exponential parameter y ,

$$\mathbb{E}e^{yZ} \leq \exp \left\{ Cy \sum_{p>y} p^{-2} \right\} = e^{O(1)}, \quad \Pr(Z > \eta) \leq e^{-\eta y + O(1)}. \quad (39.25)$$

Write $h(\mathcal{J}) = \sum_{k \in \mathcal{J}} \varphi(k)/k^2$. The elementary bounds

$$h(\mathcal{K}(K)) \asymp \log K, \quad \sum_{\substack{k \leq K \\ p|k}} \frac{\varphi(k)}{k^2} \ll \frac{\log K}{p} \quad (39.26)$$

show by the union bound that, for a sufficiently small fixed η , $Z(c) \leq \eta$ implies $h(\mathcal{J}_c) \geq c_1 \log K$. Also $S_y = 1$ makes c coprime to 24, and by definition it is coprime to $L_{\mathcal{J}_c}$.

Apply Theorem 9.4, the source-notes Theorem 34.8, to this arbitrary subfamily $\mathcal{J}_c$. Its lower proof lies in the canonical boxes (39.2), and its good triples are contained in the c -free unpruned family. Conditional on the multiplier coordinates, the distinct ℓ -coordinates are independent by Lemma 10.1. Therefore, when $Z(c) \leq \eta$,

$$\Pr(H_X = 0 | c) \leq \prod_{\ell} \left(1 - \frac{f_c^{\text{good}}(\ell)}{\ell} \right) \leq \exp\{-c_2 t^2 h(\mathcal{J}_c)\} \leq e^{-c_3 t^3}. \quad (39.27)$$

The bad multiplier coordinates cost at most $e^{-\eta B t^3 + O(1)}$ by (39.25). Increasing B if necessary and averaging (39.27) proves (39.23). $\square$

This is the load-bearing use of Theorem 9.4. It replaces Janson by supplying cubic active prime-coordinate mass uniformly for every reduced surviving multiplier fibre. The only fibres not reduced are those which set too much large-prime multiplier mass to zero, and (39.25) makes those fibres exponentially rare after the polynomial-sized coprimality cutoff. The dependency on that provisional theorem is therefore explicit and unavoidable in the present argument.

10.5. A low-degree majorant with the honest ledger. For even r , put

$$Q_r(h) = \sum_{j=0}^r (-1)^j \binom{h}{j}.$$

The exact identity

$$Q_r(0) = 1, \quad Q_r(h) = \binom{h-1}{r} \geq 0 \quad (h \geq 1) \quad (39.28)$$

shows that

$$\nu_X(n) = S_y(n) Q_r(H_X(n)) \quad (39.29)$$

is nonnegative and is at least one on every exceptional prime $n > \max(K, y)$.

Theorem 10.6 (source Theorem 39.6: modified c -free critical-window assembly; proved, CLAIMED/PROVISIONAL). *There are constants $c, C_0, D_B > 0$, depending at most on κ , such that if r is the least even integer at least $D_B t^3$, then (39.29) has exact CRT mean*

$$\mathbb{E}_{\text{CRT}} \nu_X \leq e^{-ct^3}. \quad (39.30)$$

Here and below D_B is the Bonferroni-degree constant, distinct from the source Lemma 16.2's ω -cutoff constant D in (39.3). Its congruence expansion has the explicit bounds

$$\begin{aligned} \text{degree} &\leq r + \pi(y) = O(t^3), \\ \log d_{\text{term}} &\leq O(y) + r(t + \log K) = O(t^4), \\ \log \sum_{\text{terms}} |c_{\text{term}}| &= O(rt) = O(t^4). \end{aligned} \quad (39.31)$$

Every nonzero term is one plain congruence class. Consequently, whenever $\log N \geq C_0 t^4$,

$$\sum_{n \leq N} \nu_X(n) \ll N e^{-ct^3}. \quad (39.32)$$

Proof. From (39.28), for $h \geq 1$, $\binom{h-1}{r} \leq \binom{h}{r+1}$. Lemma 10.5 and Theorem 10.4 therefore give

$$\begin{aligned} \mathbb{E}(Q_r(H_X) \mid S_y = 1) &\leq \Pr(H_X = 0 \mid S_y = 1) + \frac{\mathbb{E}((H_X)_{r+1} \mid S_y = 1)}{(r+1)!} \\ &\leq e^{-c_1 t^3} + \frac{(Ct^3)^{r+1}}{(r+1)!} \leq e^{-ct^3}, \end{aligned} \quad (39.33)$$

when D_B is sufficiently large. Multiplication by $\Pr(S_y = 1) \leq 1$ proves (39.30).

Expand (39.22) and the elementary symmetric polynomials in Q_r . Lemma 10.1 makes every compatible atom set use distinct ℓ 's; CRT then makes its further intersection with $d \mid P_y$ either empty or one class. If $A_X = |\mathcal{A}_X|$, the crude choice count from (39.2) gives $\log A_X = O(t)$, and hence

$$2^{\pi(y)} \sum_{j \leq r} \binom{A_X}{j} = \exp\{O(rt)\}. \quad (39.34)$$

The modulus is at most $P_y(KX)^r$, proving (39.31). On $[1, N]$, each class count is its CRT mean times N , plus $O(1)$. Choosing C_0 larger than the constants in (39.31) makes the aggregate rounding error $\exp\{O(rt)\}$ at most, say, $N^{1/2}$, which is negligible beside $N e^{-ct^3}$. This proves (39.32). $\square$

The literal source hypothesis $H'_{\text{PF}}(k\ell; \text{good})$ demanded coefficient sum $e^{O(J)}$, with $J \asymp t^3$, and a majorant for its larger aggregate avoider predicate. Theorem 10.6 proves *neither* clause: its direct even-Bonferroni expansion has ledger $e^{O(Jt)}$, and it only covers the c -free avoider, hence all exceptional primes. These are deliberate modifications, not suppressed losses. The ledger still closes because the critical window itself has logarithmic size $\asymp Jt$. Thus the literal $e^{O(J)}$ clause remains OPEN as stated, but is no longer needed for the exceptional set if Theorem 10.6 survives review.

10.6. Exceptional denominators.

Theorem 10.7 (source Theorem 39.7: unconditional 3/4 exceptional-set bound; CLAIMED/PROVISIONAL). *There is a constant $c > 0$ such that*

$$E_{\text{all}}(N) \ll N \exp\{-c(\log N)^{3/4}\}. \quad (39.35)$$

The same bound holds for exceptional primes.

Internal argument; provisional status retained. Put $L = \log N$, take $t = \alpha L^{1/4}$, and set $X = e^t$. Choose the fixed $\alpha > 0$ small enough that $L \geq C_0 t^4$. Lemma 10.1 and (39.29) majorize every exceptional prime larger than $\max(K, y)$; that omitted range is negligible. Equations (39.32) and $t^3 = \alpha^3 L^{3/4}$ give the prime bound.

For completeness, we spell out the semigroup transfer used for all denominators. Every prime factor of an exceptional integer $n > 1$ is itself exceptional: if $p \mid n$ and $4/p = 1/x_1 + 1/x_2 + 1/x_3$, scaling the three denominators by n/p represents $4/n$. Thus the exceptional integers are contained in the multiplicative semigroup generated by exceptional primes, including the empty product. The integer 1 contributes only $O(1)$.

Write $g(u) = u^{3/4}$ and $x = e^L$. Fix x_0 large enough that the prime bound gives, for every $x \geq x_0$,

$$E_{\text{prime}}(x) \ll x \exp\{-c_0 g(\log x)\}.$$

Fix $0 < \eta < c_0$ and, for large L , put $\delta = \eta g(L)/L$, so $0 < \delta < 1/4$. Rankin's inequality over the semigroup gives

$$E_{\text{all}}(x) \leq x^{1-\delta} \prod_{\substack{p \leq x \\ p \text{ exceptional}}} (1 - p^{-1+\delta})^{-1}.$$

The finitely many exceptional primes $p \leq x_0$ contribute $O(1)$, uniformly in δ . Partial summation and the endpoint estimate $\delta L - c_0 g(L) = -(c_0 - \eta)g(L) < 0$ give

$$\sum_{\substack{p \leq x \\ p \text{ exceptional}}} p^{-1+\delta} \ll 1 + \int_{\log x_0}^L \exp\{\delta u - c_0 g(u)\} du.$$

Because $g(u)/u = u^{-1/4}$ is decreasing, for $u \leq L$ one has $\delta u \leq \eta g(u)$. The last integral is therefore bounded by $\int_{\log x_0}^{\infty} e^{-(c_0 - \eta)g(u)} du < \infty$. The prime-power terms in the logarithm of the Euler product also satisfy

$$\sum_{p \leq x} \sum_{j \geq 2} \frac{p^{-j(1-\delta)}}{j} \leq \sum_p \sum_{j \geq 2} \frac{p^{-3j/4}}{j} = O(1).$$

Hence the Euler product is $O(1)$ and $E_{\text{all}}(x) \ll x^{1-\delta} = x \exp\{-\eta g(\log x)\}$, proving (39.35). $\square$

Theorem 10.7 is CLAIMED/PROVISIONAL. It uses the likewise provisional Theorem 9.4. It does not use the source hypotheses $H_{k\text{BV}}$, H'_{PF} , or Elliott–Halberstam, but “unconditional” does not remove the need for external verification.

10.7. Remarks on verification status. The three weakest steps identified by the maximum-severity self-review are reproduced faithfully here.

- (1) The new residue-resolved upper bound (39.11) must be rederived by a referee with every local factor present. It needs simultaneous uniformity in $k \leq X^\kappa$, every divisor $g \mid k$, and every unit residue; the proof uses Brun–Titchmarsh for the upper bound, so it does not hide a polynomially small Bombieri–Vinogradov main term. This is the most load-bearing new analytic lemma.
- (2) Lemma 10.5 uses Theorem 9.4 for the data-dependent subfamily $\mathcal{J}_c$ and uses the canonical lower boxes, not merely the larger family in the statement of the source formula (34.18). The quantifiers of Theorem 9.4 do allow every subfamily containing 1, and its proof furnishes the lower mass in those boxes, but this inheritance is new and needs an independent line-by-line check together with Theorem 9.4 itself.
- (3) The ordered-moment induction (39.14)–(39.19) is where lcm collapse and residue consistency cancel at every prime power. A missed factor there would be exponentiated through $r \asymp t^3$. The displayed conditional factor is $\varphi(p^e)$ for $p \leq y$ and p^e for $p > y$; after the $1/\varphi(g)$ residue share, only $p/(p-1)$ remains at an unconditioned shared prime. This exact replay, including unequal prime-power exponents, is mandatory in external review.

Internally, the quantifier chain is complete: Theorem 9.4 is uniform in $\mathcal{J}_c$; (39.25) removes the nonreduced fibres; (39.18) controls the even Bonferroni tail; and (39.31)–(39.34) pay the full $e^{O(t^4)}$ coefficient ledger before selecting $t = \alpha(\log N)^{1/4}$. The direct Janson proposal is rejected because its dependency sum is $\Theta(\mu^2)$, while the original $e^{O(J)}$ formulation remains open. Theorem 10.7 should not be cited as established outside this campaign until an independent expert has checked the three steps above and the priority claim against the primary literature.

A fresh search found no post-1970 improvement of Vaughan’s $N/\exp(c(\log N)^{2/3})$ exceptional-set bound. Pomerance–Weingartner’s 2025 preprint [13, §4] explicitly cites Vaughan 1970 as the state of the art and reproduces the same $2/3$ exponent when its general theorem is specialized to $m = 4$. Access to Vaughan’s paper was blocked, but DOI [10.1112/S0025579300002886](https://doi.org/10.1112/S0025579300002886) was confirmed. If Theorem 10.7 survives external review, it would therefore be the first exponent improvement found since 1970. This is a search result, not a literature guarantee; the CLAIMED/PROVISIONAL label stands.

10.8. Internal verification pedigree. The counts in this subsection concern internal checks, not independent external validation. Theorem 9.4 has had three independent internal checks. In the third, a fresh reviewer rederived the power-sized extension of the box argument, expanded the second incidence moment, paid all boundary errors, and rebuilt the low-congestion Bombieri–Vinogradov assembly. The reviewer checked Shiu’s 1980 paper [17] directly: the interval and modulus inequalities hold with Shiu’s parameters $\alpha = \beta = 1/4$, the residue classes are reduced, and both multiplicative-function hypotheses apply in the fixed-relative-length boxes with $H = K^{10}$. The use of $1 \in \mathcal{J}$, same-prime distinctness, the canonical lower boxes, and the data-dependent family $\mathcal{J}_c$ in Lemma 10.5 were separately replayed. That check was CONFIRMED-AFTER-REPAIRS; the provisional register did not change.

The downstream fixed- c -free chain has, in addition to its original construction, one attested-blind parallel construction and two paper fidelity passes. For the blind

construction, the unit read only the task brief and the source-note analogues of Sections 16 and 34 at repository commit `c489912b`, and froze its complete S1–S7 derivation at commit `347a570` before opening the source analogue of Section 39. It did not inspect the earlier verification program, although the mandatory baseline run printed that program’s summary line. Git verifies the snapshots, texts, and commit order; it cannot verify what was visible on screen or enforce read isolation. The blindness claim is therefore an explicit self-attestation, not an independently audited fact. Reconciliation classified all seven steps as CONFIRMED-SAME-METHOD; the variants (modulus k versus $\text{lcm}(g, \text{rad } k)$ in Shiu, a pointwise Euler bound, and direct weighted-fibre averaging) made no quantifier or range change. Its separate exact-rational stress test checked moments through order six, 14,210 atoms, and 1,128,378 compatible pairs. Those finite checks are bug hunts, not asymptotic evidence, and the blind unit treated Theorem 9.4 as an allowed input.

Wave 15 then ran three separate maximum-severity hostile reviews, one each for the general-numerator transfer, the slice-conspiracy analysis, and the Kloosterman endpoint. Each was repaired before inclusion here. The slice review repaired the raw/primitive boundary. The endpoint review checked the actual DFI and Bettin–Chandee statements and discovered Theorem 20.2, which was then independently verified internally.

Wave 16 added an attested-blind general- m construction. The unit read only the prescribed source §§16, 34, 39 and the archived Pomerance–Weingartner text; `blind43.md` logs its reads, and git records its S1–S7 freeze at `a0bd2940c322` before source §43 was opened. This is a self-attestation, not an audit. The identity, aggregate supply, atoms, and profiles converged; the cutoff required repair, while the moment window and assembly diverged and supplied (43.36). A separate maximum-severity adjudication rechecked the quarantine, moments, ledger, range, semigroup transfer, and crossover and returned STRENGTHENING CONFIRMED. It retained Theorem 9.4, the source §39.7 qualifications, and the new headline as CLAIMED/PROVISIONAL; none of these passes was external refereeing or priority certification.

Wave 17 added maximum-severity hostile reviews of the implication-antichain analysis and the conditional slice frontier. The first returned CONFIRMED-AFTER-REPAIRS: it checked the exact inclusion direction, semiprime-atom retention, every quantifier in the grid, the extension ratio one, and the direct reduced-moment lower bound. The second returned SOUND-AFTER-REPAIRS: it independently rederived the prime-factor criterion and checked the genus, raw/primitive, admissibility, standard- hypothesis, and streamed-census boundaries. These are internal reviews and computational bug hunts, not external validation of either provisional record.

Wave 18’s separate hostile reviews of the witness/effectivity and per-slice sections both returned SOUND-AFTER-REPAIRS: they repaired prime scope and the induced-character/dyadic-conductor ledger, and replayed the three local classes, exact exponent, elementary remainder, fixed-refinement corollary, and ineffective uniform scope. Wave 19’s stacked-sieve review returned the same verdict after repairs to the overlap threshold, empty Page branch, growing-dimensional sieve ledger, and varying-cutoff inclusion; it confirmed Theorem 53.3 as proved and effective. The attested-blind protocol in Section 14 was adjudicated CONVERGED-WITH-DIVERGENCES: exact tails and effectivity converged, but its ancillary- $o(1)$ overclaim was rejected and no label changed. Blocks (ax)–(az) check respectively

the W tail, the fixed-slice sieve, and stacked geometry. These three hostile reviews, the adjudication, and all finite checks remain internal, not external validation.

Wave 20 added hostile reviews of the logarithmic lower tail and the m -uniform effectivity perimeter. The first returned CONFIRMED-AFTER-REPAIR; the repair enlarged verifier coverage but did not change Section 17. The second returned SOUND-AFTER-REPAIRS; it narrowed the favorable-sign claim to the full Layer-1 family and the structured c -free fibres, inserted the worst-case fibre subtraction, and kept every inherited provisional label. Block (ba) checks the two residue-one lower-tail constructions, and block (bb) checks the general- m finite harvest and favorable-sign mass. A wave-21 hostile review of the certificate ceilings and extremal census returned SOUND-AFTER-MINIMAL-SCOPE-REPAIRS. It preserved both constants and proofs while making explicit that a generic least-prime bound must select a prime beyond an eventual certificate's finite exceptional prefix, correcting the census-memory description to a linear resident-state bound, and restoring the architecture-only pointwise pointer. Block (bc) replays the certificate data, record tables, natural-log normalizations, and the memory-bounded full scans. The internal verifier coverage used by this draft runs through (bc). These are internal reviews and bug hunts, not external validation. A separate full-source review of the auro-zero Lean file replayed its identities but did not compile it; its open-core characterization and the source paper's own conditional Conclusion are related-work checks, not validation of this paper.

11. GENERAL NUMERATORS

For $m \geq 3$, define

$$E_m(N) = \#\{n \leq N : m/n \text{ is not a sum of three positive unit fractions}\}. \quad (43.1)$$

This concerns the rational number: reducing a common factor does not change representability. We nevertheless count the unreduced pairs, as does Pomerance–Weingartner [13].

11.1. The identity and the first exceptional-set layer.

Lemma 11.1 (general-numerator multiplier identity). *Let $k, \ell \geq 1$, suppose $k\ell \equiv -1 \pmod{m}$, and let $A = (k\ell + 1)/m = uvw$ be a factorization into positive integers. If $n > 0$, $nv \equiv -u \pmod{k\ell}$, and $s = (nv + u)/(k\ell)$, then*

$$\frac{m}{n} = \frac{1}{suw} + \frac{1}{nsuv} + \frac{1}{nuvw}. \quad (43.2)$$

Moreover $(A, k\ell) = 1$, so v is invertible modulo $k\ell$.

Proof. The congruence makes s a positive integer. The numerator over the common denominator $nsuvw$ is $nv + u + s = s(k\ell + 1) = smuvw$. A common divisor of A and $k\ell$ divides $mA - k\ell = 1$. $\square$

No parity, primality, or coprimality condition on m and n occurs here. For a prime $\ell \nmid m$, the old condition $k \equiv 1 \pmod{4}$ is replaced by $k \equiv -\ell^{-1} \pmod{m}$.

Put

$$\eta_1(m) = \prod_{p|m} \frac{p}{p+1}, \quad \eta_2(m) = \prod_{p|m} \frac{p^2}{p^2+p-1}, \quad C_2 = \prod_p (1-p^{-1}) \left(1 + \frac{p-1}{p^2}\right). \quad (43.3)$$

Lemma 11.2 (exact thinning factors). *For fixed m and every reduced $r \pmod{m}$,*

$$\begin{aligned} \sum_{\substack{k \leq K \\ k \equiv r \pmod{m}}} \frac{\varphi(k)}{k^2} &\sim \frac{\eta_1(m)}{\varphi(m)\zeta(2)} \log K, \\ \sum_{\substack{k \leq K \\ k \equiv r \pmod{m}}} \frac{\varphi(k)^2}{k^3} &\sim \frac{C_2 \eta_2(m)}{\varphi(m)} \log K. \end{aligned} \quad (43.4)$$

If $\log m = O(\log K)$, the aggregate sums over $(k, m) = 1$ are, up to constants depending only on that O -constant, $\eta_1(m) \log K$ and $\eta_2(m) \log K$. Also

$$\frac{\varphi(m)}{m} \leq \eta_1(m) \leq \zeta(2) \frac{\varphi(m)}{m}, \quad \eta_2(m) \asymp \frac{\varphi(m)}{m}. \quad (43.5)$$

Proof. The Euler series for $\varphi(k)/k$ is $\zeta(s+1)/\zeta(s+2)$; removing multiples of $p \mid m$ deletes the local factor $1 + 1/p$. For $b(k) = (\varphi(k)/k)^2$, the local harmonic factor is

$$1 + \sum_{e \geq 1} \frac{(1 - p^{-1})^2}{p^e} = 1 + \frac{p-1}{p^2}.$$

Character orthogonality divides the principal pole by $\varphi(m)$ and, for fixed m , each nonprincipal twist is regular there. The same convolution argument without characters proves the uniform aggregate bounds. Finally, divide each product in (43.3) by $\varphi(m)/m$; the resulting Euler products converge absolutely. $\square$

The fixed-class asymptotics in (43.4) are deliberately not asserted uniformly for growing m . In particular, no Siegel-uniform estimate and no equidistribution among short residue classes is used below. Only the aggregate nonnegative family $(k, m) = 1$ is used.

Define the multiplicative function

$$F_m(p^a) = \begin{cases} 1, & p \mid m, \\ p/(p-1), & p \nmid m. \end{cases}$$

For $(u, v) = 1$ one has the exact identity

$$\frac{1}{\varphi(muv)} = \frac{F_m(u)F_m(v)}{\varphi(m)uv}. \quad (43.6)$$

This avoids both a false extra factor $1/m$ and the false restriction $(uv, m) = 1$.

Lemma 11.3 (general- m class mass). *Fix $B > 0$. Let $K_0 \leq K \leq (\log X)^B$, $H = K^{10}$, $m \leq (\log X)^B$, and $\mathcal{J} \subseteq \{k \leq K : (k, m) = 1\}$ contain 1. Put $h(\mathcal{J}) = \sum_{k \in \mathcal{J}} \varphi(k)/k^2$ and suppose $(c, L_{\mathcal{J}}) = 1$. For primes $X^{1/2} < \ell \leq X$, let $f_{m,c}(\ell)$ count the distinct residues $-uv^{-1} \pmod{\ell}$ furnished by*

$$\begin{aligned} k &\in \mathcal{J}, \\ H < u, v \leq \ell^{1/3}, \quad (u, v) &= (uv, k) = 1, \quad \omega(uv) \leq D \log \log X, \\ muv &\mid k\ell + 1, \quad k \mid u + cv. \end{aligned} \quad (43.7)$$

Then, with constants depending only on B ,

$$\sum_{X^{1/2} < \ell \leq X} \frac{f_{m,c}(\ell)}{\ell} \asymp_B \frac{(\log X)^2 h(\mathcal{J})}{\varphi(m)}. \quad (43.8)$$

Each counted residue, combined with the congruence modulo k , is forced by Lemma 11.1.

Proof. For fixed k , the lattice congruence is still $u + cv \equiv 0 \pmod{k}$. Thus the box count, boundary ratio $O(K^2/H)$, low- ω Rankin tail, and Shiu modulus condition $k < U^{1/2}, V^{1/2}$ are unchanged. The condition $k \equiv -\ell^{-1} \pmod{m}$ is not a second lattice condition: after (k, u, v) is fixed, all congruences are carried by the one reduced prime progression

$$\ell \equiv -k^{-1} \pmod{muv}. \quad (43.9)$$

The modulus is the product muv , not $\text{lcm}(m, uv)$; common prime powers, including powers of 2, are already present with their summed exponent. In the canonical boxes $uv \leq x^{1/3}$, so a fixed log-power m remains below the Bombieri–Vinogradov level. Each modulus has at most $K2^{\omega(uv)}$ descriptions, absorbed by choosing the logarithmic saving in terms of B . Brun–Titchmarsh gives the upper half. Identity (43.6) and the same two-variable harmonic calculation give exactly the factor $1/\varphi(m)$ on both sides. Shiu’s auxiliary modulus remains k ; no condition $(uv, m) = 1$ is inserted.

For deduplication, a collision at one ℓ gives $\ell \mid uv' - u'v$ and $|uv' - u'v| < \ell^{2/3} < \ell$, hence reducedness gives $(u, v) = (u', v')$. The two atoms then fix $k \pmod{muv}$, and $muv > mH^2 > K$ gives $k = k'$. Finally $n \equiv c \pmod{L_{\mathcal{J}}}$ and a hit modulo ℓ combine to $nv \equiv -u \pmod{k\ell}$. $\square$

Theorem 11.4 (first general-numerator layer; CLAIMED/PROVISIONAL). *For every fixed $m \geq 3$ there is an absolute $c_0 > 0$ such that, for all sufficiently large N ,*

$$E_m(N) \ll_m N \exp \left\{ -c_0 \left(\frac{\eta_1(m)}{\varphi(m)} \right)^{1/3} (\log N)^{2/3} (\log \log N)^{1/3} \right\}. \quad (43.10)$$

The prime-denominator and all-denominator forms are both uniform, for every fixed $\varepsilon > 0$, when $3 \leq m \leq (\log N)^{2-\varepsilon}$, after an ε -dependent lower threshold. This theorem has a complete internal proof but remains provisional externally because it would improve the published general- m benchmark.

Internal argument; provisional status retained. Take $K = \lfloor \delta \log N \rfloor$, all $k \leq K$ coprime to m , and $M_0 = \text{lcm}_{k \leq K, (k, m)=1} k$. Since $\log M_0 \leq (1 + o(1))K$, each reduced subsequence has length $N^{1-O(\delta)}$. Apart from the $O(\log m)$ primes dividing m , every prime denominator above K occurs in one. Lemmas 11.2 and 11.3 give uniformly in its residue c the prime class mass

$$\asymp \lambda_m (\log X)^2 \log K, \quad \lambda_m = \frac{\eta_1(m)}{\varphi(m)}.$$

The upper mass has the same factor, so the larger-sieve Rankin truncation used in Theorem 9.1 closes with

$$\log X = \alpha \left(\frac{\log N}{\lambda_m \log \log N} \right)^{1/3}.$$

This gives (43.10) for primes. In the stated range, $\log X = o(\log N)$ and K, m are fixed powers of $\log X$; the sieve length and BV/Shiu levels follow with a saving chosen in terms of ε , not m .

If $a \mid n$ and m/a is representable, scaling its denominators by n/a represents m/n . Thus every divisor, in particular every prime factor, of an exceptional n is exceptional. The semigroup argument in Theorem 10.7 applies with $g_m(u) = \lambda_m^{1/3} u^{2/3} (\log u)^{1/3}$.

For the uniform transfer put $\Theta_1 = \varphi(m)/\eta_1(m) \asymp m$, fix the final margin $0 < \varepsilon_0 < 2$, choose $0 < \varepsilon' < \varepsilon_0/(3 - \varepsilon_0)$, and start the prime estimate at $\log x_0 \asymp \Theta_1^{1/(2-\varepsilon')}$. Its Rankin parameter is $\delta \asymp \Theta_1^{-1/3}(\log N)^{-1/3}(\log \log N)^{1/3}$, and for $\Theta_1 \ll (\log N)^{2-\varepsilon_0}$,

$$\delta \log x_0 \ll (\log N)^{-\gamma}(\log \log N)^{1/3} = o(1), \quad \gamma = \frac{\varepsilon_0(1 + \varepsilon') - 3\varepsilon'}{3(2 - \varepsilon')} > 0.$$

Declaring every smaller prime exceptional costs $\exp\{O(\log \Theta_1)\}$, absorbed by the displayed saving. Above x_0 , $g_m(u)/u$ decreases and $g_m(\log x_0)$ is a positive power of Θ_1 , so partial summation is uniform. This gives the same $3 \leq m \leq (\log N)^{2-\varepsilon_0}$ range for all denominators. $\square$

11.2. The provisional cubic transfer. Fix $0 < \kappa < 1/240$, put $t = \log X$, $K = \lfloor X^\kappa \rfloor$, $H = K^{10}$, and use the dyadic blocks $x < \ell \leq 2x$ in $(X^{1/2}, X]$. The general- m c -free atoms are

$$\begin{aligned} A &= (k, \ell, u, v), \quad k \leq K, \quad (k, m) = 1, \quad H < u, v \leq x^{1/6}, \\ (u, v) &= (uv, k) = 1, \quad \omega(uv) \leq D \log \log X, \quad muv \mid k\ell + 1, \\ E_A &= \{n : n \equiv -uv^{-1} \pmod{k\ell}\}. \end{aligned} \quad (43.11)$$

The proof of Lemma 10.1, with $4uv$ replaced by muv , shows that the coupling $k \mid u + cv$ is automatic after revealing $c \equiv n \pmod{L_{m,K}}$, distinct compatible atoms have distinct ℓ 's, and

$$\Pr \left(\bigcap_{i \leq j} E_{A_i} \right) = \frac{\mathbf{1}_{\text{compatible}}}{\text{lcm}(k_1, \dots, k_j) \prod_i \ell_i}. \quad (43.12)$$

The two relevant inequalities only improve: $x^{1/3} < \ell$ makes (u, v) unique, and $muv > K$ then makes k unique.

Lemma 11.5 (general- m mass profiles). *Uniformly for $m \leq t^B$, $g \mid k$, and reduced $a \pmod{g}$,*

$$W_{k,a}(g) \ll_B \frac{t^2}{\varphi(m)\varphi(g)} \frac{\varphi(k)^2}{k^3}, \quad W_k \asymp_B \frac{t^2}{\varphi(m)} \frac{\varphi(k)^2}{k^3}. \quad (43.13)$$

Consequently

$$\mu_{m,X} \asymp_B \frac{\eta_2(m)}{\varphi(m)} t^3. \quad (43.14)$$

Proof. Identity (43.6) extracts $1/\varphi(m)$. In boxes put $q_0 = \text{lcm}(g, \text{rad } k)$. Since $(k, m) = 1$, there are still $\varphi(q_0)^2/\varphi(g)$ compatible reduced class pairs. Shiu applies to $F_m \leq n/\varphi(n)$ because $q_0 \leq k < U^{1/10}, V^{1/10}$, and gives

$$\sum_{\text{boxes}} \frac{F_m(u)F_m(v)}{uv} \ll \frac{t^2}{\varphi(g)} \left(\frac{\varphi(k)}{k} \right)^2.$$

Brun–Titchmarsh on modulus muv proves the upper profile. For the BV lower profile, replacing $n/\varphi(n)$ by $F_m(n)$ changes the coprime-pair factor at $p \mid m$ by $(p^2 - 1)/(p^2 + 1)$; the product is bounded away from zero. The Rankin truncation and allocation multiplicity are unchanged. Summation and the second aggregate formula in Lemma 11.2 prove (43.14). $\square$

Theorem 11.6 (general- m pruned cubic slice; CLAIMED/PROVISIONAL). *For fixed B , uniformly for $m \leq t^B$, every $\mathcal{J} \subseteq \{k \leq K : (k, m) = 1\}$ containing 1, and every reduced fibre $(c, L_{\mathcal{J}}) = 1$, the restriction $r_{\mathcal{J}}(u, v; c) \leq t^4$ leaves*

$$\sum_{X^{1/2} < \ell \leq X} \frac{f_{m,c}^{\text{good}}(\ell)}{\ell} \asymp_B \frac{t^2 h(\mathcal{J})}{\varphi(m)}. \quad (43.15)$$

For the full coprime family this is $\asymp \eta_1(m)t^3/\varphi(m)$. This theorem inherits the provisional status of Theorem 9.4.

Proof ledger; provisional status retained. The incidence second moment is unchanged after restricting the outer k -set to $(k, m) = 1$: it sees only $k \mid u + cv$, and its lcm boundary is at most K^2 . Markov pruning therefore removes $o(t^2 h(\mathcal{J}))$. Identity (43.6) leaves at least $1/\varphi(m)$ of the retained harmonic mass. The retained multiplicity is $2^{\omega(uv)}t^4$, while BV runs over $muv \leq t^B x^{1/3}$; increasing its fixed logarithmic saving by B makes the error $o(x \log x h(\mathcal{J})/\varphi(m))$. Deduplication is (43.12), and the upper bound is Lemma 11.5. These are all entries at which m changes the proof of Theorem 9.4. $\square$

For completeness, the prime-power moment cancellation also survives. Condition on $(n, P_y) = 1$ and add a compatible atom with multiplier k to a set whose multiplier lcm is R . With $g = (k, R)$, the exact density ratio is $q_y(k)b_y(g)/(k\ell)$, while compatibility fixes one unit residue modulo g . Lemma 11.5 supplies $1/\varphi(g)$, leaving

$$\frac{b_y(g)}{\varphi(g)} = \prod_{p \mid g, p > y} \frac{p}{p-1}. \quad (43.16)$$

Primes dividing m do not divide k . At an unmodified such prime, deleting the multiplier Euler factor gives $p^2/(p^2 + p - 1)$; if $p \leq y$, it instead gives the smaller $p/(p+1)$. Thus the total deletion factor is at most $\eta_2(m)$, and

$$\sum_{\substack{k \leq K \\ (k, m) = 1}} \frac{\varphi(k)^2}{k^3} q_y(k) \prod_{p \mid (k, R), p > y} \frac{p}{p-1} \ll \eta_2(m) \log K. \quad (43.17)$$

The ordered induction, including unequal prime-power exponents, gives

$$\mathbb{E}(H_{m,X})_j \leq \left(C \frac{\eta_2(m)}{\varphi(m)} t^3 \right)^j \leq (Ct^3)^j. \quad (43.18)$$

Theorem 11.7 (source Theorem 43.8; CLAIMED/PROVISIONAL). *For every fixed $m \geq 3$, and uniformly for each fixed $\varepsilon > 0$ in $3 \leq m \leq (\log N)^{3/4-\varepsilon}$,*

$$E_m(N) \ll N \exp \left\{ -c_0 \frac{\eta_1(m)}{\varphi(m)} (\log N)^{3/4} \right\}. \quad (43.28)$$

This holds for primes and all denominators and inherits the provisional status of Theorems 9.4 and 10.7.

It is the direct §39 replay with $y, r \asymp t^3$; it remains valid, but is superseded by the thinned-window theorem below.

11.3. Pomerance–Weingartner comparison. Pomerance–Weingartner’s Theorem 1.3 [13] states that there is an absolute $C > 0$ such that, for $4 \leq m \leq (\log N)^2$,

$$E_m(N) \leq N \exp \left\{ -C \left(\frac{(\log N)^2}{\varphi(m)} \right)^{1/3} \right\}. \quad (43.32)$$

Their prime mass is $(\log X)^2/\varphi(m)$, so no $(\log \log N)^{1/3}$ is hidden in (43.32). For the optimized provisional layer, write $L = \log N$ and compare the exponent scales

$$R = \left(\frac{\eta_2(m)}{\varphi(m)} \right)^{1/4} L^{3/4}, \quad P = \frac{L^{2/3}}{\varphi(m)^{1/3}}. \quad (43.33)$$

Then

$$\frac{R}{P} = \{\eta_2(m)^3 \varphi(m) L\}^{1/12}, \quad R \geq P \iff L \geq \frac{1}{\eta_2(m)^3 \varphi(m)}. \quad (43.34)$$

Unknown absolute constants prevent reading (43.34) as a literal finite threshold. In the common stated domain $4 \leq m \leq L^2$, the formal new scale is larger once (43.34) holds, and it eventually does for each fixed m . Theorem 11.8 continues through every fixed-gap range $m \leq L^{3-\varepsilon}$, where the quoted published theorem is not stated once $m > L^2$; at $m \asymp L^3$ the new saving stops growing, so no claim is made there or beyond. This scale comparison does not erase the provisional-versus-published evidentiary distinction. The first layer is uniform for primes and all denominators through $m \leq L^{2-\varepsilon}$ and has exponent ratio $(\eta_1(m) \log L)^{1/3}$ relative to (43.32).

11.4. The attested-blind thinned window. The protocol and adjudication are recorded in Subsection 10.8. In particular, the blind construction’s finite relative-mass inequality was adjudicated line by line, but neither that internal verdict nor the attestation upgrades the inherited status.

Theorem 11.8 (source Theorem 43.12; CLAIMED/PROVISIONAL). *Put $L = \log N$. For every fixed $\varepsilon > 0$, uniformly for $3 \leq m \leq L^{3-\varepsilon}$,*

$$E_m(N) \ll_{\varepsilon} N \exp \left\{ -c_{\varepsilon} \left(\frac{\eta_2(m) L^3}{\varphi(m)} \right)^{1/4} \right\}. \quad (43.35)$$

The same formula holds for exceptional primes. It inherits the provisional Theorem 9.4 through Theorem 11.6, together with every maximum-severity qualification in Subsection 10.7 (source §39.7); the internal wave-16 adjudication changes the exponent and range, not the external status.

Internal argument; provisional status retained. Put

$$\theta_m = \frac{\eta_2(m)}{\varphi(m)}, \quad \Theta_m = \theta_m^{-1} \asymp m, \quad M = \theta_m t^3.$$

The shrinking-window input is a finite relative-mass inequality. If $H_m(K) = \sum_{k \leq K, (k, m)=1} \varphi(k)/k^2$, then for every prime $p \nmid m$,

$$\sum_{\substack{k \leq K, (k, m)=1 \\ p|k}} \frac{\varphi(k)}{k^2} = \sum_{e \geq 1} \frac{1 - p^{-1}}{p^e} \sum_{\substack{a \leq K/p^e \\ (a, pm)=1}} \frac{\varphi(a)}{a^2} \leq \frac{H_m(K)}{p}. \quad (43.36)$$

For $p \mid m$ the left side is zero. Reveal the multiplier coordinates, put $\mathcal{J}_c = \{k \leq K : (k, m) = (k, c) = 1\}$, and set $Z(c) = \sum_{y < p \leq K, p \mid c, p \mid L_{m,K}} 1/p$. The union bound and (43.36) give $h(\mathcal{J}_c) \geq H_m(K)(1 - Z(c))$. Independence above y gives

$$\mathbb{E}e^{yZ} \leq \exp \left\{ Cy \sum_{p > y} p^{-2} \right\} = e^{O(1)}, \quad \Pr\{Z > \eta_0\} \leq e^{-\eta_0 y + O(1)} \quad (43.37)$$

for a fixed $0 < \eta_0 < 1$.

Take an integer $y \asymp B_0 M$, choosing the absolute B_0 after the good-fibre constant. On $Z \leq \eta_0$, Theorem 11.6 gives active prime-coordinate mass

$$\frac{t^2 h(\mathcal{J}_c)}{\varphi(m)} \gg \frac{\eta_1(m)}{\varphi(m)} t^3 \asymp M,$$

so the conditioned void is e^{-cM} ; (43.37) makes the bad-fibre cost the same size. The estimate is uniform because $y \sum_{p > y} p^{-2} = O(1)$ and $\eta_1 \asymp \eta_2$. Equation (43.18) supplies every factorial moment with base CM . Taking r to be the least even integer at least $D_0 M$, with fixed sufficiently large D_0 , Stirling and the exact even-Bonferroni identity give

$$\mathbb{E}_{\text{CRT}}\{S_y Q_r(H_{m,X})\} \leq e^{-cM} + \frac{(CM)^{r+1}}{(r+1)!} \leq e^{-c'M}. \quad (43.38)$$

The resulting expansion has the genuinely thinned ledger

$$\begin{aligned} \deg &= O(M), \\ \log d_{\text{term}} &\leq O(y) + r(t + \log K) = O(Mt), \\ \log \sum |c_{\text{term}}| &= O(rt) = O(Mt). \end{aligned} \quad (43.39)$$

Here $\log |\mathcal{A}_{m,X}| = O(t)$, and the atom modulus is $k\ell$, not $mk\ell$: m selects atoms but contributes no ledger factor. Thus finite-interval rounding is absorbed when

$$L \geq C_0 M t = C_0 \theta_m t^4. \quad (43.40)$$

Choose $t = \alpha(L/\theta_m)^{1/4}$ with fixed sufficiently small α . Then

$$M \asymp \theta_m^{1/4} L^{3/4}, \quad (43.41)$$

which proves (43.35) for primes.

The fixed-gap range supplies every analytic constraint. Since $\Theta_m \asymp m \leq L^{3-\varepsilon}$, $t/L \asymp (\Theta_m/L^3)^{1/4} \ll L^{-\varepsilon/4}$ and $M \gg L^{\varepsilon/4}$. Also $m \leq t^4$ after a harmless fixed enlargement, $K = e^{\kappa t} \gg m$, $mu\nu$ remains exponentially below the Bombieri–Vinogradov level, $2 \leq y < X^{1/2}$, and $\max(m, K, y) = e^{o(L)}$. Shrinking y exposes more primes, but (43.16)–(43.18) already sum their local factors uniformly.

For the all-denominator transfer fix a final margin $0 < \varepsilon_0 < 3$, choose $0 < \varepsilon' < \varepsilon_0/(4 - \varepsilon_0)$, and start the prime theorem at $u_0 = \log x_0 \asymp \Theta_m^{1/(3-\varepsilon')}$. With $g_m(u) = \Theta_m^{-1/4} u^{3/4}$ and $\delta \asymp \Theta_m^{-1/4} L^{-1/4}$,

$$\delta u_0 \ll L^{-\gamma}, \quad \gamma = \frac{\varepsilon_0(1 + \varepsilon') - 4\varepsilon'}{4(3 - \varepsilon')} > 0. \quad (43.42)$$

Thus declaring all smaller primes exceptional costs only $\exp\{O(\log \Theta_m)\} = e^{o(g_m(L))}$. Above x_0 , $g_m(u)/u$ decreases and $g_m(u_0) = \Theta_m^{\varepsilon'/(4(3-\varepsilon'))}$ beats the partial-summation Jacobian uniformly. The semigroup argument of Theorem 10.7 preserves the exponent and the full $L^{3-\varepsilon_0}$ fixed-gap range. $\square$

No pointwise equidistribution among the classes $k \pmod{m}$ is used, and no growing-saving claim is made at $m \asymp L^3$ or beyond.

12. POINTWISE TYPE-I SLICES

12.1. Exact vanishing and finite conspiracies. For an odd prime p , put

$$\mathcal{B}_p = \left\{ (c, k) : 1 \leq k \leq \lfloor 2p/3 \rfloor, 1 \leq c \leq \left\lfloor \frac{2p+k}{4k} \right\rfloor, (p, ck) = 1 \right\}.$$

For $(c, k) \in \mathcal{B}_p$, let $h = 4ck$, $N_{c,k} = p^2 + 4ck^2$, and define the raw ordered Type-I slice

$$M_{c,k}(p) = \#\{D : D \mid N_{c,k}, D \equiv -p \pmod{h}\}. \quad (44.1)$$

Each such divisor gives the ordered row $a = (D + p)/h$, $b = (N_{c,k}/D + p)/h$ in $p(a + b) = k(4abc - 1)$, and every row arises this way.

If $N_{c,k} = \prod_j \ell_j^{e_j}$, character orthogonality gives the exact vanishing law

$$M_{c,k}(p) = \frac{1}{\varphi(h)} \sum_{\chi \pmod{h}} \overline{\chi(-p)} \prod_j (1 + \chi(\ell_j) + \cdots + \chi(\ell_j)^{e_j}), \quad (44.2)$$

$$M_{c,k}(p) = 0 \iff \nexists (0 \leq u_j \leq e_j) : \prod_j \ell_j^{u_j} \equiv -p \pmod{4ck}.$$

The second line is an exponent-box condition, not merely subgroup membership; vanishing need not be caused by one character factor. This is the raw count. The primitive count additionally requires $(a, b) = 1$ and can vanish when the raw count does not: for $(p, c, k) = (241, 7, 3)$ the two target divisors give $(a, b) = (3, 66), (66, 3)$, so $M_{7,3}(241) = 2$ but the primitive mass is zero.

The deterministic obstruction is a moving genus character. Write $c = st^2$ with s squarefree, let $\Delta_s < 0$ be the fundamental discriminant of $\mathbb{Q}(\sqrt{-s})$, and put

$$\chi_s(n) = \left(\frac{\Delta_s}{n} \right). \quad (48.1)$$

Its conductor divides $4s$.

Theorem 12.1 (moving genus obstruction; source Theorem 48.1). *Let $p \equiv 1 \pmod{24}$ and $(c, k) \in \mathcal{B}_p$. Then*

$$\chi_s(p) = 1 \implies M_{c,k}(p) = 0. \quad (48.2)$$

The restriction of χ_s to the hard progression is identically one if and only if

$$s \in \{1, 2, 3, 6\}. \quad (48.3)$$

For every other squarefree s , the hard primes with $\chi_s(p) = 1$ have relative Dirichlet density $1/2$, apart from the finitely many ramified primes; admissibility for fixed (c, k) removes only the additional finite set dividing ck .

Proof. Every prime $\ell \mid p^2 + 4s(tk)^2$ is odd and prime to stk , and $(p(2tk)^{-1})^2 \equiv -s \pmod{\ell}$. Hence $\chi_s(\ell) = 1$, and every divisor D of the norm has $\chi_s(D) = 1$. Since the conductor divides $4s \mid 4ck$, a target divisor would instead satisfy $\chi_s(D) = \chi_s(-p) = -\chi_s(p)$, proving (48.2). The primitive quadratic character is constant on the kernel of reduction modulo 24 exactly when its conductor divides 24. The negative fundamental discriminants with that property are $-4, -8, -3, -24$, which give precisely (48.3). Otherwise the restriction is a nonprincipal quadratic character,

so half of the compatible reduced classes have value one; Dirichlet's theorem gives the density assertion. $\square$

Thus all eight slices $ck \leq 4$ and the 80 slices $ck \leq 30$ with $s \in \{1, 2, 3, 6\}$ vanish on every hard prime. For each of the other 31 slices, Theorem 12.1 still forces a positive-density vanishing subfamily, but it is only a sufficient source of zeros.

The remaining slices can conspire beyond fixed congruence data: $193 \equiv 1033 \pmod{840}$, both the $(5, 1)$ and $(7, 1)$ slices vanish at 193, but the $(5, 1)$ slice is positive at 1033. Conversely, every prime $p \equiv 97 \pmod{120}$ has a positive $(5, 1)$ slice via $D = 3$; this is the already known multiplier progression, not a new supply.

An exact finite census over the 385 primes $p < 30000$, $p \equiv 1 \pmod{24}$, gives

C	4	5	10	15	20	25	30
$\#\{(c, k) : ck \leq C\}$	8	10	27	45	66	87	111
$\#\{p : \text{all slices vanish}\}$	385	220	161	63	41	30	15.

This is a finite exact factorization census, not an asymptotic theorem. The 15 primes for which all 111 slices vanish include 2521; its known twelve ordered Type-I rows all lie outside $ck \leq 30$.

12.2. Fixed divisors, residue-one escape, and depth. The positive progression just displayed is one instance of an exact law.

Theorem 12.2 (fixed-divisor progression law; source Theorem 48.4). *Fix (c, k) , put $h = 4ck$, and fix a positive integer d prime to h . Let $L_d = \text{lcm}(24, h, d)$. A reduced class $r \pmod{L_d}$ is a hard-prime class on which this fixed d guarantees $M_{c,k}(p) > 0$ if and only if*

$$r \equiv 1 \pmod{24}, \quad r \equiv -d \pmod{h}, \quad r^2 \equiv -4ck^2 \pmod{d}. \quad (48.16)$$

Every prime in such a class for which $(c, k) \in \mathcal{B}_p$ has

$$e = \frac{p^2 + 4ck^2}{d}, \quad a = \frac{p+d}{h}, \quad b = \frac{p+e}{h}, \quad p(a+b) = k(4abc - 1). \quad (48.17)$$

Within the proof shape “one specified divisor has the target grade,” these conditions are also necessary; classes modulo multiples of L_d are exactly refinements of them.

Proof. The last congruence in (48.16) is $d \mid p^2 + 4ck^2$, and the middle one is $d \equiv -p \pmod{h}$. Conversely those two facts give a target divisor in (44.1). Since the norm is $p^2 \pmod{h}$, its cofactor e has the same target grade, so (48.17) is integral; direct expansion proves its final identity. The first congruence is exactly the hard-prime restriction. $\square$

Theorem 12.3 (residue-one escape; source Theorem 48.5). *No class in (48.16) contains $1 \pmod{L_d}$. Consequently, for every Q , the union of all fixed-divisor slice guarantees with $L_d \leq Q$ misses the infinite prime class*

$$p \equiv 1 \pmod{\Lambda_Q}, \quad \Lambda_Q = \text{lcm}\{24, L_d : L_d \leq Q \text{ occurs above}\}. \quad (48.20)$$

Proof. If residue one were guaranteed, then $d \equiv -1 \pmod{h}$ and $d \mid 1 + 4ck^2 = 1 + hk$. The cofactor $e = (1 + hk)/d$ would also be $-1 \pmod{h}$, so $d, e \geq h - 1$, whereas

$$(h-1)^2 - (1 + hk) = h(h-k-2) > 0, \quad h-k-2 = k(4c-1) - 2 \geq 1,$$

a contradiction. There are finitely many triples with $L_d \leq Q$, and Dirichlet's theorem supplies infinitely many primes in (48.20). $\square$

This is the slice analogue of the residue-one compactness wall for the paper's fixed identity classes. It does not control a divisor chosen from the moving factorization of each prime, and “not guaranteed” does not mean “vanishing.” The later Theorem 17.3 instead quantifies the related residue-one *genus* construction of Theorem 13.5; it proves actual slice vanishing through a logarithmic box, not a consequence of this fixed-divisor non-guarantee.

For scale only, let $ck_{\min}(p)$ be the first positive admissible slice whose squarefree c -core is not in $\{1, 2, 3, 6\}$ and put $D(p) = ck_{\min}(p) - 1$. The exact streamed census finds $ck_{\min}(p) \leq 77$ for all 385 hard primes below 30000, with maximum $D = 76$ at $p = 12289$; the 15 full- $ck \leq 30$ conspiracies have first positive product between 31 and 77. The optional census finds $ck_{\min}(p) \leq 103$ for all 1181 hard primes below 10^5 , with maximum $D = 102$ at $p = 92401$. These are finite computations, not a growth law or pointwise progress. Proving $ck_{\min}(p) < \infty$ for every hard prime would already prove the Type-I form of Erdős–Straus and hence the conjecture; it remains open.

13. CONDITIONAL POINTWISE ROUTES THROUGH THE SLICE FRAME

Scope and verdict. This section isolates a one-prime-factor event sufficient for a Type-I solution and states the corresponding conditional theorem. Its hypothesis is deliberately labelled BESPOKE AND UNPROVED: none of Bateman–Horn, Linnik, GRH for Dirichlet, Hecke, or ray-class L -functions, Elliott–Halberstam, or Duke-type equidistribution has this pointwise factorization conclusion in its standard form. The audit makes the quantifier failure exact. Two proved walls sharpen the earlier slice analysis: prime values of the norm have the wrong grade, and any bounded collection of squarefree cores is simultaneously genus-forced on infinitely many hard primes. This section's deliverable is a rigorous conditional reduction and a map of its missing pointwise input. The later stacked sieve, Theorem 16.4, proves an almost-all form but still does not settle that pointwise quantifier or claim that a standard conjecture secretly settles Erdős–Straus.

13.1. The good prime-factor class and exact reduction. Fix a hard prime p , an admissible slice $(c, k) \in \mathcal{B}_p$, and write

$$c = st^2, \quad s = \text{sf}(c), \quad h = 4ck, \quad N = N_{c,k} = p^2 + 4ck^2. \quad (50.1)$$

Inflate the primitive genus character χ_s to $(\mathbb{Z}/h\mathbb{Z})^\times$; its conductor divides $4s \mid h$. Put

$$K_s(h) = \ker \chi_s, \quad \mathcal{G}_p(c, k) = \{-p \pmod{h}\} \cap K_s(h). \quad (50.2)$$

Every prime $q \mid N$ lies in $K_s(h)$, while

$$\mathcal{G}_p(c, k) = \begin{cases} \{-p \pmod{h}\}, & \chi_s(p) = -1, \\ \emptyset, & \chi_s(p) = 1, \end{cases} \quad (50.3)$$

because $\chi_s(-1) = -1$. Thus the moving-genus theorem is exactly the empty-good-set case. On an unforced slice there is still only one good ray grade, not half the factor grades; a split prime is the wrong kind of divisor unless it is also $-p \pmod{h}$.

Theorem 13.1 (one-prime-factor slice criterion; source Theorem 50.1). *Let p be an odd prime and $(c, k) \in \mathcal{B}_p$. If a prime q satisfies*

$$q \mid p^2 + 4ck^2, \quad q \pmod{4ck} \in \mathcal{G}_p(c, k), \quad (50.4)$$

then the (c, k) -slice is positive and gives a Type-I Erdős–Straus representation. Explicitly, with

$$e = \frac{p^2 + 4ck^2}{q}, \quad a = \frac{p+q}{4ck}, \quad b = \frac{p+e}{4ck}, \quad (50.5)$$

one has $a, b \in \mathbb{Z}_{>0}$ and

$$p(a+b) = k(4abc - 1), \quad \frac{4}{p} = \frac{1}{ack} + \frac{1}{bck} + \frac{1}{pabc}. \quad (50.6)$$

Primality of q does not imply $(a, b) = 1$, and no primitivity is needed. For example, $q = 11$ at $(p, c, k) = (241, 7, 3)$ gives $(a, b) = (3, 66)$.

Proof. Admissibility gives $(p, ck) = 1$, hence $(N, h) = 1$ and $(q, h) = 1$. The target grade is $q \equiv -p \pmod{h}$, while $N \equiv p^2 \pmod{h}$; therefore $e = Nq^{-1} \equiv -p \pmod{h}$. This proves the integrality and positivity in (50.5). Substitution of $q = ha - p$, $e = hb - p$, and $qe = N$, followed by cancellation of p^2 , gives $4ck^2 = h^2ab - hp(a+b)$. Division by $h = 4ck$ gives the first identity in (50.6), and the unit-fraction identity follows over the common denominator $pabc$. Reduction of the first identity modulo p gives $4abc \equiv 1 \pmod{p}$, so exactly the denominator $pabc$ is divisible by p . This is Type I. $\square$

The raw/primitive boundary is intentional: the selected prime is a raw target divisor, and a nonprimitive row still proves the displayed identity. The canonical decomposition can recover a primitive Type-I tuple if wanted.

Hypothesis 13.2 (slice prime factor $H_{\text{SPF}}(A)$; bespoke and unproved). *Fix $A > 0$. There is a constant $p_0 = p_0(A)$ such that every prime $p > p_0$, $p \equiv 1 \pmod{24}$, admits positive integers c, k and a prime q with*

$$\boxed{ck \leq (\log p)^A, \quad q \mid p^2 + 4ck^2, \quad q \equiv -p \pmod{4ck}.} \quad (50.7)$$

This is a quantitative sufficient hypothesis, not the logically weakest one-prime-factor statement: deleting the log-power bound is weaker. It puts no bound on q , prescribes no slice, imposes no primitivity, and needs no extra genus condition, which is automatic from (50.3). It is stronger than slice positivity, since a product of wrong-grade prime factors can hit the target even when no one prime does. Theorem 17.3 proves that $H_{\text{SPF}}(A)$ is false for every $0 < A < 1$; exactly the exponents $A \geq 1$ remain open.

Theorem 13.3 (conditional on $H_{\text{SPF}}(A)$; $H_{\text{SPF}}(A)$ unproved; source Theorem 50.2). *If $H_{\text{SPF}}(A)$ holds for some A , then every sufficiently large hard prime has a Type-I Erdős–Straus representation.*

Proof. For sufficiently large p , (50.7) gives $ck < p$, $3k \leq 2p$, and $4ck \leq 2p + k$. Hence $(p, ck) = 1$ and $(c, k) \in \mathcal{B}_p$. The congruence in (50.7) makes q the unique good class in (50.2), so Theorem 13.1 applies. $\square$

If q is fixed first, (50.4) is exactly the pair of root progressions

$$p \equiv -q \pmod{4ck}, \quad p \equiv \pm 2k\sqrt{-c} \pmod{q}, \quad p \equiv 1 \pmod{24}. \quad (50.8)$$

The fixed-divisor law preassigns q and obtains a fixed CRT class; (50.7) asks the actual factorization attached to each p to supply some q after p is known. The residue-one escape walls bounded collections of the former, not the latter.

13.2. Prime values, bounded cores, and least nonresidues.

Lemma 13.4 (prime norms are wrong-grade; source Lemma 50.3). *For a hard prime p and any admissible (c, k) , if $N_{c,k}$ is prime then the slice vanishes. More generally, a good prime factor in (50.4) is $3 \pmod{4}$, its complementary divisor is also $3 \pmod{4}$, and hence $N_{c,k}$ must be composite.*

Proof. The target grade is $-p \equiv 3 \pmod{4}$, whereas $N_{c,k} \equiv 1 \pmod{4}$. If N is prime, both divisors $1, N$ have grade $1 \pmod{4}$. Under (50.4), both q and N/q have the target grade by Theorem 13.1. $\square$

Assessment 50.4 (Bateman–Horn). Standard Bateman–Horn gives an asymptotic in the variable for a fixed finite list of irreducible integer polynomials. It neither controls every external coefficient p uniformly down to a $(\log p)^A$ interval nor prescribes a prime factor of a composite value. Applied literally to $p^2 + 4ck^2$ as a prime-value polynomial, success triggers Lemma 13.4, not (50.7). A uniform assertion that every p has a logarithmic-box value with a factor in the moving grade $-p \pmod{4ck}$ would imply Theorem 13.3, but merely restates H_{SPF} ; it is not standard Bateman–Horn. The same quantifier defect applies to Hardy–Littlewood prime tuples. Standard smooth-value conjectures control factor sizes, not the bounded exponent box of grades in $(\mathbb{Z}/h\mathbb{Z})^\times$.

Theorem 13.5 (bounded-core genus escape; source Theorem 50.5). *For every fixed B , infinitely many hard primes satisfy*

$$\chi_s(p) = 1 \quad \text{for every squarefree } s \leq B. \quad (50.9)$$

For each such prime, every admissible slice whose squarefree c -core is at most B vanishes, for all k . Consequently no absolute bound for a least active core, and no proof using only boundedly many cores with arbitrary k , can hold.

Proof. Let $R_B = 24 \prod_{\substack{\ell \leq B \\ \ell \text{ prime}}} \ell$. Dirichlet’s theorem gives infinitely many primes $p \equiv 1 \pmod{R_B}$. For every odd prime $\ell \leq B$, quadratic reciprocity and $p \equiv 1 \pmod{4}$ give $(\ell/p) = (p/\ell) = 1$; the factor 24 gives $(2/p) = (-1/p) = 1$. Multiplicativity proves (50.9), and Theorem 12.1 makes every asserted slice zero. $\square$

Theorem 13.6 (GRH gives an active slice only; source Theorem 50.6). *Conditional on the unproved Generalized Riemann Hypothesis, every sufficiently large hard prime p has a squarefree $s \ll (\log p)^2$ with $\chi_s(p) = -1$; hence $(c, k) = (s, 1)$ is an admissible unforced slice in a logarithmic box. This does not assert that the slice is positive.*

Proof. The standard Ankeny consequence of GRH gives an integer $n \ll (\log p)^2$ with $(n/p) = -1$. Its squarefree kernel s has the same Legendre symbol and is no larger. Since $p \equiv 1 \pmod{8}$, $\chi_s(p) = (-s/p) = (s/p) = -1$. Admissibility is automatic for large p ; the residual exponent-box condition in (44.2) remains untouched. $\square$

Thus least-nonresidue technology supplies permission for a slice, not the divisor in it. Linnik similarly finds a small prime in $q \equiv -p \pmod{h}$ but gives no reason for that prime to divide the fixed integer $N_{c,k}$. Asking for both properties is a bespoke form of (50.7), not an existing Linnik hypothesis.

13.3. The GRH/Chebotarev compositum wall. With the notation (50.1), put

$$K = \mathbb{Q}(\sqrt{-s}), \quad \alpha = p + 2tk\sqrt{-s}, \quad N_{K/\mathbb{Q}}(\alpha) = N_{c,k}. \quad (50.10)$$

For a prime $q \nmid 2ck$,

$$q \mid N_{c,k} \iff \text{some } \mathfrak{q} \mid q \text{ in } K \text{ divides } (\alpha) \iff p \equiv \pm 2k\sqrt{-c} \pmod{q}. \quad (50.11)$$

In particular q splits in K , but the converse is much weaker: (50.11) selects a prime-ideal divisor of the one specified principal ideal (α) . Ring-class Chebotarev controls primes represented by form classes, not the factorization of a fixed represented integer. Combining (50.11) with the ray condition gives (50.8), a condition on p modulo q when q is fixed, not a Frobenius condition on p in one fixed field as q varies. Effective Chebotarev under GRH does not turn the infinite disjunction over actual divisors of (α) into a pointwise factorization theorem.

A quantitative check exposes the scale. Restrict $ck \leq L$ and fixed candidate primes $q \leq Z$. Before deduplication, (50.8) supplies at most two classes modulo $4ckq$, so

$$\begin{aligned} \mu_{\text{raw}}(L, Z) &\leq \frac{1}{2} \sum_{ck \leq L} \frac{1}{ck} \sum_{\substack{q \leq Z \\ q \equiv 3 \pmod{4} \\ q \text{ prime}}} \frac{1}{q} \\ &= \left(\frac{1}{8} + o(1) \right) (\log L)^2 \log \log Z. \end{aligned} \quad (50.12)$$

This is a class-mass envelope, not a coverage theorem. Although there are $L \log L + O(L)$ slices in the box, all k over one core share one genus bit.

Assessment 50.7 (the compositum wall). A literal joint-Chebotarev implementation introduces cyclotomic conductors containing $\text{lcm}(1, \dots, 4L)$ and candidate primes through Z ; its naive compositum degree is exponential on the $L + Z$ scale. The same GRH-effective bookkeeping as the earlier Chebotarev audit limits a direct compositum to $L + Z = O(\log N)$ when $p \leq N$ is counted. At $L, Z \asymp \log N$, even an ideal independent-events model applied to (50.12) predicts only

$$\exp\{-O((\log \log N)^2 \log \log \log N)\} \quad (50.13)$$

for the survivor proportion. Allowing an average sieve to take $Z = N^\theta$ changes the model exponent only to $O((\log \log N)^3)$. Both leave $N^{1-o(1)}$ possible survivors and are much weaker in shape than the CLAIMED/PROVISIONAL bound $N \exp\{-c(\log N)^{3/4}\}$. GRH supplies no independence theorem for the overlapping root classes, so (50.13) is an audit ceiling, not a conditional exceptional-set theorem. There is no GRH exceptional-set improvement to record here. Elliott–Halberstam and Kloosterman-refined levels of distribution change average ranges but retain an exceptional set and do not imply (50.7) in standard form.

13.4. The exact genus pairing and the Duke wall. For a product cutoff L , the exact active-box mass is

$$T_L^-(p) = \sum_{\substack{ck \leq L, (c,k) \in \mathcal{B}_p \\ \chi_{\text{sfc}(c)}(p) = -1}} \frac{1}{\varphi(4ck)} \sum_{\chi \pmod{4ck}} \overline{\chi(-p)} \prod_{\ell^e \parallel p^2 + 4ck^2} (1 + \chi(\ell) + \dots + \chi(\ell)^e). \quad (50.14)$$

Inflate χ_s to modulus h . Every factor $\ell \mid N$ has $\chi_s(\ell) = 1$, while an active slice has $\chi_s(-p) = 1$. Therefore

$$\text{the terms indexed by } \chi \text{ and } \chi\chi_s \text{ are equal.} \quad (50.15)$$

For a genus-forced slice they are opposites and cancel. Restricting to active slices removes that one cancellation but leaves every character of the residual quotient of $K_s(h)$; it does not make the principal character dominant. The principal/genus pair contributes

$$P_L^-(p) = \sum_{\substack{ck \leq L, (c,k) \in \mathcal{B}_p \\ \chi_{\text{sf}(c)}(p) = -1}} \frac{2\tau(p^2 + 4ck^2)}{\varphi(4ck)} \leq p^{o(1)} (\log L)^2 \log \log(3L) \quad (50.16)$$

when $L = p^{o(1)}$. It is positive when the active box is nonempty, but the remaining ray characters can cancel it exactly. At $p = 2521$, for example, active cores 11, 17, 19, 22, 23, 29 occur in $ck \leq 30$, yet every unforced slice in that box vanishes.

Assessment 50.8 (Duke/class-group GRH). Summing (50.14) does not create a fixed theta series: the discriminant, ray modulus, norm, and active-core selection all move. The Hurwitz–Kronecker identity restores class numbers only after removing the ray projector and changing to a weighted all-divisor mass. Equation (50.15) removes exactly the genus character and no other part of that projector. GRH for class-group or ray-class L -functions controls prime or ideal sums, not the target coefficient of the finite Euler product of one norm. A Duke-type theorem would therefore need a new uniform statement for the whole moving family (50.14), with error $o(P_L^-)$; this is a genuinely new hypothesis, not a recognized consequence of GRH for the relevant L -functions.

13.5. Conditional frontier and census. The audit is summarized by

input	standard output	missing
Bateman–Horn	prime polynomial values on average in the variable	a composite value with a g
Linnik	a prime $q \equiv -p \pmod{h}$	$q \mid$
GRH/Chebotarev	split/ray primes in fixed fields	$\mathfrak{q} \mid (\alpha)$ for
least nonresidue under GRH	$\chi_s(p) = -1$	the residual ex
EH/GEH/Kloosterman	average distribution in p	an empty ex
Duke/class-group GRH	unprojected or averaged form mass	the moving

(50.17)

The proved obstructions are Lemma 13.4, Theorem 13.5, the fixed-divisor escape Theorem 12.3, and the exact pairing (50.15). The claims about named hypotheses are assessments of their standard conclusions, not logical independence theorems. Within the pointwise frontier (50.7), Theorem 17.3 refutes every $0 < A < 1$ and leaves exactly $A \geq 1$ open, while Corollary 16.5 proves that the failure set is triple-logarithmically small. Replacing “prime factor” by “divisor” returns the open unrestricted depth criterion.

For finite comparison define

$$ck_{\text{pr}}(p) = \min\{ck : (c, k) \in \mathcal{B}_p, \exists \text{ prime } q \mid p^2 + 4ck^2, q \equiv -p \pmod{4ck}\}. \quad (50.18)$$

Computational 50.9 (exact stated range; informational). A streamed exact computation finds $ck_{\text{pr}}(p)$ for all 385 hard primes below 30000, checks the complete

exponent box at every tested norm, and reconstructs (50.5)–(50.6) for each good-prime event. Its distribution is

ck_{pr}	5	7	10	11	13	14	17	19	21	22	23
#	156	30	36	35	13	15	16	4	15	9	6
ck_{pr}	26	28	29	31	33	34	35	37	38	39	42
#	10	4	3	1	2	3	1	1	4	2	5
ck_{pr}	43	46	55	62	66	67	69	70	77	78	
#	1	1	2	1	4	1	1	1	1	1	

(50.19)

and the strict records are

p	73	193	241	1201	2521	4729	7489	9601
ck_{pr}	7	10	21	34	38	66	70	78.

(50.20)

The prime-factor minimum equals the unrestricted $ck_{\min}$ for 311 primes and is larger for 74. The largest gap is 55 at $p = 23689$: $ck_{\min} = 11$ but $ck_{\text{pr}} = 66$, witnessed by $(c, k, q) = (33, 2, 77951)$. The largest ck_{pr} is 78 at $p = 9601$. The optional memory-bounded scan over the 1181 hard primes below 10^5 has maximum 282 at $p = 83689$. These observations support no pointwise growth law; the later almost-all theorem does not use them. The verification factors one norm at a time and stores at most one residue set of size $4ck$.

14. WITNESS-MODULUS TAILS AND THE EFFECTIVITY REPAIR

For a prime, or any positive integer, p , define

$$W(p) = W_{\Pi}(p) = \min k\ell, \quad (51.1)$$

where the minimum is over positive k, ℓ, u, v, w satisfying

$$k\ell \equiv 3 \pmod{4}, \quad \frac{k\ell + 1}{4} = uvw, \quad p \equiv -uv^{-1} \pmod{k\ell}. \quad (51.2)$$

Put $W(p) = +\infty$ if there is no such tuple; no general finiteness assertion is made. Lemma 3.4 says exactly that $W(p) \leq T$ supplies an explicit representation with multiplier modulus at most T . The definition allows composite ℓ , although the proof uses a prime subfamily.

14.1. Parameterized truncation and the master tail. Write $t = \log X$, let $K \geq K_0$, and suppose that the canonical dyadic boxes (39.2)–(39.3) furnish, uniformly in every $\mathcal{J} \subseteq \{k \leq K : k \equiv 1 \pmod{4}\}$ containing 1 and every reduced compatible fibre c ,

$$\sum_{X^{1/2} < \ell \leq X} \frac{f_{c;\mathcal{J}}(\ell)}{\ell} \asymp t^2 h(\mathcal{J}), \quad h(\mathcal{J}) = \sum_{k \in \mathcal{J}} \frac{\varphi(k)}{k^2}. \quad (51.3)$$

Put

$$\mu = t^2 \log K. \quad (51.4)$$

Lemma 14.1 (truncated c -free assembly; proved internally, with the provisional review status of the fixed- c -free chain). *Assume (51.3), $KX \leq T$, $\log K \ll t$, and*

$$\log N \geq C\mu t. \quad (51.5)$$

Then, with effective constants whenever the supply constants in (51.3) are effective,

$$\#\{p \leq N : p \text{ prime}, W(p) > T\} \ll N \exp\{-c\mu\}. \quad (51.6)$$

Proof. Use the atoms (39.2)–(39.4), all of modulus at most $KX \leq T$. Lemma 10.1's geometry needs only $u, v \leq X^{1/6}$ and $4uv > 4H^2 > K$. The profile and exact CRT/Euler induction replay as

$$W_{k,a}(g) \ll \frac{t^2}{\varphi(g)} \frac{\varphi(k)^2}{k^3}, \quad \mathbb{E}(H_X)_m \leq (C\mu)^m. \quad (51.7\text{--}51.8)$$

for every needed order. With $y = B\mu$, the bad fibres cost $\exp\{-\eta B\mu + O(1)\}$, while the others have $h(\mathcal{J}_c) \gg \log K$; (51.3) therefore gives $\Pr(H_X = 0 \mid S_y = 1) \leq e^{-c\mu}$. The fixed-polylog lower proof of Lemma 5.2 and the cubic proof of Theorem 9.4 both supply exactly these canonical dyadic boxes.

Take the least even $r \geq D_B\mu$. The same Bonferroni factorial tail has CRT mean $O(e^{-c\mu})$ and ledger

$$\deg = O(\mu), \quad \log d_{\text{term}} = O(\mu t), \quad \log \sum |c_{\text{term}}| = O(\mu t). \quad (51.11)$$

Since $y < X^{1/2}$ and the atom count has logarithm $O(t)$, (51.5) absorbs rounding and the primes at most $\max(K, y)$. $\square$

Theorem 14.2 (master witness-modulus tail; proved internally with the exact status below). *There are effectively computable positive constants such that the following hold. Every set counted with variable p in this theorem and Corollary 14.3 is restricted to primes.*

1. **Fixed-polylogarithmic supply, without Theorem 9.4.** Uniformly when

$$3 \leq T, \quad \log N \geq C\{1 + (\log T)^3 \log(2 + \log T)\}, \quad (51.12)$$

one has

$$\#\{p \leq N : W(p) > T\} \ll N \exp\{-c(\log T)^2 \log(2 + \log T)\}. \quad (51.13)$$

This is unconditional in the no-unproved-hypothesis sense and does not inherit Theorem 9.4; it does inherit the campaign's provisional-review qualification on the Section 10 moment and Bonferroni machinery.

2. **Cubic supply.** Fix $0 < \kappa < 1/240$. Uniformly when

$$3 \leq T, \quad \log N \geq C_\kappa\{1 + (\log T)^4\}, \quad (51.14)$$

one has

$$\#\{p \leq N : W(p) > T\} \ll_\kappa N \exp\{-c_\kappa(\log T)^3\}. \quad (51.15)$$

This statement inherits Theorem 9.4 and is CLAIMED/PROVISIONAL.

Both estimates are uniform throughout the simpler common range

$$3 \leq T \leq \exp\{c(\log N)^{1/4}\}. \quad (51.16)$$

Proof. For (1), set $X = T^{1/2}$ and $K = \lfloor (\log X)^5 \rfloor$, with fixed adjustments below the asymptotic threshold. Then $KX \leq T$, $\log K \asymp \log t$, and the canonical-box proof of (21) supplies (51.3). Thus $\mu \asymp t^2 \log t$ and (51.5) is (51.12). For (2), use the same X and $K = \lfloor X^\kappa \rfloor$. Theorem 9.4 supplies (51.3), $\mu \asymp t^3$, and (51.5) becomes (51.14). Bounded T is absorbed by enlarging the implied constants. $\square$

At $T = \exp\{\alpha(\log N)^{1/4}\}$, (51.15) is $N \exp\{-c(\log N)^{3/4}\}$, exactly the prime part of (39.35) up to constants. This is a consistency check, not a new record.

Corollary 14.3 (polylogarithmic tails; same status as Theorem 14.2). *For each fixed $A > 0$,*

$$\#\{p \leq N : W(p) > (\log N)^A\} \ll_A N \exp\{-c_A(\log \log N)^2 \log \log \log N\}, \quad (51.17)$$

unconditionally and without Theorem 9.4. The cubic supply gives the additional CLAIMED/PROVISIONAL estimate

$$\#\{p \leq N : W(p) > (\log N)^A\} \ll_A N \exp\{-cA^3(\log \log N)^3\}. \quad (51.18)$$

The $(\log \log N)^3$ exponent shape at polylogarithmic depth has the near-simultaneous independent antecedent Dahan [3, Theorem 4.17] for a different two-parameter (u, a) search-depth statistic without a genus gate; his printed $o(1)$ is explicitly ineffective. Equations (51.17)–(51.18) instead bound the minimal congruence modulus W , and Lemmas 14.4–14.5 give effective constants subject to the labels displayed here. This priority distinction is part of the statement, not an assertion that the statistics are interchangeable.

The formulas are interpreted beyond a fixed large threshold. The same bounds hold with cutoff $(\log p)^A$: split at $N^{1/2}$, use comparable logarithms on the upper half, and absorb the lower half. Define

$$H_{\text{MOD}}(A) : \quad W(p) \leq (\log p)^A \quad \text{for every sufficiently large prime } p. \quad (51.19)$$

Theorem 17.1 proves that this hypothesis is false for every $0 < A < 1$; exactly the exponents $A \geq 1$ remain open. For any A for which it holds, this unproved pointwise hypothesis implies eventual prime solvability by Lemma 3.4; the upper tail does not make its failure set empty. It is not equivalent to $H_{\text{SPF}}(A)$: the former is the Type-II multiplier statistic (51.1), while the latter is the Type-I factor-in-one-specified-grade assertion (50.7). Neither hypothesis implies the other as stated, although each implies an eventual representation. Thus (39.35) and H_{MOD} are respectively a tail and a pointwise question for the same W ; H_{SPF} is a different frontier.

14.2. Effective class supply away from one conductor. The lower half of the fixed-polylogarithmic supply uses Bombieri–Vinogradov, not merely the elementary box calculation. Large composite progression moduli do not remove a possible Siegel zero, because their characters may be induced from a small primitive conductor. The needed effective form is the following.

Lemma 14.4 (effective Bombieri–Vinogradov away from one conductor; standard cited input with dependency proof). *For every $A > 0$ there are effectively computable B, C, x_0 such that, for $x \geq x_0$ and $Q \leq x^{1/2}/(\log x)^B$, there is either no exceptional conductor or one real primitive conductor $r_x \geq 3$ for which*

$$\sum_{\substack{q \leq Q \\ (a,q)=1 \\ r_x \nmid q}} \max \left| \psi(x; q, a) - \frac{x}{\varphi(q)} \right| \leq \frac{Cx}{(\log x)^A}. \quad (51.20)$$

The same statement holds for dyadic π -differences by effective partial summation.

Proof. Lenstra–Pomerance [11, Lemma 11.2, pp. 1258–1260] give computable absolute constants, one excluded $s(x) \in [(\log x)^{1/2}, e^{\sqrt{\log x}}]$, maxima over $2 \leq y \leq x$ and reduced residues, and error $Cx^{1/2}Q(\log x)^5 + Cxe^{-c\sqrt{\log x}}$ over $s(x) \nmid q$. Restricting to $Q \leq x^{1/2}(\log x)^{-B}$, and enlarging smaller Q to $x^{1/3} \log x$, proves (51.20); the maximum in y fixes one conductor through a dyadic interval.

For completeness, set $R = (\log x)^{B_0}$. Away from Page, the effective explicit formula gives $|\psi(y, \chi^*)| \ll ye^{-c\sqrt{\log x}}$ uniformly on $[x, 2x]$, while $\sum_{q \leq Q, r|q} 1/\varphi(q) \ll \log(2Q)/\varphi(r)$. At most $\varphi(r)$ primitive characters have conductor r , so all $r \leq R$ cost $O(R \log(2Q))$; the exponential saving absorbs this. Vaughan's identity and the large sieve handle larger conductors. Fixed conductors, including 4, are checked effectively, and selecting Page at $2x$ prevents two endpoint deletions. $\square$

Lemma 14.5 (exceptional-conductor deletion tolerance; proved). *In the boxes used for (21), let $r > 1$ be a primitive real character conductor. Apart from the explicitly harmless conductor 4, there is a prime $p \mid r$ (or $p = 2$ when the required extra factor is the third power of 2) such that*

$$p \nmid uv \implies r \nmid 4uv. \quad (51.21)$$

Uniformly in c, k , the pairs satisfying (51.21) retain a fixed positive proportion of the harmonic box mass. The low- ω cutoff may be chosen so that it and (51.21) retain a fixed positive proportion simultaneously. The same remains true after the low-congestion pruning of Theorem 9.4.

Proof. A primitive real conductor is the absolute value of a fundamental discriminant. Choose an odd prime factor, or $p = 2$ for conductor 8; conductor 4 is handled effectively directly. Fix k . If $p \mid k$, the condition $(uv, k) = 1$ already excludes p . If $p \nmid k$ and $p \leq Ck$, the box count with the extra reduced conditions modulo p retains the local proportion

$$\frac{(1 - 1/p)^2}{1 - 1/p^2} = \frac{p-1}{p+1} \geq \frac{1}{3}, \quad (51.22)$$

and $H = K^{10}$ absorbs the boundary for combined modulus at most Ck^2 . If $p > Ck$, direct counting of the discarded pairs with $p \mid u$ or $p \mid v$ costs $O(1/(pk))$ against main term $\asymp \varphi(k)/k^2$, a ratio $O(k/(p\varphi(k))) = O(1/C)$; endpoint errors are absorbed as before. Choose C large. Increasing the fixed Rankin cutoff makes the high- ω tail smaller than any prescribed fixed fraction, and the high-incidence triples have $o(1)$ of the original mass by Lemma 9.3. Both deletions therefore leave the same order of mass. $\square$

Theorem 14.6 (effectivity of the repaired chains; proved internally, record-adjacent and therefore CLAIMED/PROVISIONAL). *The constants in the source class-mass conclusions (16.8) and (34.18)–(34.19), in source Theorems 16.4 and 39.7 (paper Theorems 9.1 and 10.7), and in Theorem 14.2 may all be taken effectively computable. The record statement retains every correctness and priority qualification already attached to Theorem 10.7.*

Proof. In each dyadic prime block apply Lemma 14.4; if its exceptional conductor exists, retain only triples with $r_x \nmid 4uv$. Lemma 14.5 preserves the full order of box mass, uniformly in every subfamily and fibre. This retained subfamily proves the original lower class-mass conclusion. All downstream choices in the larger sieve, exact CRT moments, Chernoff estimate, Bonferroni polynomial, integer class counting, and Rankin semigroup transfer are finite and effective. $\square$

The perimeter is deliberately narrow. The first sentence of Theorem 9.4 also contains the source Theorem 34.8 ancillary $o(1)$ bound for the absolute progression-error sum over *all* low-congestion triples. Excluded-conductor deletion effectivizes

the class supply (34.18)–(34.19), not that all-triples $o(1)$ as printed. The exceptional-set chain uses the class supply only. Thus Theorem 14.6 is not an effectivity upgrade of every clause of Theorem 9.4.

14.3. Pomerance–Weingartner, priority, and blind adjudication. Pomerance–Weingartner [13, §4] prove for their $k = 1$ forced classes

$$\sum_{\ell \leq X} \frac{f(\ell)}{\ell} \asymp (\log X)^2. \quad (51.24)$$

Stopping their larger-sieve argument at prime modulus T gives

$$\#\{p \leq N : W_{\text{PW}}(p) > T\} \ll N \exp\{-c(\log T)^2\}, \quad \log N \geq C(\log T)^3, \quad (51.25)$$

where W_{PW} measures that subfamily. It is also a weaker tail for W . Their lower class mass explicitly uses Bombieri–Vinogradov after a polylogarithmic multiplicity bound, their upper mass uses Brun–Titchmarsh, and their N -side is the elementary larger sieve and Rankin truncation. Thus PW already contains the square-log truncated tail; the campaign-derived one-conductor deletion makes the fixed- $m = 4$ supply effective, but PW do not print that repair, and no uniform varying- m effectivity is claimed when the exceptional conductor divides m . The resulting tail exponents are $(\log T)^2$ for the proved PW $k = 1$ truncation, $(\log T)^2 \log \log T$ for the unconditional fixed-polylog multipliers with the Section 10 review qualification, and $(\log T)^3$ for the CLAIMED/PROVISIONAL power multipliers. Vaughan’s 1970 paper remains inaccessible to this campaign; this is a comparison with PW’s reconstruction, not a direct audit of Vaughan. Searches found no paper explicitly claiming an effective Vaughan-strength constant or a Siegel-free proof, but that is a search result, not a literature guarantee. PW Theorem 1.3 says only “There is an absolute positive constant C ,” not that it or the threshold is computable; their Section 4 says “We now use the Bombieri–Vinogradov theorem.” They do not perform the effectivity repair. Dahan’s statistic and cubic-shape priority are distinguished immediately after (51.18).

The attested-blind protocol used branch `blind51`; base `4dcf3b7e1e10e31f529f5e2862a2aa23c0820bb6`; freeze `7ecd7d514ae379e9c90d6a7b292d11e80840b360`; and artifact `blind51.md`, SHA-256 `b0a4a27dba52fb563b4667427f2200cc98e670c27e8fff390aa846b1039c8b7c`. Read isolation is self-attested. Adjudication was CONVERGED-WITH-DIVERGENCES: both tails, windows, the Section 10 replay, conductor repair, (39.35), and PW truncation converged, but the blind side overclaimed effectivity of the ancillary all-triples $o(1)$. The paper’s narrower perimeter is correct; no label changed.

15. PER-SLICE VANISHING FREQUENCY AND THE EMPTY THIRD LAYER

The moving genus obstruction is the last fixed-congruence obstruction for one fixed slice. On every reduced hard class where the genus permits positivity, an upper-bound sieve makes residual vanishing relatively sparse. This is a proved, nonprovisional per-slice almost-all result, not a pointwise bound or a stack over growing slices. Here “hard” still means prime and $1 \pmod{24}$.

15.1. The fixed-slice sieve. Fix positive c, k and put

$$c = st^2, \quad s = \text{sf}(c), \quad h = 4ck, \quad Q = \text{lcm}(24, h). \quad (52.1)$$

Theorem 15.1 (per-slice sieve upper bound; proved). *Suppose $s \notin \{1, 2, 3, 6\}$. Fix a reduced class $a \pmod{Q}$ such that*

$$a \equiv 1 \pmod{24}, \quad (a, h) = 1, \quad \chi_s(a) = -1. \quad (52.2)$$

Then

$$\#\{p \leq X : p \text{ prime}, p \equiv a \pmod{Q}, M_{c,k}(p) = 0\} \ll_{c,k} \frac{X}{(\log X)^{1+2/\varphi(h)}}. \quad (52.3)$$

The finitely many p for which $(c, k) \notin \mathcal{B}_p$ are included in the implied constant, which is uniform over the finitely many classes $a \pmod{Q}$ satisfying (52.2).

Proof. The contrapositive of Theorem 13.1, followed by the genus calculation for every good $q \nmid Q$, is

$$\begin{aligned} M_{c,k}(p) = 0 &\implies p^2 + 4ck^2 \text{ has no prime divisor } q \equiv -a \pmod{h}, \\ \chi_s(q) = \chi_s(-a) = 1, \quad \left(\frac{-4ck^2}{q}\right) &= \left(\frac{\Delta_s}{q}\right) = 1, \\ n^2 &\equiv -4ck^2 \pmod{q}. \end{aligned} \quad (52.4\text{--}52.7)$$

The last congruence has exactly two distinct nonzero roots because $4, k^2, t^2$ are nonzero squares modulo q .

Sieve $\mathcal{A} = \{n \leq X : n \equiv a \pmod{Q}\}$ by removing zero at every $\ell \nmid Q$ and the two roots at every good q . For squarefree d prime to Q , CRT and $\rho(\ell) \leq 3$ give

$$\#\mathcal{A}_d = \frac{X\rho(d)}{Qd} + O(\rho(d)), \quad \sum_{d \leq D} \mu^2(d)\rho(d) \ll D(\log D)^2. \quad (52.8\text{--}52.10)$$

The exact dimension is $\kappa = 1 + 2/\varphi(h) < 2$. With fixed sufficiently large $u = u(\kappa)$, $D = X^{1/2}$, and $z = D^{1/u} = X^\eta$, the upper fundamental lemma gives

$$S(\mathcal{A}, z) \ll_\kappa \frac{X}{Q} V(z) + \sum_{d \leq D} \mu^2(d) |r_d|. \quad (52.12)$$

This avoids asserting that the endpoint sieve parameter 2 works for every fractional dimension. The local product is

$$V(z) = \prod_{\substack{\ell \leq z \\ \ell \nmid Q}} (1 - 1/\ell) \prod_{\substack{q \leq z \\ q \equiv -a \pmod{h} \\ q \nmid Q}} \frac{1 - 3/q}{1 - 1/q}. \quad (52.13)$$

Mertens in the fixed ray says $\sum_{q \leq z, q \equiv -a \pmod{h}} q^{-1} = \varphi(h)^{-1} \log \log z + C(h, -a) + o(1)$, whence $V(z) \asymp_{c,k,a} (\log z)^{-1-2/\varphi(h)}$. The elementary remainder is $O(X^{1/2}(\log X)^2)$ and the $O(z)$ small primes are negligible, proving (52.3). $\square$

The only analytic inputs are the upper-bound fundamental lemma, ordinary Mertens, and Mertens in one fixed reduced progression. No Bombieri–Vinogradov, Siegel–Walfisz, or prime level of distribution is hidden: primality is majorized by the zero-class sieve. For explicitly fixed (c, k, a) these constants are effective in principle.

Corollary 15.2 (the congruence-level third layer is empty; proved). *Under Theorem 15.1, let M' be any fixed multiple of Q and let $a' \pmod{M'}$ be a reduced refinement of $a \pmod{Q}$. The set*

$$\{p : p \equiv a' \pmod{M'}, M_{c,k}(p) = 0\} \quad (52.16)$$

does not contain all sufficiently large primes in that progression. Consequently no prime-bearing arithmetic progression inside an unforced slice-class is an identically vanishing progression.

Proof. The fixed-progression prime number theorem gives

$$\#\{p \leq X : p \equiv a' \pmod{M'}\} \sim \frac{X}{\varphi(M') \log X}. \quad (52.17)$$

The vanishing set in the refinement is bounded by (52.3), and the extra factor $(\log X)^{-2/\varphi(h)}$ tends to zero. $\square$

Thus the universal cores 1, 2, 3, 6 and the moving genus half in Theorem 12.1 are the complete congruence-level vanishing laws for a fixed slice. The complementary half has no identically vanishing reduced subprogression. This does not exclude infinitely many residual vanishers or a positive slice guaranteed by a fixed divisor.

Proposition 15.3 (polylogarithmic uniformity with an epsilon loss; proved, ineffective). *Fix $\theta > 0$ and $0 < \epsilon < 1$. Uniformly for slices and reduced classes satisfying (52.1)–(52.2) and*

$$h \leq (\log X)^\theta, \quad (52.18)$$

one has

$$\#\{p \leq X : p \equiv a \pmod{Q}, M_{c,k}(p) = 0\} \ll_{\theta, \epsilon} \frac{X}{\varphi(Q)(\log X)^{1+2(1-\epsilon)/\varphi(h)}}. \quad (52.19)$$

The implied constant and threshold are ineffective. No uniformity for arbitrary $h = X^{o(1)}$ is asserted.

Proof. Retain the two extra root classes only for good primes in

$$y = \exp\{(\log X)^\epsilon\} < q \leq z = X^\eta. \quad (52.20)$$

For $q \geq y$, the hypothesis gives $h \leq (\log q)^{\theta/\epsilon}$, so Siegel–Walfisz with this fixed parameter and partial summation yield uniformly

$$\sum_{\substack{y < q \leq z \\ q \equiv -a \pmod{h}}} \frac{1}{q} = \frac{(1-\epsilon) \log \log X + O(1)}{\varphi(h)} + o(1). \quad (52.21)$$

Ordinary Mertens with primes dividing Q omitted contributes $\ll Q/(\varphi(Q) \log z)$. The coarse uniform dimension-three upper sieve and the elementary remainder $O(X^{1/2}(\log X)^2)$ now give (52.19). The usual all-moduli Siegel–Walfisz constants are ineffective, which is exactly the label in the statement. $\square$

15.2. Finite residual structure and its limits. The memory-bounded verifier block (ay) factors one norm at a time and checks all 31 unforced slices with $ck \leq 30$ over the 385 hard primes below 30000; the optional full scan uses all 1181 below 10^5 . The residual frequencies broadly track the fixed-slice log-power scale, but this is finite calibration only.

The 74 strict gaps $ck_{\text{pr}} > ck_{\text{min}}$ in the first range contain 98 canonical target divisors at the minimal positive product; the full range has 221 events and 308 divisors. Individual divisor shapes are 68, 24, 6 below 30000 and 199, 93, 16 below 10^5 for respectively two distinct primes, at least three prime factors, and a prime power. Thus the majority mechanism is a product of two individually wrong-grade

primes, but products of three or more factors and exponent caps are genuine. For example $D = 155 = 5 \cdot 31$ occurs at $(p, c, k) = (241, 11, 1)$, $D = 75 = 3 \cdot 5^2$ at $p = 409$ on that slice, and $D = 27 = 3^3$ at $(p, c, k) = (5953, 5, 1)$.

After conditioning on both genus signs being negative, define

$$R(A, B) = \frac{\Pr(M_A = 0, M_B = 0)}{\Pr(M_A = 0) \Pr(M_B = 0)}. \quad (52.26)$$

Most tested pairs are near one, but stable anticorrelations remain: $(5, 1)$ with $(5, 2)$ gives .529, .577 and $(5, 1)$ with $(10, 1)$ gives .714, .699 in the two ranges. Hence approximate independence is a useful first calibration for many pairs, not a blanket theorem. No asymptotic constant, simultaneous-factorization law, lower bound for residual vanishing, or pointwise conclusion follows. In particular, the sieve is upper-bound only and does not cross the prime-pair parity barrier.

16. THE STACKED SLICE SIEVE AND ALMOST-ALL H_{SPF}

Distinct slice labels can define the same norm: $(5, 2)$ and $(20, 1)$ both have $ck^2 = 20$. Their root classes then coincide whenever both ray conditions accept the prime. Quotienting duplicate radicands gives the correct fixed-box exponent. A canonical squarefree-core subfamily has no duplicates and proves, for every fixed $A < 1$, the effective headline

$$\#\mathcal{F}_A(X) \ll_A \frac{X}{\log X} \exp \left\{ -c_A \frac{(\log \log X)^{3/2}}{\sqrt{\log \log \log X}} \right\}, \quad (53.1)$$

where $\mathcal{F}_A(X)$ is the set of hard primes at which $H_{\text{SPF}}(A)$ fails. It is an almost-all theorem only. The square-root loss relative to the provisional Type-II cubic tail is caused by the Type-I genus-permission gate.

16.1. Duplicate-radical geometry and the fixed stack. For fixed $T \geq 1$, put

$$Q_T = \text{lcm}(24, \{4ck : ck \leq T\}). \quad (53.2)$$

For integral $T \geq 6$, $Q_T = 4\text{lcm}(1, \dots, T)$ and Chebyshev's bounds give $\log Q_T = \psi(T) + O(1) = O(T)$. Let $C \pmod{Q_T}$ be reduced and $C \equiv 1 \pmod{24}$. Call (c, k) with $ck \leq T$ active at C when $\chi_{\text{sf}(c)}(C) = -1$. The raw sum

$$m_{\text{raw}}(C) = \sum_{\substack{ck \leq T \\ (c, k) \text{ active at } C}} \frac{2}{\varphi(4ck)} \quad (53.3)$$

is not the local-density mass. Put $d = ck^2$ and define

$$\mathcal{H}_d(C) = \{4ck : ck \leq T, ck^2 = d, (c, k) \text{ active at } C\}. \quad (53.4)$$

For reduced $b \pmod{Q_T}$ set

$$J_C(b) = \#\{d : \exists h \in \mathcal{H}_d(C), b \equiv -C \pmod{h}\},$$

$$m_{\#}(C) = \frac{2}{\varphi(Q_T)} \sum_{b \in (\mathbb{Z}/Q_T\mathbb{Z})^\times} J_C(b). \quad (53.5)$$

A radicand is thereby counted once whenever at least one of its rays accepts the prime.

Lemma 16.1 (root overlap; proved). *Put $d_i = c_i k_i^2$. If a root class of*

$$x^2 \equiv -4d_1 \pmod{q} \quad (53.6)$$

coincides with a root class for d_2 , then

$$q \mid 4(d_1 - d_2). \quad (53.7)$$

Consequently, if $d_1 \neq d_2$, all root pairs belonging to slices $c_i k_i \leq T$ are disjoint at every odd prime $q > T^2$, and no root is zero there. If $d_1 = d_2$, the root sets coincide identically.

Proof. Square a common root and subtract. At odd q this gives $q \mid d_1 - d_2$. Since $d_i = (c_i k_i) k_i \leq T^2$, a nonzero difference has absolute value less than T^2 . A zero root similarly forces $q \mid d_i$; equality of radicands gives identical congruences. $\square$

For (5, 2) and (20, 1) the moduli are 40 and 80. On an active class the modulus-80 ray lies inside the modulus-40 ray, so the corrected contribution is $2/\varphi(40) = 1/8$, not the raw $3/16$.

Let $E_C(X, z)$ count primes $p \leq X$, $p \equiv C \pmod{Q_T}$, for which there is no prime $q \leq z$ and active (c, k) , $ck \leq T$, with

$$q \mid p^2 + 4ck^2, \quad q \equiv -C \pmod{4ck}. \quad (53.8)$$

Theorem 16.2 (fixed-box stacked sieve; proved, effective). *For fixed T there is an effective $\eta_T > 0$ such that, uniformly over the finitely many reduced hard classes $C \pmod{Q_T}$,*

$$E_C(X, X^{\eta_T}) \ll_T \frac{X}{(\log X)^{1+m_\#(C)}}. \quad (53.9)$$

The same estimate holds for primes having no good prime factor of any size in the box.

Proof. Take $T_0 > \max(Q_T, T^2)$ and sieve $\mathcal{A}_C = \{n \leq X : n \equiv C \pmod{Q_T}\}$. Remove zero at ordinary primes and accepted root pairs at $q > T_0$. The genus calculation supplies two roots, and Lemma 16.1 makes distinct radicands disjoint, so

$$\rho_C(q) = 1 + 2J_C(q \bmod Q_T), \quad \#(\mathcal{A}_C)_d = \frac{X\rho_C(d)}{Q_T d} + O_T(\rho_C(d)) \quad (53.11\text{--}53.12)$$

for squarefree d prime to Q_T , with multiplicative ρ_C . Effective fixed-class Mertens gives

$$\sum_{T_0 < q \leq z} \frac{2J_C(q \bmod Q_T)}{q} = m_\#(C) \log \log z + O_T(1), \quad V_C(z) \asymp_T (\log z)^{-1-m_\#(C)}. \quad (53.13\text{--}53.14)$$

Apply Friedlander–Iwaniec [8, Theorem 6.9] with $D = X^{1/2}$ and $z = D^{1/u_T} = X^{\eta_T}$ for sufficiently large fixed u_T :

$$S(\mathcal{A}_C, z) \ll_T \frac{X}{Q_T} V_C(z) + \sum_{d \leq D} \mu^2(d) O_T(\rho_C(d)). \quad (53.15)$$

Since $\rho_C(\ell) = O_T(1)$, the remainder and small primes are negligible. $\square$

For one active slice, $m_\# = 2/\varphi(4ck)$, so Theorem 15.1 is exactly the one-slice specialization. If a class has no active slice, $m_\# = 0$ and only the ordinary one-dimensional bound remains. Such escape classes exist for every fixed T : $C = 1 \pmod{Q_T}$ is one, and Theorem 13.5 supplies infinitely many corresponding primes.

16.2. Exact genus bits and deterministic permission mass. The full raw box has the record-only Selberg–Delange asymptotic

$$\begin{aligned}\mathfrak{M}(T) &= \sum_{ck \leq T} \frac{2}{\varphi(4ck)} = \sum_{n \leq T} \frac{2\tau(n)}{\varphi(4n)} \sim C_{\text{sl}}(\log T)^2, \\ C_{\text{sl}} &= \frac{1}{2} \prod_{p \text{ odd}} (1 - p^{-1})^2 \left(1 + \frac{p(2p-1)}{(p-1)^3}\right) > 0.\end{aligned}\tag{53.16–53.17}$$

Only the effective bound $\mathfrak{M}(T) \ll (1 + \log T)^2$ is used below. Retain

$$\mathcal{S}_T(C) = \{(s, k) : s \text{ squarefree}, sk \leq T, \chi_s(C) = -1\}, \quad m_0(C) = \sum_{(s,k) \in \mathcal{S}_T(C)} \frac{2}{\varphi(4sk)}.\tag{53.19–53.20}$$

Their radicands sk^2 are distinct. Quadratic reciprocity gives exactly

$$\chi_s(C) = \prod_{\substack{q|s \\ q \geq 5}} \left(\frac{C}{q}\right).\tag{53.21}$$

As C ranges uniformly over reduced hard classes modulo Q_T , the bits (C/q) for distinct primes $5 \leq q \leq T$ are independent uniform signs. This is a finite Chinese-remainder theorem identity, not a heuristic: the $-1, 2, 3$ factors are pinned by $C \equiv 1 \pmod{24}$, and each odd prime-power coordinate is independent.

Lemma 16.3 (permission-pair mass; proved). *Suppose $5 \leq q \leq T/2$ is prime and $\chi_q(C) = -1$. Then*

$$m_0(C) \gg \frac{\log^2(T/q)}{q}.\tag{53.22}$$

The constant is absolute. The same estimate, with another absolute constant, survives deleting every slice whose modulus $4sk$ is divisible by one prescribed primitive real-character conductor r , provided $q \nmid r$.

Proof. Pair each squarefree $u \leq T/q$, $(u, q) = 1$, with qu . Equation (53.21) makes exactly one core active. For $k \leq T/(qu)$ its weight is at least $1/(2quk)$, and Möbius summation gives

$$m_0(C) \gg \frac{1}{q} \sum_{\substack{u \leq T/q \\ u \text{ squarefree} \\ (u, q) = 1}} \frac{1}{u} \log \frac{T/q}{u} \gg \frac{\log^2(T/q)}{q}.\tag{53.24}$$

If an exceptional conductor has odd prime factor ℓ , restrict to $\ell \nmid uk$; if it is 8, restrict the relevant cores and k to odd integers. These restrictions retain fixed Euler factors and ensure $r \nmid 4sk$. Conductor 4 is effectively controlled directly. $\square$

Using only core q would provide one logarithm. Pairing u with qu provides the second regardless of every later genus bit.

16.3. The effective growing stack. Fix $0 < A < 1$ and put

$$\begin{aligned}T &= (\log X)^A, & L &= \log \log X, & K_T &= C_0(1 + \log^2 T), \\ D &= X^{1/3}, & z &= \exp \left\{ \frac{\log D}{BK_T} \right\}.\end{aligned}\tag{53.25}$$

where C_0, B are sufficiently large absolute constants. Then

$$\log z \asymp \frac{\log X}{L^2}, \quad \log \log z = L + O(\log L), \quad \log \log Q_T = AL + O(1), \quad (53.26)$$

and $Q_T < z$ for every fixed $A < 1$ once X is large.

Theorem 16.4 (growing stacked slice sieve; proved, effective). *For every fixed $0 < A < 1$ there are effectively computable $c_A > 0$ and X_A such that the following holds for $X \geq X_A$. Let $\mathcal{E}_A(X)$ be the hard primes $p \leq X$ for which there do not exist an active (c, k) with $ck \leq (\log X)^A$ and a prime $q \leq z$ satisfying $q \mid p^2 + 4ck^2$ and $q \equiv -p \pmod{4ck}$. Then*

$$\#\mathcal{E}_A(X) \ll_A \frac{X}{\log X} \exp \left\{ -c_A \frac{L^{3/2}}{\sqrt{\log L}} \right\}. \quad (53.27)$$

The proof uses only the squarefree- c subfamily, so the assertion for all slices follows a fortiori.

Proof. Partition into the $\varphi(Q_T)/\varphi(24)$ reduced hard classes modulo $Q_T = \exp\{O(T)\} = X^{o(1)}$. For ray moduli $4sk \leq 4T$, effective Landau–Page-repaired Siegel–Walfisz leaves at most one primitive real conductor $r = r_X$, $|r| \leq 4T$. If it exists retain only $r \nmid 4sk$; otherwise delete nothing. Uniformly for retained moduli and $x \geq Q_T$, the effective explicit formula is $\psi(x; h, a) = x/\varphi(h) + O(xe^{-c\sqrt{\log x}})$.

Write $m_r(C)$ for the retained mass and start root primes above Q_T . Distinct canonical radicands and partial summation give, uniformly in C ,

$$\sum_{\substack{Q_T < q \leq z \\ q \equiv -C \pmod{4sk}}} q^{-1} = \frac{\log \log z - \log \log Q_T}{\varphi(4sk)} + o(1/\varphi(4sk)).$$

The errors over all $O(T \log T)$ labels and the quadratic local terms are $o(1)$. Hence ordinary roughness and the added roots give

$$V_C(z) \ll \frac{Q_T}{\varphi(Q_T) \log z} e^{-m_r(C)(\log \log z - \log \log Q_T) + o(1)} \ll_A \frac{L^{O(1)}}{\log X} e^{-c_A m_r(C)L}. \quad (53.30)$$

The load-bearing uniform-in-dimension input is Friedlander–Iwaniec [8, Theorem 6.9]. Under

$$\prod_{w \leq \ell < y} (1 - g_C(\ell))^{-1} \leq K_0 \left(\frac{\log y}{\log w} \right)^{\kappa_C}, \quad (53.31)$$

if $s = \log D / \log z > 9\kappa_C + 1$, its main multiplier is $1 + O(e^{9\kappa_C - s} K_0^{10})$ with an absolute constant. Here the interval prime estimates give absolute K_0 , $\kappa_C \ll 1 + m_r(C) \leq K_T$, and $s = BK_T$; choose C_0 and then $B > 9$ with room. The class remainders, crudely dominated with $R_T = O(T \log T)$, and the resulting class average are

$$\begin{aligned} Q_T \sum_{d \leq D} \mu^2(d) O(\rho_C(d)) &\leq X^{1/3 + o(1)}, \\ \#\mathcal{E}(X) &\ll_A \frac{XL^{O(1)}}{\log X} \frac{1}{\#\mathcal{C}_T} \sum_{C \in \mathcal{C}_T} e^{-c_A m_r(C)L}. \end{aligned} \quad (53.32-53.33)$$

This sieves integers in each class; no prime number theorem modulo growing Q_T is used.

Set $y = L^{3/2}\sqrt{\log L}$ and ignore the $O(L)$ primes dividing r . Exact genus-bit independence handles delayed permission, while Lemma 16.3 handles every other class:

$$\begin{aligned} 2^{-\pi(y)+O(L)} &\leq e^{-cy/\log y} \leq e^{-cL^{3/2}/\sqrt{\log L}}, \\ m_r(C) &\gg_A L^2/y, \quad e^{-c_A m_r(C)L} \leq e^{-c_A L^3/y} = e^{-c_A L^{3/2}/\sqrt{\log L}}. \end{aligned} \quad (53.35\text{--}53.36)$$

Absorb $L^{O(1)}$ in the exponent. Every threshold after the one-conductor deletion is effective. $\square$

The balance is exact: delaying the first negative bit beyond y costs $\exp\{-cy/\log y\}$, while permission by y costs $\exp\{-c_A L^3/y\}$. The endpoint $A = 1$ is excluded both because $Q_T = X^{o(1)}$ is then lost and because the remaining Mertens length $(1 - A)L + O(\log L)$ vanishes.

Corollary 16.5 (almost-all H_{SPF} ; proved, effective). *For each fixed $0 < A < 1$, if $\mathcal{F}_A(X)$ is the set of hard primes $p \leq X$ at which the event (50.7) fails, then*

$$\#\mathcal{F}_A(X) \ll_A \frac{X}{\log X} \exp\left\{-c_A \frac{(\log \log X)^{3/2}}{\sqrt{\log \log \log X}}\right\}. \quad (53.37)$$

Proof. A failure of (50.7) has no good prime of any size. Fix $A' < A$. On $[X/2, X]$, eventually $(\log X)^{A'} \leq (\log p)^A$, so each failure is counted by Theorem 16.4 with A' . Summation over dyadic intervals gives (53.37), with constants renamed in terms of A . $\square$

16.4. The genus-permission asymmetry. The honest comparison is: Type I has failure exponent $(\log \log X)^{3/2}/\sqrt{\log \log \log X}$ and requires independent genus bits; fixed-polylog Type II has exponent $(\log \log X)^2 \log \log \log X$ and no permission gate; cubic Type II has exponent $(\log \log X)^3$, no gate, and remains CLAIMED/PROVISIONAL. Theorem 13.5's escape classes are exactly the fixed-box extreme of the Type-I row. The Type-II stack reaches cubic scale because it has no permission gate; Type I pays the square root while waiting for the first negative genus bit. Dahan [3, Theorem 4.17] obtains the cubic shape for his different stacked-shift two-parameter (u, a) depth, again without a genus gate, and labels its error ineffective. His statistic and the Type-I statistic here are not interchangeable; the Landau–Page deletion makes (53.27) effective.

Corollary 16.5 does not prove the pointwise hypothesis, bound $D(p)$, remove fixed-box escape progressions, or cross the parity barrier.

17. THE LOWER TAIL AND THE TWO-SIDED FRONTIER

The upper tails above leave exceptional sets. Residue-one progressions, combined with effective Linnik, prove that those sets are genuinely nonempty infinitely often below logarithmic exponent one. These are escape theorems, not nonsolvability theorems.

17.1. Type II: the minimal multiplier modulus. For real $T \geq 3$, put

$$M(T) = \text{lcm}(1, 2, \dots, \lfloor T \rfloor), \quad \log M(T) = \psi(\lfloor T \rfloor). \quad (54.1)$$

The full least common multiple is needed because (51.1) ranges over every product $k\ell$, with no primality condition on ℓ .

Theorem 17.1 (Type-II logarithmic lower tail; proved, effective; source Theorem 54.1). *Every prime $p \equiv 1 \pmod{M(T)}$ satisfies*

$$W(p) > T. \quad (54.2)$$

There is an effectively computable absolute constant C_0 such that, for all sufficiently large T , one can choose such a prime with

$$p \leq C_0 M(T)^{5.2} = \exp\{(5.2 + o(1))T\}. \quad (54.3)$$

Consequently these primes are unbounded and

$$\limsup_{\substack{p \rightarrow \infty \\ p \text{ prime}}} \frac{W(p)}{\log p} \geq \frac{1}{5.2} = 0.192307 \dots > 0.19. \quad (54.4)$$

Proof. Suppose data in (51.2) have $m = k\ell \leq T$. Then $m \equiv 3 \pmod{4}$, $uvw = (m+1)/4$, and $p \equiv 1 \pmod{m}$. If they witnessed p , multiplying $1 \equiv -uv^{-1} \pmod{m}$ by v would give $m \mid u+v$. But

$$2 \leq u+v \leq uv+1 \leq \frac{m+1}{4} + 1 < m;$$

this includes $m = 3$, where the upper bound is $2 < 3$. Thus (54.2) holds. The class $1 \pmod{M(T)}$ is reduced. Xylouris's effective Linnik theorem [19] gives its least prime at most $C_0 M(T)^{5.2}$, and the effective prime number theorem gives $\psi(T) = T + o(T)$. Any prime in the class is at least $M(T) + 1$, so primes selected along unbounded T are unbounded. Equations (54.2)–(54.3) then give (54.4). $\square$

Corollary 17.2 (pointwise Type-II frontier; proved; source Corollary 54.2). *For every $0 < A < 1$, $H_{\text{MOD}}(A)$ is false. Thus, in its defined positive-exponent range, exactly $A \geq 1$ remains open.*

Proof. Along Theorem 17.1's primes, $W(p) > (1/(5.2 + o(1))) \log p > (\log p)^A$. $\square$

If

$$E_A(N) = \{p \leq N : p \text{ prime, } W(p) > (\log N)^A\},$$

then Corollary 14.3 and Theorem 17.1 give the two-sided summary

$$\begin{aligned} \#E_A(N) &= o(\pi(N)) & (A > 0), \\ \#E_A(N) &\geq 1 \text{ for infinitely many } N & (0 < A < 1). \end{aligned} \quad (54.5)$$

The first line retains the internal-review qualification of (51.17), and the additional cubic upper bound (51.18) remains CLAIMED/PROVISIONAL. For fixed T , the exhibited progression has relative Dirichlet density $1/\varphi(M(T)) = \exp\{-(1+o(1))T\}$ among the primes. Thus the lower side asserts only nonemptiness infinitely often; the true size of $E_A(N)$ is open.

17.2. Type I: simultaneous genus forcing. Put

$$R(T) = \text{lcm} \left(24, \prod_{\substack{\ell \leq T \\ \ell \text{ prime}}} \ell \right), \quad \log R(T) = \vartheta(T) + \log 4. \quad (54.6)$$

The identity holds for $T \geq 3$.

Theorem 17.3 (Type-I logarithmic lower tail; proved, effective; source Theorem 54.3). *For every $T \geq 3$ and every prime $p \equiv 1 \pmod{R(T)}$,*

$$\chi_s(p) = 1 \quad (1 \leq s \leq T, \text{ } s \text{ squarefree}), \quad ck_{\text{pr}}(p) > T, \quad ck_{\text{min}}(p) > T. \quad (54.7)$$

In fact $M_{c,k}(p) = 0$ for every admissible slice with $ck \leq T$. There are unboundedly many such primes satisfying

$$p \leq C_0 R(T)^{5.2} = \exp\{(5.2 + o(1))T\}, \quad \limsup_{\substack{p \rightarrow \infty \\ p \equiv 1 \pmod{24}}} \frac{ck_{\text{min}}(p)}{\log p} \geq \frac{1}{5.2} > 0.19. \quad (54.8)$$

The convention $ck_{\text{min}}(p) = +\infty$ is allowed.

Proof. For every odd prime $\ell \leq T$, quadratic reciprocity and $p \equiv 1 \pmod{4}$ give

$$\left(\frac{\ell}{p}\right) = \left(\frac{p}{\ell}\right) = 1. \quad (54.9)$$

The congruence modulo 24 also gives $(2/p) = (-1/p) = (3/p) = 1$. The sign and factor 4 in the fundamental discriminant Δ_s therefore have symbol one at p ; multiplicativity proves $\chi_s(p) = 1$ for every squarefree $s \leq T$. If $ck \leq T$, then $s = \text{sf}(c) \leq T$, so (50.3) is empty and Theorem 12.1 gives $M_{c,k}(p) = 0$.

All positive pairs $ck \leq T$ are admissible. Bertrand's postulate, with $3 \leq T < 7$ checked directly, gives $R(T) > 4T$, hence $p > 4T$; this supplies the coprimality and size conditions in $\mathcal{B}_p$. Apply effective Linnik to $1 \pmod{R(T)}$ and use $\vartheta(T) = T + o(T)$. Since every prime in the class is at least $R(T) + 1$, the argument for (54.4) proves (54.8). $\square$

Corollary 17.4 (pointwise Type-I frontier; proved; source Corollary 54.4). *For every $0 < A < 1$, $H_{\text{SPF}}(A)$ is false; exactly $A \geq 1$ remains open. More strongly, infinitely many hard primes have no positive slice at all with*

$$ck \leq (1/(5.2 + o(1))) \log p. \quad (54.10)$$

For each $0 < A < 1$, these primes make the failure set in Corollary 16.5 nonempty at infinitely many heights, while (53.37) gives its effective almost-all upper bound. This quantifies Theorem 13.5's genus-forced zeros; it is strictly stronger than the fixed-divisor non-guarantee in Theorem 12.3.

The classical effectivity register is exact. Xylouris [19] proves the usable exponent 5.2, refining Heath-Brown [10]; the implied constant and threshold are effectively computable, with no ineffective Siegel lower bound. Rosser-Schoenfeld [14] give the convenient explicit bounds $\psi(x) < 1.03883x$ and $\vartheta(x) < 1.01624x$. The sharper $(1 + o(1))x$ used above comes from the classical effective prime number theorem. No numerical value of C_0 is extracted here.

17.3. The two-sided frontier. The census puts much of the observed mass at small constant depths, while (51.17) says that for every fixed $\varepsilon > 0$, all but $o(\pi(N))$ primes have $W(p) \leq (\log p)^\varepsilon$. Theorem 17.1 supplies the effective opposite extreme. The resulting proved sandwich is

$$\begin{aligned} \text{typical: } & W(p) \leq (\log p)^\varepsilon \quad \text{for almost all primes, every fixed } \varepsilon > 0, \\ \text{extremal: } & W(p) \geq c \log p \quad \text{for infinitely many primes, effectively.} \end{aligned} \quad (54.11)$$

The finite W census, Dahan's different two-parameter depth statistic [3], and the different statistic measured in the auro-zero file suggest logarithmic, logarithm-logarithm, and logarithm-squared normalizations worth plotting, but the statistics

are not interchangeable and no finite envelope supplies a growth law. The exact pointwise frontier is only

$$\boxed{\underbrace{A < 1}_{\text{proved impossible pointwise}} \mid \underbrace{A \geq 1}_{\text{still open pointwise}}} . \quad (54.12)$$

The same proved lower edge holds for H_{SPF} . The extended census and its deliberately separate finite-search panel appear in (56.16); in particular, its symbol $\lesssim 2$ is not a proved upper bound or a conjectured limiting exponent. Covering every $m \leq T$ by residue one forces divisibility by $M(T)$, whose logarithm is $T + o(T)$, so the full-lcm escape route is intrinsically logarithmic. Theorems 18.1–18.2 sharpen this observation: logarithmic order is optimal inside the complete-class-plus-generic-least-prime architecture, with a factor-two gap in the proved Type-II modulus constant and exact saturation in Type I. They do not close any part of the open pointwise range in (54.12). None of these lower-tail primes is asserted to be an Erdős–Straus exception.

18. THE CONGRUENCE-CERTIFICATE CEILING AND EXTREMAL CENSUS

This section closes the elementary two-sided certificate question left by Sections 14 and 17. A complete congruence class certifying escape through depth T must itself have exponentially large modulus. The Type-II statement is the prime-progression extension of the all-integer cylinder wall in Theorem 8.2, not a new independent supply theorem. The Type-I statement uses Corollary 15.2 to turn every unforced prime core into a required conductor coordinate. The theorems are asymptotic and effective. The censuses below are exact only in their stated ranges and imply no growth law.

18.1. Type II: every certificate carries the 3 (mod 4) primes. Call $a \pmod{Q}$ a *prime congruence certificate* for $W > T$ if $(a, Q) = 1$ and every sufficiently large prime $p \equiv a \pmod{Q}$ satisfies $W(p) > T$. An all-integer certificate has the analogous meaning with every sufficiently large positive integer in the class; it need not be reduced. Put

$$P_3(T) = \prod_{\substack{\ell \leq T \\ \ell \text{ prime}, \ell \equiv 3 \pmod{4}}} \ell. \quad (56.1)$$

Theorem 18.1 (Type-II congruence-certificate ceiling; proved, effective; source Theorem 56.1). *If $a \pmod{Q}$ is either a prime or an all-integer congruence certificate for $W > T$, then*

$$P_3(T) \mid Q, \quad \log Q \geq \vartheta(T; 4, 3) = \left(\frac{1}{2} + o(1)\right) T. \quad (56.2)$$

Consequently, for every fixed $\epsilon > 0$, there is an effectively computable T_ϵ such that $Q \geq \exp\{(1/2 - \epsilon)T\}$ for $T \geq T_\epsilon$.

Proof. Fix a prime $\ell \leq T$, $\ell \equiv 3 \pmod{4}$, and put $A = (\ell + 1)/4$. The divisor $D = 1$ in Lemma 4.1 is not merely a formal intrinsic class: its Lemma 3.4 datum is

$$k = 1, \quad \ell = \ell, \quad (u, v, w) = (1, A, 1), \quad -uv^{-1} \equiv -A^{-1} \equiv -4 \pmod{\ell}. \quad (56.3)$$

Thus every positive integer $n \equiv -4 \pmod{\ell}$ has $W(n) \leq \ell \leq T$.

Suppose $\ell \nmid Q$. The Chinese remainder theorem gives infinitely many positive integers satisfying both $n \equiv a \pmod{Q}$ and $n \equiv -4 \pmod{\ell}$, contradicting an all-integer certificate. In the prime version their joint class modulo $Q\ell$ is reduced: it is reduced modulo Q by hypothesis and $-4 \not\equiv 0 \pmod{\ell}$. Dirichlet's theorem therefore gives infinitely many primes in it, again contradicting the certificate. Hence every factor in (56.1) divides Q . The final estimate is the effective prime number theorem in the fixed progression $3 \pmod{4}$. This is exactly the mechanism of Theorem 8.2, with the prime-bearing conclusion added. $\square$

The theorem uses only one harvested class at each prime modulus. It does *not* prove that the full $\text{lcm}(1, \dots, T)$ of Section 17 is necessary for an arbitrary certificate. Composite moduli can meet a progression in only one coset modulo (Q, M) , and the full harvested set need not cover all such cosets. Thus replacing the constant $1/2$ in (56.2) by 1 would require an additional argument.

18.2. Type I: every prime core is forced into the conductor. A *hard Type-I congruence certificate through T* is a reduced class $a \pmod{Q}$ with

$$24 \mid Q, \quad a \equiv 1 \pmod{24}, \quad (56.4)$$

such that every sufficiently large prime $p \equiv a \pmod{Q}$ has $ck_{\min}(p) > T$. Equivalently, every admissible slice with $ck \leq T$ vanishes on every sufficiently large prime in the class.

Theorem 18.2 (Type-I congruence-certificate ceiling; proved, effective; source Theorem 56.2). *Every hard Type-I congruence certificate through T satisfies*

$$\prod_{\substack{5 \leq \ell \leq T \\ \ell \text{ prime}}} \ell \mid Q, \quad \log Q \geq \vartheta(T) - \log 6 = T + o(T). \quad (56.5)$$

In particular, for every $\epsilon > 0$, effectively for all sufficiently large T , one has $Q \geq \exp\{(1 - \epsilon)T\}$.

Proof. Let $5 \leq \ell \leq T$ be prime and suppose $\ell \nmid Q$. The primitive quadratic character attached to the core $s = \ell$ is nonconstant modulo its prime coordinate. Concretely, choose a nonzero quadratic nonresidue $b \pmod{\ell}$, and use the Chinese remainder theorem to refine the certificate to

$$p \equiv a \pmod{Q}, \quad p \equiv b \pmod{\ell}. \quad (56.6)$$

This is a reduced class modulo $Q\ell$, hence contains infinitely many primes. For every such hard prime, quadratic reciprocity and $p \equiv 1 \pmod{4}$ give

$$\chi_{\ell}(p) = \left(\frac{\Delta_{\ell}}{p}\right) = \left(\frac{-\ell}{p}\right) = \left(\frac{p}{\ell}\right) = -1. \quad (56.7)$$

The slice $(c, k) = (\ell, 1)$ is therefore unforced and is admissible for all sufficiently large primes in (56.6). Since $Q\ell$ is a multiple of $\text{lcm}(24, 4\ell)$, Corollary 15.2 says that this reduced refinement cannot be identically vanishing: arbitrarily far out it contains a prime with $M_{\ell,1}(p) > 0$. That prime has $ck_{\min}(p) \leq \ell \leq T$, contradicting the certificate. Therefore $\ell \mid Q$ for every prime $5 \leq \ell \leq T$. Taking logarithms and using the effective prime number theorem proves (56.5). $\square$

The primitivity point in this proof is additive, not multiplicative: the progression modulo a conductor fills residue classes modulo every conductor coordinate omitted from Q . Equation (56.6) exhibits both character signs directly. No claim that a multiplicative subgroup generated by Q controls the progression is used.

Corollary 18.3 (congruence/Linnik recipe; proved within the stated recipe; source (56.8)). *Suppose a certificate construction at depth T uses $Q = \exp\{(\alpha + o(1))T\}$, and a least-prime theorem with exponent L selects a certified prime $p \leq Q^{L+o(1)}$. Here “certified” includes that p lies beyond any finite exceptional prefix allowed by the certificate definition; a least-prime bound alone does not ensure this for an arbitrary eventual certificate. Then*

$$W(p) > T \quad \text{or} \quad ck_{\min}(p) > T, \quad T \geq \frac{1 + o(1)}{\alpha L} \log p. \quad (56.8)$$

Theorems 18.1 and 18.2 force respectively $\alpha \geq 1/2$ and $\alpha \geq 1$. Thus the coefficient available from this generic recipe is at most $2/L$ in Type II and at most $1/L$ in Type I. The residue-one choices of Theorems 17.1 and 17.3 have $\alpha = 1$ in both frames. They therefore saturate the Type-I certificate rate and are within a factor two of the proved Type-II ceiling; in both frames they establish the optimal logarithmic order for a certificate-plus-least-prime argument, up to the least-prime exponent. With the conservative effective Linnik value $L = 5.2$, their coefficient is $1/5.2$.

Remark 18.4 (GRH least-prime consequence; conditional; source (56.9)). *Conditional on the unproved Generalized Riemann Hypothesis, the standard least-prime bound permits $L = 2 + \epsilon$. The same residue-one constructions therefore give, respectively along unbounded prime sequences,*

$$W(p) > (1/2 - \epsilon) \log p, \quad ck_{\min}(p) > (1/2 - \epsilon) \log p, \quad (56.9)$$

after harmlessly changing ϵ . This statement is conditional; (56.2) and (56.5) are unconditional and effective.

Assessment 56.2a (the congruence wall). The proved conclusion is that a *uniform complete-class* certificate through T has exponential modulus. Consequently, inside the Section 17 architecture—complete congruence certificate, modulus $\exp\{\Theta(T)\}$, then a generic least-prime theorem—a superlogarithmic lower bound would have to abandon at least one ingredient. The natural alternative is a non-congruence certificate on a sifted set of primes, in the critical-window territory of Section 8. The ceiling does not logically prohibit every congruence-informed construction: it says nothing about an accidentally tiny first prime in a chosen class, or a modulus much larger than $\exp\{\Theta(T)\}$. Accordingly this is an architecture wall, not a theorem that every possible extremal proof beyond $C \log p$ must be noncongruential.

18.3. The exact W census. Computational 56.3 (exact stated ranges; informational). Verifier block (bc) harvests the complete intrinsic class set of Lemmas 3.4 and 4.1, $\{-4D \pmod{M} : D \mid ((M+1)/4)^2\}$, for every $M \leq 3000$, $M \equiv 3 \pmod{4}$. It then scans every hard prime in increasing order and stops at the first harvested modulus. All 9,732 hard primes below 10^6 resolve within the cap. Their strict records are

$$\begin{array}{r|ccccccc} p & 73 & 193 & 1201 & 2521 & 3361 & 33289 & 90841 \\ W(p) & 7 & 15 & 31 & 47 & 99 & 155 & 167 \\[10pt] p & 144169 & 167521 & 225289 & 361321 & 915961 & 954409 & \\ W(p) & 191 & 259 & 279 & 287 & 303 & 335 & \end{array} \quad (56.10)$$

With `ES_FULL_SCAN=1`, the streamed, chunked scan covers all 82,887 hard primes below 10^7 . The only later records are

$$(p, W(p)) = (1853329, 383), \quad (2031121, 2495), \quad (56.11)$$

and the running maximum remains 2495 through the endpoint. These values are exact minima in the harvested family because every smaller eligible modulus was tested; the cap is not hit.

The requested normalized maxima, using natural logarithms, are INFO-ONLY:

p -range	$\max W / \log p$	$\max W / (\log p \log \log p)$	$\max W / (\log p)^2$
$p < 10^6$	24.330286 (954409, 335)	9.277839 (954409, 335)	1.836625 (225289, 279)
$p < 10^7$	171.783468 (2031121, 2495)	64.198698 (2031121, 2495)	11.827479 (2031121, 2495)

(56.12)

Each parenthesis is $(p, W(p))$. At $p = 10^7$, the unconditional Section 17 coefficient gives $(1/5.2) \log p = 3.0996 \dots$. This comparison must not mix registers: Section 17 guarantees rare residue-one escapes at a height bounded by $\exp\{(5.2 + o(1))T\}$, whereas (56.10)–(56.12) record whatever finite primes happen to be extremal below a moderate endpoint. For example, the generic height attached to $T = 2495$ is astronomically beyond this scan. The census neither samples that constructed sequence nor estimates a tail density, and the large isolated jump in (56.11) supports no growth law.

18.4. The Type-I extremal extension. Computational 56.4 (exact stated ranges; informational). The same block factors one norm at a time, skips the exact genus-forced signs before factorization, and streams unresolved primes in product order. For the 3,202 hard primes below $3 \cdot 10^5$, the strict ck_{pr} records are

$$\begin{array}{c|ccccccc} p & 73 & 193 & 241 & 1201 & 2521 & 4729 & 7489 \\ ck_{\text{pr}} & 7 & 10 & 21 & 34 & 38 & 66 & 70 \end{array} \quad (56.13)$$

$$\begin{array}{c|ccccccc} p & 9601 & 31081 & 51769 & 83689 & 113161 & 171481 \\ ck_{\text{pr}} & 78 & 110 & 249 & 282 & 378 & 461 \end{array}$$

The unrestricted conspiracy-depth records in the same range are

$$\begin{array}{c|ccccccc} p & 73 & 193 & 241 & 769 & 1321 & 2281 \\ D(p) & 6 & 9 & 10 & 12 & 20 & 25 \end{array} \quad (56.14)$$

$$\begin{array}{c|ccccccc} p & 2521 & 9601 & 12289 & 55441 & 92401 \\ D(p) & 37 & 66 & 76 & 82 & 102 \end{array}$$

With `ES_FULL_SCAN=1`, all 9,732 hard primes below 10^6 are scanned. The later ck_{pr} records are (319489, 878), (538561, 898); the sole later depth record is (414241, 217). Every prime resolves by these maxima. The normalized maxima are again INFO-ONLY:

statistic and range	$\max X / \log p$	$\max X / (\log p \log \log p)$	$\max X / (\log p)^2$
$ck_{\text{pr}}, p < 3 \cdot 10^5$	38.250190 (171481, 461)	15.366153 (171481, 461)	3.173703 (171481, 461)
$ck_{\text{pr}}, p < 10^6$	69.273069 (319489, 878)	27.277261 (319489, 878)	5.465556 (319489, 878)
$D, p < 3 \cdot 10^5$	8.920846 (92401, 102)	3.661213 (92401, 102)	0.857113 (12289, 76)
$D, p < 10^6$	16.777222 (414241, 217)	6.553922 (414241, 217)	1.297121 (414241, 217)

(56.15)

Here X in the header denotes the statistic in the first column, and each parenthesis is $(p, X(p))$. These are census maxima, not estimates of H_{SPF} , ck_{min} , or a population exponent. The block's persistent state is linear in the hard-prime count: it keeps the hard-prime tuple, unresolved sets, per-prime minimum maps, and a small core cache. It keeps only one bounded residue set for the current norm. Prime generation is chunked; no prime-by-slice factor table or interval-sized residue array is retained.

18.5. The remaining window. The exact logical frontier and the finite plotting window can be displayed together only if their labels remain separate:

$$\underbrace{A < 1}_{\substack{\text{impossible pointwise} \\ \text{Theorems 17.1, 17.3}}} \quad | \quad \underbrace{1 \leq A \lesssim 2}_{\substack{\text{Assessment: finite-search window only} \\ \log p, \log p \log \log p, (\log p)^2 \text{ are the census anchors}}} \quad | \quad \underbrace{A \geq 1}_{\substack{H_{\text{MOD}}(A), H_{\text{SPF}}(A) \\ \text{still open pointwise.}}} \quad (56.16)$$

The symbol $\lesssim 2$ in the middle is deliberately not a proved upper bound, a conjectured limiting exponent, or even a monotonic trend. It only names the normalization panel in (56.12) and (56.15). The proved lower wall is $A = 1$; no finite census supplies the other side. The pointwise-frontier statements in Sections 14 and 17 remain unchanged: Theorems 18.1 and 18.2 constrain only the complete-class certificate architecture.

19. GENERAL-NUMERATOR TAILS AND THE EFFECTIVITY PERIMETER

This section truncates the general-numerator machinery; it does not extend its range. The fixed-polylogarithmic row has the internal review qualification of Lemma 14.1 and Section 10, but uses neither Theorem 9.4 nor Theorem 11.6. The cubic row and record recovery inherit them and remain CLAIMED/PROVISIONAL. Effectivity below means computability of constants conditional on those correctness labels. Put

$$\lambda_m = \frac{\eta_1(m)}{\varphi(m)}, \quad \theta_m = \frac{\eta_2(m)}{\varphi(m)}; \quad (55.1)$$

both are $\asymp 1/m$ up to absolute constants.

19.1. The minimal general-numerator witness modulus. For $m \geq 3$ and positive p , define

$$W_m(p) = \min k\ell, \quad (55.2)$$

where the minimum is over positive k, ℓ, u, v, w satisfying

$$k\ell \equiv -1 \pmod{m}, \quad \frac{k\ell + 1}{m} = uvw, \quad pv \equiv -u \pmod{k\ell}; \quad (55.3)$$

put $W_m(p) = +\infty$ if no datum exists. The coordinate ℓ need not be prime. Lemma 11.1 turns $W_m(p) \leq T$ into a representation of m/p , and at $m = 4$ this is exactly (51.1)–(51.2).

Let $t = \log X$, $KX \leq T$, and use the canonical boxes from the proof of Theorem 11.6. Their two masses are

$$M_1 = \lambda_m t^2 \log K, \quad M_2 = \theta_m t^3 \quad (K = X^\kappa). \quad (55.4)$$

Lemma 19.1 (m -uniform truncated assembly; proved internally with the Section 10 review qualification; source Lemma 55.1). *Fix $B > 0$. Uniformly for $m \leq t^B$, if K is a sufficiently large fixed power of t , $KX \leq T$, and*

$$\log N \geq C_B t(1 + M_1), \quad (55.6)$$

then

$$\#\{p \leq N : p \text{ prime}, W_m(p) > T\} \ll_B N \exp\{-c_B M_1\}. \quad (55.5)$$

For fixed $0 < \kappa < 1/240$, if $K = X^\kappa$, $KX \leq T$, and

$$\log N \geq C_{B,\kappa} t(1 + M_2), \quad (55.8)$$

then

$$\#\{p \leq N : p \text{ prime}, W_m(p) > T\} \ll_{B,\kappa} N \exp\{-c_{B,\kappa} M_2\}. \quad (55.7)$$

The second assertion is CLAIMED/PROVISIONAL through Theorem 11.6. Every constant is effective in the precise sense of Subsection 19.2.

Proof. The full coprime multiplier family has mass comparable to M_1 ; replacing η_1 by η_2 in the upper profile and moments changes only an absolute factor. The cubic family has mass M_2 by (43.20), (43.27), and the relative-mass argument (43.36)–(43.38). Choose the small-prime quarantine and even Bonferroni degree of order M_i when it grows, and fixed degree otherwise. The exact ledger is

$$\deg = O(1 + M_i), \quad \log d_{\text{term}}, \log \sum |c_{\text{term}}| = O_B\{t(1 + M_i)\}. \quad (55.9)$$

Conditions (55.6) and (55.8) pay this ledger. Every retained atom has modulus at most $KX \leq T$; primes dividing m and fixed initial ranges are absorbed effectively. $\square$

Theorem 19.2 (general- m master tails; status as in Lemma 19.1; source Theorem 55.2). *Fix $B > 0$. Uniformly for $m \leq (\log T)^B$, put*

$$\mathcal{M}_1(m, T) = \lambda_m (\log T)^2 \log(2 + \log T), \quad \mathcal{M}_2(m, T) = \theta_m (\log T)^3. \quad (55.10)$$

Set $\mathcal{P}_{m,T}(N) = \#\{p \leq N : p \text{ prime}, W_m(p) > T\}$. With effective constants depending only on the displayed fixed parameters,

$$\begin{aligned} \mathcal{P}_{m,T}(N) &\ll_B N e^{-c_B \mathcal{M}_1(m,T)}, & \log N &\geq C_B \{\log T + (\log T) \mathcal{M}_1(m,T)\}, \\ \mathcal{P}_{m,T}(N) &\ll_{B,\kappa} N e^{-c_{B,\kappa} \mathcal{M}_2(m,T)}, & \log N &\geq C_{B,\kappa} \{\log T + (\log T) \mathcal{M}_2(m,T)\}. \end{aligned} \quad (55.11)$$

The first line uses only the Layer-1 supply. The second is CLAIMED/PROVISIONAL. No uniformity is asserted when m exceeds every fixed power of $\log T$.

Proof. Take $X = T^{1/2}$. In the first row choose $K = t^{D(B)}$ large enough for the sign selection below; then $KX \leq T$ and $\log K \asymp_B \log(2 + t)$. In the second, rescale X by a harmless fixed power so $K = X^\kappa$ and $KX \leq T$. Lemma 19.1 gives both rows; bounded T is absorbed in the constants. $\square$

At the top of the cubic window, $\log N \asymp \theta_m (\log T)^4$, so the saving is $\theta_m^{1/4} (\log N)^{3/4}$, exactly (43.35). In the fixed-gap range $m \leq (\log N)^{3-\varepsilon}$ the extra $\log T$ term is smaller and the $m \leq t^4$ ledger closes. The first row similarly recovers the scale $\lambda_m^{1/3} (\log N)^{2/3} (\log \log N)^{1/3}$ and the fixed-gap range $m \leq (\log N)^{2-\varepsilon}$. These are consistency checks, not range or status upgrades.

Corollary 19.3 (general- m polylogarithmic tails; source Corollary 55.3). *Fix $A, B > 0$ and $m \leq (\log \log N)^B$. With p restricted to primes,*

$$\begin{aligned} \#\{p \leq N : W_m(p) > (\log N)^A\} &\ll_{A,B} N \exp\{-c_{A,B} \lambda_m (\log \log N)^2 \log \log \log N\}, \\ \#\{p \leq N : W_m(p) > (\log N)^A\} &\ll_{A,B} N \exp\{-c_{A,B} \theta_m (\log \log N)^3\}. \end{aligned} \quad (55.12)$$

The second line is CLAIMED/PROVISIONAL. The almost-all result does not prove the pointwise hypothesis

$$H_{\text{MOD},m}(A) : W_m(p) \leq (\log p)^A \quad \text{for every sufficiently large prime } p. \quad (55.13)$$

19.2. The conductor coupled to m . The prime progression in the general- m supply is

$$q = muv, \quad \ell \equiv -k^{-1} \pmod{q}, \quad (55.14)$$

not modulo uv or $\text{lcm}(m, uv)$. Thus if a Page conductor r divides m , every modulus is divisible by r and deletion would erase the entire supply. Retaining the exceptional term gives the effective form

$$\begin{aligned} \mathcal{E}_r(y; q, a) &= \psi(y; q, a) - \frac{y}{\varphi(q)} + \frac{\mathbf{1}_{r|q} \chi_r(a) y^\beta}{\beta \varphi(q)}, \\ \sum_{q \leq Q} \max_{(a, q)=1} \max_{x \leq y \leq 2x} |\mathcal{E}_r(y; q, a)| &\ll_A \frac{x}{(\log x)^A}. \end{aligned} \quad (55.15)$$

Here $Q \leq x^{1/2}/(\log x)^{B_A}$; constants and thresholds are effective, and the exceptional term is absent if there is no Page character. This is the displayed-exceptional-character variant obtained by retaining the one primitive real character in the proof of Lenstra–Pomerance [11, Lemma 11.2]; their lemma itself prints the excluded-conductor form.

Lemma 19.4 (m -uniform Page repair; proved; source Lemma 55.4). *Uniformly for $m \leq t^B$, effective aggregate lower class mass holds for exactly the full coprime multiplier family used by the Layer-1 chain and the structured fibres $\mathcal{J}_c$ used by the c -free chain. This is not a per-multiplier lower profile and not an arbitrary-sparse-subfamily claim. The conductor cases are:*

1. *With no exceptional conductor, retain every box.*
2. *If $r \nmid m$, choose a prime a with $v_a(r) > v_a(m)$ and impose $a \nmid uv$. Then $r \nmid muv$, including when r and m share other prime-power factors.*
3. *If $r \mid m$, retain exactly the multipliers satisfying*

$$\chi_r(-k^{-1}) = -1. \quad (55.16)$$

For every progression actually retained, the exceptional term in (55.15) then adds to the ordinary main term. This family retains a fixed positive fraction of the aggregate mass needed downstream.

The low- ω deletion and low-congestion pruning may be imposed simultaneously.

Proof. In case 2, omitting a from uv leaves the a -adic exponent of muv below that of r ; Lemma 14.5's local proportion and large- a estimate apply pointwise in k .

Suppose $r \mid m$. The lifted nonprincipal real character takes each sign on half the units modulo m . Uniformly in reduced $b \pmod{m}$, Möbius expansion gives

$$\sum_{\substack{Y < k \leq 2Y \\ k \equiv b \pmod{m}}} \frac{\varphi(k)}{k} = \frac{Y}{m} \prod_{p \nmid m} (1 - p^{-2}) + O(\log(2Y)). \quad (55.17)$$

After summing favorable classes and dividing by $k \asymp Y$, intervals with $Y \geq Cm^2$ contribute $\gg \eta_1(m)$. If F_r is the set in (55.16), $H_m(K) = \sum_{k \leq K, (k, m)=1} \varphi(k)/k^2$, and $K \geq m^D$ for a fixed large D , then effectively

$$h(F_r) \geq c_0 \eta_1(m) \log K \geq c_1 H_m(K). \quad (55.17a)$$

The cubic family has $K = e^{\kappa t} \gg m^D$; the Layer-1 tail uses $K = t^{D(B)}$.

For the structured fibre $\mathcal{J}_c = \{k : (k, m) = (k, c) = 1\}$, (43.36) gives the adversarially safe subtraction

$$h(F_r \cap \mathcal{J}_c) \geq h(F_r) - H_m(K) \sum_{p|c, p|L_{m,K}} \frac{1}{p} \geq (c_1 - Z(c))H_m(K). \quad (55.17b)$$

Thus the same Chernoff event as (43.37), now with $Z(c) \leq c_1/2$, leaves a fixed positive favorable mass without asserting character balance in an arbitrary sparse family. If an incidence statement formally requires $1 \in \mathcal{J}$, adjoin it only for bookkeeping and omit its triples from the BV lower sum when its sign is unfavorable. Increasing the low- ω cutoff and applying the existing low-congestion pruning preserve the order of mass. $\square$

This quantifier boundary is load-bearing: an arbitrary sparse $\mathcal{J}$ could contain only the unfavorable sign. The full Layer-1 family and the fibres $\mathcal{J}_c$ are exactly the two downstream families for which Lemma 19.4 proves enough positive mass.

Theorem 19.5 (effectivity perimeter; proved internally, status-preserving; source Theorem 55.5). *“Effective” means that constants and thresholds are computable uniformly in the theorem’s displayed fixed-gap range. The exact perimeter is*

statement	effective?	inheritance	
Layer-1 tails	yes	Section 39 review no 34.8 or 43.7	(55.18)
thinned-cubic tails	yes, conditional on correctness	CLAIMED/PROVISIONAL 34.8 and 43.7	
Theorem 43.12	yes, conditional on correctness	CLAIMED/PROVISIONAL 34.8, 43.7, and Section 39.7	

In every row, $r \nmid m$ uses deletion in uv , while $r \mid m$ uses the favorable k -half. The cubic rows effectivize only the surviving class mass, followed in Theorem 43.12 by the existing effective ledger and semigroup transfer.

Proof. Lemma 19.4 repairs the only Siegel-sensitive lower prime-progression input. Shiu, Brun–Titchmarsh, the box and incidence bounds, exact CRT moments, Chernoff, Bonferroni, integer class counting, and the Rankin semigroup transfer are effective. At the cubic top, (55.8) is the existing (43.40) ledger, so no parameter or range changes. $\square$

The table does not effectivize the inherited ancillary absolute-error assertion in Theorem 11.6 over *all* low-congestion triples. Restoring deleted unfavorable triples restores the unresolved exceptional term; no row of (55.18) uses that assertion.

19.3. Comparison and finite companion. Truncating Pomerance–Weingartner’s general- m , $k = 1$ supply gives mass $\asymp (\log T)^2/\varphi(m)$. The ratios of the two masses in (55.10) to that benchmark are

$$\eta_1(m) \log \log T, \quad \eta_2(m) \log T. \quad (55.19)$$

Thus PW’s pure $\varphi(m)^{-1}$ dependence is formally stronger below the crossovers where either ratio is one, and the multiplier exponents are formally stronger above. For every fixed B , the below-crossover regions eventually lie outside the proved window $m \leq (\log T)^B$; this separates the unrestricted formal comparison from actual uniformity. The cubic side is provisional, PW is proved, and unknown constants preclude a finite crossover. At the optimized N -window the standing comparison remains (43.33)–(43.34), including PW’s $\varphi(m)^{1/3}$ uniformity. No pointwise theorem or range extension follows.

Computational 55.6 (exact finite regression; informational). Verifier block (bb) harvests all classes (55.3) for $m \in \{3, 5, 6, 7\}$ and $k\ell \leq 1500$. For primes $p \leq 10^5$ prime to m , the counts with $W_m(p) > 25, 100, 400$ and those still unharvested are

m	$\#p$	three tail counts	unharvested at 1500
3	9591	(0, 0, 0)	0
5	9591	(875, 61, 5)	1
6	9590	(3136, 663, 77)	17
7	9591	(2222, 438, 35)	10

(55.20)

The block independently checks the intrinsic class description, reproduces the $m = 4$ classes from (ax), and verifies

$$\eta_2(3) = \frac{9}{11}, \quad \eta_2(5) = \frac{25}{29}, \quad \eta_2(6) = \frac{36}{55}, \quad \eta_2(7) = \frac{49}{55}. \quad (55.21)$$

For the Page toys $(m, r) = (5, 5), (6, 3)$, both signs occur and the favorable side retains between one third and three quarters of the finite harmonic weight. This is a finite shadow of (55.17a), not evidence for its asymptotic uniform assertion.

20. RESIDUE-DISPERSION FRONTIER

Put

$$L = \log X, \quad z = \frac{L^3 \log \log L}{\log L}, \quad \Lambda = \frac{L^3}{\log L}.$$

For a composite z -rough $M \leq X$, $M \equiv 3 \pmod{4}$, choose the least positive divisor $D \mid ((M+1)/4)^2$ in each fibre of $D \mapsto -4D \pmod{M}$, and delete a representative when its class is implied by a prime-modulus atom. Call the remaining set $\mathcal{D}^\circ(M)$. With the exact conditional weight

$$\kappa(M) = \prod_{\substack{q \mid M, \, z < q \leq Y \\ q \equiv 3(4)}} \frac{q}{q - F(q)}, \quad n_{M,p}(a) = \#\{D \in \mathcal{D}^\circ(M) : -4D \equiv a \pmod{p}\},$$

the pair-level dispersion quantity is exactly

$$\mathcal{V}_X = \sum_{z < p \leq X} p \sum_{a \bmod p} \left(\sum_{\substack{M \leq X, \, M \text{ composite}, \, P^-(M) > z \\ M \equiv 3(4), \, p \mid M}} \frac{\kappa(M)}{M} n_{M,p}(a) \right)^2. \quad (40.5)$$

This is the residue-dispersion reformulation; canonical deduplication, prime powers, and the prime-implied deletion are part of the formula.

Write $D = R^2/s$, with $s \mid \text{rad}(R)$ and $M = 4Rk - 1$. For $p \nmid 4R$, let $c_p(R) \in \{1, \dots, p-1\}$ be the inverse of $4R$ modulo p . The condition $p \mid M$ gives $k = c_p(R) + jp$. Splitting into the endpoint $j = 0$ and regular tail $j \geq 1$ gives vectors $u_{p,a}^{\text{end}}$ and $u_{p,a}^{\text{reg}}$. The regular-tail theorem is proved, with the full canonical and retention restrictions allowed:

$$\mathcal{V}_X^{\text{reg}} := \sum_{z < p \leq X} p \sum_a (u_{p,a}^{\text{reg}})^2 \ll L^2 \log L + \frac{L^7}{z \log z} = o(\Lambda^2). \quad (40.14)$$

Indeed the contribution of fixed D is at most $CL/(pR_0(D))$; after expansion, $p \mid D - E$, while $\sum_D L/R_0(D) \ll L^3$ and $\sum_D (L/R_0(D))^2 \ll L^2$. For $D \neq E$ there are at most $2L/\log z$ prime divisors above z , and for $D = E$ Mertens' estimate costs $O(\log L)$, giving (40.14).

The exact remaining pair target is therefore

$$\boxed{\mathcal{V}_X^{\text{end}} = \sum_{z < p \leq X} p \sum_a (u_{p,a}^{\text{end}})^2 = O(\Lambda^2).} \quad (40.18)$$

Equivalently, in divisor variables, it is the explicit short-cofactor estimate

$$\sum_{z < p \leq X} p \sum_{a \bmod p} \left(\sum_{\substack{R, s, c, q: s \mid \text{rad}(R), 1 \leq c < p \\ pq = 4Rc - 1 \leq X, P^-(pq) > z, pq \equiv 3(4) \\ R^2/s \equiv -4^{-1}a \pmod{p}, q > 1 \text{ (so } pq \text{ composite)} \\ R^2/s \text{ is the retained representative at } pq}} \frac{\kappa(pq)}{pq} \right)^2 \ll \Lambda^2. \quad (40.19)$$

Every term has $q > z$ and $q < 4R$.

20.1. Endpoint sparsity and proved ranges. For $(p, 4R) = 1$, let $q_p(R)$ be the unique integer in $[1, 4R)$ with

$$pq_p(R) \equiv -1 \pmod{4R}. \quad (42.1)$$

Let $I_p(R, s)$ assert all the conditions in (40.19), including $s \mid \text{rad}(R)$, $q_p(R) > z$, $pq_p(R) \leq X$, roughness, and canonical retention. Then endpoint incidences are exactly the pairs with $I_p(R, s) = 1$, and

$$q = q_p(R), \quad z < q < 4R, \quad 1 \leq c = \frac{pq + 1}{4R} < p, \quad \frac{z}{4} < R \leq \frac{X + 1}{4}. \quad (42.2)$$

Thus there is one cofactor for each (p, R) , not one for each p ; the moving multiplicity $2^{\omega(R)}$ in s remains.

Put $A = (pq + 1)/4 = Rc$. The complementary divisor to $D = R^2/s$ is $A^2/D = c^2s$, and hence

$$D \equiv \overline{16c^2s} \pmod{p}, \quad -4D \equiv -\overline{4c^2s} \pmod{p}. \quad (42.3)$$

Two endpoint incidences over p collide exactly when $c^2s \equiv c'^2s' \pmod{p}$. Distinct incidences cannot have equal integer shifts: $R^2/s = R'^2/s'$ forces the common value's radical root $R = R_0(D) = R'$, then $s = s'$, and (42.1) gives the same q . The endpoint energy therefore has only genuine nonzero divisibility coincidences off the diagonal.

Theorem 20.1 (fixed endpoint fibres). *For every fixed squarefree s , uniformly in s ,*

$$\sum_{z < p \leq X} p \sum_{a \bmod p} (u_{p,a}^{\text{end},(s)})^2 \ll L^3 + L^2 \log L = o(\Lambda^2). \quad (42.4)$$

A prescribed set of K_0 squarefree fibres has energy $O(K_0^2 L^3)$ and therefore closes when $K_0 = O(L^{3/2}/\log L)$.

Proof. Drop canonical restrictions by positivity and use $\kappa \leq 2$. For $1 \leq c < p$, put

$$B_{p,c}^{(s)} = \sum_{\substack{z < q \leq X/p \\ pq \equiv -1 \pmod{4c} \\ s \mid \text{rad}((pq+1)/(4c))}} \frac{1}{q}.$$

If $p \nmid s$, the map $c \mapsto -\overline{4c^2s}$ has fibres of size at most two, and therefore

$$p \sum_a (u_{p,a}^{\text{end},(s)})^2 \leq \frac{8}{p} \sum_{c < p} (B_{p,c}^{(s)})^2.$$

The q 's lie in one progression of spacing $4c$. Separating its first member and writing $\ell_p = 1 + \log^+(X/(pz))$ gives

$$(B_{p,c}^{(s)})^2 \leq \sum_q \frac{1}{q^2} + O\left(\frac{\ell_p}{cz} + \frac{\ell_p^2}{c^2}\right).$$

After multiplication by $1/p$ and summation, the first terms cost $O(L^3)$: with $M = pq$ and $A = (M+1)/4$, their sum is at most $\sum_{M \leq X} \tau(A)/M \ll L^3$ by Theorem 4.2. Mertens' estimates bound the errors by $O(L^2/z + L^2 \log L)$, proving (42.4). Cauchy over the prescribed fibres gives the last assertion. $\square$

For every fixed $B > 3$, the endpoint also satisfies

$$\sum_{z < p \leq L^B} p \sum_a (u_{p,a}^{\text{end}})^2 = O_B(\Lambda^2), \quad (42.5)$$

and there are no endpoint incidences for $p > X/z$. Indeed Theorem 7.2 gives $t_p^{\text{end}} \leq 2A(b_p + u_p)$; uniformly for $p \leq L^B$, $pb_p = O(\Lambda)$ and $pu_p = L^{o(1)} = o(\Lambda)$. Residue Cauchy and Mertens prove (42.5). The super-polylogarithmic moving- s range remains nonempty and has harmonic length $\Theta(\log L)$.

20.2. Kloosterman-fraction audit. For an endpoint incidence put $m = 4c^2s$ and aggregate its cofactors by

$$G_{m,p} = \sum_q \frac{\kappa(pq)}{q}, \quad (45.1)$$

where the sum includes every condition in (40.19); set it to zero unless $m/4 = c^2s$ with s squarefree. This decomposition is unique. Additive Parseval and (42.3) give the exact matrix identity

$$\mathcal{V}_X^{\text{end}} = \sum_{z < p \leq X/z} \sum_{h \bmod p} \left| \frac{1}{p} \sum_{m \geq 1} G_{m,p} e_p(-h\tilde{m}) \right|^2. \quad (45.2)$$

The support has $(m, pq) = 1$, $z < q \leq X/p$, and $m \leq c(pq+1) < p(X+1)$. The only valid reciprocity orientations are

$$e_p(-h\tilde{m}) = e_{pq}(-hq\tilde{m}) = e_m(h\tilde{p})e\left(-\frac{h}{mp}\right). \quad (45.3)$$

In particular, q is never the phase modulus.

The DFI theorem, checked directly against the obtained 1997 article [4], allows arbitrary *separate* sequences α_m, β_n and gives

$$\left| \sum_{\substack{m \sim M, n \sim N \\ (m,n)=1}} \alpha_m \beta_n e(a\tilde{m}/n) \right| \ll_\varepsilon \|\alpha\|_2 \|\beta\|_2 (|a| + MN)^{3/8} (M+N)^{11/48+\varepsilon} \quad (45.4)$$

for every nonzero integer a , with no range hypothesis. Its balanced power-saving window is $N^{5/6+\varepsilon} \ll M \ll N^{6/5-\varepsilon}$. Bettin–Chandee [1] similarly allow three

separate sequences and prove, for nonzero real ϑ ,

$$\left| \sum_{\substack{a \sim A, m \sim M, n \sim N \\ (m, n) = 1}} \nu_a \alpha_m \beta_n e(\vartheta a \bar{m}/n) \right| \ll_{\varepsilon} \|\nu\|_2 \|\alpha\|_2 \|\beta\|_2 \left(1 + \frac{|\vartheta|A}{MN}\right)^{1/2} \times \{(AMN)^{7/20+\varepsilon} (M+N)^{1/4} + (AMN)^{3/8+\varepsilon} (AN+AM)^{1/8}\}. \quad (45.5)$$

Neither statement allows the joint coefficient matrix $G_{m,p}$. An exact singular-value decomposition introduces its nuclear norm. Worse, (45.2) asks for the complete row-by-row h -norm: summing DFI over h loses the full number of frequencies, while dualizing for Bettin–Chandee introduces an arbitrary (h, p) matrix. Their scalar savings do not pay these projective-norm losses. A long interval for q does not help because q stays inside $G_{m,p}$, and near $p = X/z$ that interval can be arbitrarily short. Thus the actual DFI PDF and the Bettin–Chandee theorem do not prove (40.19); the missing input is an operator/large-sieve estimate for the joint matrix, not a stronger scalar bilinear form.

Theorem 20.2 (small-fibre moving- s closure). *Let $\mathcal{E}(W_0)$ contain all endpoint incidences with $m = 4c^2s \leq W_0$. If $4 \leq W_0 \leq z/2$, then*

$$\mathcal{V}(\mathcal{E}(W_0)) = \sum_{z < p \leq X/z} \frac{1}{p} \sum_{4 \leq m \leq W_0} G_{m,p}^2 \ll W_0(L^3 + L^2 \log L). \quad (45.6)$$

Consequently $W_0 = \lfloor L^3/(\log L)^2 \rfloor$ gives $\mathcal{V}(\mathcal{E}(W_0)) \ll \Lambda^2$ over the entire endpoint prime range. This is the injective prefix used by the wider closure below.

Proof. Here $m, m' < p$ and $|m - m'| < p$, so distinct supported cells are distinct modulo p ; inversion preserves distinctness. Orthogonality of the full h -sum in (45.2) therefore leaves no cross-cell terms and proves the equality in (45.6). For each unique decomposition $m = 4c^2s$, the cells with common s lie in the fixed- s family of Theorem 20.1. Its positive diagonal gives

$$\sum_p \frac{1}{p} \sum_m |G_{m,p}^{(s)}|^2 \ll L^3 + L^2 \log L.$$

There are at most $W_0/4$ possible squarefree s , which proves the upper bound. The specialization uses $W_0 = o(z)$ and $W_0(L^3 + L^2 \log L) \ll L^6/(\log L)^2 = \Lambda^2$. $\square$

The injective prefix is joined by a supported-cell argument. If $\mathcal{G}_C$ denotes cells $m = 4c^2s$ with $c \geq C$, progression occupancy of their cofactors gives $G_{4c^2s,p} \leq 2/z + L/(2c)$ and hence

$$\mathcal{D}(\mathcal{G}_C) \ll \frac{L^4}{C \log z} + \frac{L^4}{z(\log z)^2}.$$

A residue bucket modulo p contains at most $T/p + 1$ cells with $m \leq T$; thus Cauchy gives $\mathcal{V}(\mathcal{A}(T, C)) \leq (1 + T/z) \mathcal{D}(\mathcal{G}_C)$, where $\mathcal{A}(T, C) = \{m \leq T, c \geq C\}$.

Theorem 20.3 (prefix/wedge consequences of source Theorem 46.2). *Put*

$$W_1 = \left\lfloor \frac{zL^2}{\log L} \right\rfloor \asymp \frac{L^5 \log \log L}{(\log L)^2}, \quad W_2 = \lfloor zW_1 \rfloor. \quad (46.8)$$

Then

$$\mathcal{V}(\mathcal{A}(W_1, 1)) = O(\Lambda^2), \quad \mathcal{V}(\mathcal{A}(W_2, z)) = O(\Lambda^2). \quad (46.10)$$

Thus the full prefix through W_1 and the wedge $c \geq z$, $m \leq zW_1$ close over the complete endpoint-prime range.

Indeed the two substitutions in the preceding diagonal bound give (46.10). Set $\mathcal{A} = \{m \leq W_1\} \cup \{c \geq z, m \leq W_2\}$, $\mathcal{B} = \{m > W_1\} \cap (\{c < z\} \cup \{m > W_2\})$. Then $\mathcal{V}(\mathcal{A}) = O(\Lambda^2)$, while the complete cell diagonal is $o(\Lambda^2)$. Positivity and the two-vector inequality give:

Corollary 20.4 (exact endpoint remainder; source Corollary 46.3). *The endpoint target (40.19) is equivalent to*

$$\sum_{z < p \leq X/z} \frac{1}{p} \sum_{\substack{m \neq m' \in \mathcal{B}_p \\ m \equiv m' \pmod{p}}} G_{m,p} G_{m',p} = O(\Lambda^2). \quad (46.13)$$

Equation (46.13) is the current pair-level open core, not all $m > W_0$. Theorem 8.8 refutes the literal raw hierarchy (40.28) and raw- H target (37.19), while Theorem 8.12 refutes the antichain hierarchy (47.16) and the ordinary reduced-count moment target. Neither refutation reaches (40.19), (37.27), H'_{PF} , or the conjecture.

Status. This is not a submission. Both 3/4 headlines remain CLAIMED/PROVISIONAL, especially through Theorem 9.4; the internal proofs and refutations above do not close the open critical-window ledger, (46.13), or H'_{PF} . The raw and antichain factorial-moment formulations are refuted and that axis is dormant; the pair endpoint and other possible pointwise minorants retain exactly the narrower open scope stated above.

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